

Mus 201 History of Western Music

Numbered Transcript

MUS 201 • Samuel Andreyev

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Lecture 1: Musical Horizons

At a glance — Musical horizons — the course begins.

- 1.1 I have to say this is really unbelievably exciting because I've been preparing for this for, I don't know, my entire life, basically. This is really exciting. Music is my entire life. This is really my life's work. I'm a composer, I'm an educator, I'm a teacher, a public speaker. Basically, everything I do is related in some manner to this central thing, which is music and communicating about music. This is really just an extraordinary opportunity. I'm grateful for all of you for coming out and listening to me talk for a little bit about this topic, which I believe is an immensely important topic.
- 1.2 I wouldn't devote my life to it if it weren't the case. I was thinking about this a little bit this morning. I'm going to tell you an anecdote here. I was being driven here in a Lyft car, and I noticed on the window there was a little sticker that said, "Tips are not required, but are greatly appreciated." I was reflecting on this. The thing is, there are all kinds of things in life that are not required, that are discretionary, like tipping, tipping a Lyft driver. It's like, if you're not completely socially adept, you might see a sticker like that and think, "Why would I give money to this guy, voluntarily?" It's like, I don't have to. It's not required. What that sticker actually means is you better tip that guy unless something went seriously wrong in your drive. That's a discretionary thing.
- 1.3 The thing is, it's like, that's what causes our society to function. It's things like that. Music is one of those things in a sense. It's not required. Nobody has to listen to music. Nobody has to engage with it. Nobody has to try to go deeper with music. But it's a discretionary thing, but I think it's such an important topic. You won't die if you don't listen to music. You as an individual are not going to starve to death. You're not going to have your life completely fall apart if you stop listening to music.
- 1.4 But I think that collectively, as a society, if we didn't have music, our society would just shrivel up. I think that that would be a catastrophe. I think that on a certain basic level, everybody knows this. Everybody knows this. It's very, very rare to find anybody in life who doesn't love music. Occasionally, you do. I've met maybe a couple of people that claim not to be interested in music. Or they'll say something like, "I love music, but I don't have time or I'm too busy."
- 1.5 This lecture series is going to be an encouragement to go farther, to dig into music as a phenomenon, to understand its origins, to understand how music works, and also to become better listeners, which is really a key thing. I was thinking about this the other day. What makes a composer? I mean, that's my main thing. I'm a composer. I write music for the stage. I write music for orchestras and ensembles and soloists and singers to perform. A composer, fundamentally, is an expert at listening. A composer is somebody who spent their entire life learning how to listen, learning how to make fine distinctions when they're listening to a piece of music. Also, obviously, you have to

have an inventive faculty. You have to have the capacity to imagine music, to write it down, to communicate it to other people, and so on.

- 1.6** But so much of it is about listening, fundamentally, and learning to listen. Listening is an art. It doesn't just happen. You can think of this as being the difference between looking and seeing. Anybody can see things. Just by virtue of having functioning eyesight, you can go out and see things. But looking is different. Looking involves intention. It's the same with listening versus hearing. There's a wonderful quote from *The Poetics of Music*, which is a series of lectures written by Igor Stravinsky that I love. He talks about the fact that hearing is not particularly special. I mean, everybody hears. A duck hears, he points that, he points out in his lectures. A cat hears. Hearing is just a passive act. It's just sound waves hitting your eardrum. You can listen unconsciously without intention, without understanding.
- 1.7** But listening requires a whole different set of faculties. That's going to be a lot of what we're going to be focusing on today, is really just learning how to listen and learning how to appreciate this phenomenon. I believe that music is one of the most profound things that we have in our culture. It reaches extremely deep. It can touch your soul. It can completely change your worldview. And what I'm going to encourage everybody to do is to see how far they can take that. We'll see where it leads.
- 1.8** So who is this course for? Well, it's really for anybody who enjoys music, who's interested in it, and who might be curious about going farther. That's really the essence of it. So this is not going to require any specialized knowledge. I'm keeping technical terms to an absolute minimum, and I'm going to try to explain them very carefully. There'll be plenty of time for questions as well. If there's anything that isn't clear or that you'd like me to expand upon, we can certainly do that at the end of every lecture. That's who this lecture series is for. I think a basic, simple prerequisite would just have to be you're interested in the topic, and that's enough.
- 1.9** Now, what I'm going to try to do is lay out something quite expansive, let's say, in its scope. We're going to talk about individual musical works. I'm going to play some examples at the keyboard and illustrate what it is I'm talking about. I'm going to talk about the history and the cultural dimension of music as well. It's actually very difficult to separate those things out. That's one thing that I'd like to emphasize a little bit before we really get into the heart of the discussion.
- 1.10** We have this idea that music is a universal language, that anybody anywhere can hear a piece of music and instantly understand it because it touches you on an emotional level or because there's this abstract power of sound. I think that's true to a certain degree. But there's also a lot that you won't necessarily be able to hear if the culture is really distant or if it's really foreign for you. And so we're going to talk a little bit about things that are, let's say, a little bit more culturally specific, things that you might need to delve into a little bit more in order to appreciate in greater depth what's going on in a piece of music.

- 1.11** For example, the first time that I heard traditional Tibetan music, I was struck by it. I was struck by how it sounded. I was really impressed by it on many levels. But I'm not from Tibet. I don't understand what goes on in the monasteries in Tibet. I don't understand the background, what it is they're doing, what the religious or the ritualistic signification of these sounds is. When I started to read about that and understand it a little bit better, then I could really appreciate it a lot more.
- 1.12** I would suggest to you that, even though you may be listening with intent, even though you may be appreciating and enjoying a piece of music, there's always going to be a certain degree that escapes you, I think. But that's also one of the great joys of music. It's a remarkable thing that you can take a piano sonata by Mozart or a song by Burt Bacharach or Bing Crosby, or it doesn't matter. You can listen to it 300 times, 1000 times, 2000 times, and not only continue to enjoy it, but to get more out of it each time you listen to it. And that, I think, is an extraordinary thing. How is that possible? Bach died in 1750. I mean, 273 years ago. How is it that we're still listening to his music today? And not only listening to it, but still learning to listen to it, still getting into the depths of everything that he had to contribute to humanity. We haven't exhausted it, not even close. It might be an inexhaustible force. I mean, that's an incredible to contemplate.
- 1.13** Here's a bit of a proviso before I start. This course cannot be comprehensive. It's impossible. It's such a gigantic topic. We have 10,000 years of music history in the West. That's a conservative estimate. By 10,000 years, I'm talking about evidence of music-making, instruments that have survived, depictions, descriptions of music, writings about music, all this sort of thing. That covers obviously a vast geographic region. I'm not going to be comprehensive. There's no pretension whatsoever that I'm going to be able to do that.
- 1.14** Not only that, but I'm going to focus on things that have been important to me and that I think that I have, hopefully, reasonably intelligent things to say about. But there are going to be huge gaps. That's just unavoidable. For example, opera is an extremely important dimension to the development of Western music. I'm not going to touch on opera, really, in this course. There's going to be, really, entire areas that I'm just not going to be able to talk about. But what I am going to do is hopefully extend out a thread that has a narrative structure to it that shows you what are some of the key moments in the development of the Western musical enterprise. And what I hope is that this will encourage you to increase your listening and also the depth of your listening. I guess we'll start there. It's difficult to compress 1000 years of musical history into a few hours, but let's see how we do.
- 1.15** The other thing I want to talk about, and this is something that I constantly advocate for is, I talk to lots of people, lots of people involved in music, lots of people not involved in music, and I meet a very, very wide range of people throughout my different activities. I travel all the time. One thing I've noticed is that there's a sense

that a lot of people are—what would you say?—reticent, maybe, about expanding the range of what they listen to. By that, I mean, there's a sense that you want to stay close to the things that you already know you enjoy. And that's understandable. That's an impulse that everybody has. I certainly do. I remember actually when I was a child, I used to want to watch the same movies over and over and over and over and over again, whatever, the Smurfs or Charlie Brown cartoons and things like this. My dad used to say, "Why don't you watch something else? Why are you watching this movie 50 times?"

- 1.16** I understand that impulse. It's like you want to stay close to something that you already know you're going to get a dopamine hit from it. You're going to enjoy it. It's going to relax you. You've already had that experience. It took me a while to get out of that loop, I think, of just sticking to things that I already knew. But I really want to encourage you to consider the fact that there are great benefits to adopting a different relationship to art. Cultivating openness, curiosity, expanding the range of what you can enjoy—it's a hugely beneficial thing. In my experience, most people tend to rely too much on the memory of things that they enjoyed in the past. That really does limit you. It limits who you can become. You could be a hugely expanded individual, and music can help that to happen. That might seem like an extravagant claim, but I really do believe that's true.
- 1.17** Why is that? Well, let's talk about the functions of music. Music, I think everybody knows on some level, is a transcendental experience. It's not a commodity. You can't really buy and sell music. I mean, you can buy an album, I guess. Hardly anyone does that anymore. You can buy a Spotify subscription. You can buy a ticket to a concert. But it's not fundamentally a commodity. It's an experience. But what music can do is, it takes you away from logic and reason. It takes you away from the everyday. It takes you away from the limitations of space and time, the limitations of what we can physically do in life. That's a really powerful thing.
- 1.18** In our daily lives, we're constrained by all sorts of things. We're constrained by gravity, we're constrained by lack of time, we're constrained by the limits of how far we can walk in a given day, how much we can do in a given day. What music can do really is, well, it gives you the opportunity to experience vicariously a kind of highly abstracted form of movement, goal-directed movement, movement that is taking you from somewhere to somewhere else. You can do that and have this experience, and you're not actually moving, obviously. That is such an important thing, I think, in everyday life, to be able to transcend the everyday, transcend the banality of whatever it is, buying groceries or anything like that. It allows you to connect with something that is much bigger than yourself.
- 1.19** In that sense, I believe that music is, fundamentally, whether one wants to acknowledge it or not, whether or not one identifies with a particular religion or not, I believe that music stems from a religious impulse, a form of religious impulse,

ultimately. That doesn't matter if you're listening to a piano sonata or death metal or EDM, whatever. It's the same. It's a transcendental experience, ultimately. Here's one of the central paradoxes of music that I find just so fascinating and actually difficult to grasp fully. Music is an embodied art form, by which I mean it's directly related to physical phenomena. OK, so where does music come from? Well, it comes from the heartbeat in your chest. It comes from the blood coursing through your veins. It comes from the rhythm of breathing, the momentum of walking, the momentum of running, these sorts of things. Things that are located in the body that are physical, corporeal experiences. This is an embodied phenomenon.

- 1.20** There's a wonderful quote by the American poet, Ezra Pound, who says that, "When music strays too far from the dance, it wilts." I think that's right. I think that's right on a fundamental level. You can define dance however you want. There are a very broad range of—there's a very broad range of different ways of moving, obviously. You can see that very easily if you listen to music of different cultures. But it's an embodied art form. At the same time, and particularly, I would argue, in the Western tradition, it's the most abstract of the arts. It's literally made of air. Music is a disturbance in the air. We have periodic waves, or not periodic waves, depending—we'll talk about that a little bit later—that create a disturbance in the air and hit our eardrum. It's like, how do you describe a phenomenon like that?
- 1.21** And music is not even really made up of things that you can easily identify. I mean, of course, a professional musician can put names to things. A professional musician can say, "Well, this is an exposition. This is a theme. This is a motif. This is a chord. This is a chord progression," and so on. They can identify those things and label them. But in a sense, the experience of it is extremely hard to describe. You can try this as an exercise, actually. Listen to your favorite piece of music and then write about your experience of listening to it. What did you experience? What was it that you just experienced? Not easy.
- 1.22** So it's an embodied art form, but it's extremely abstract at the same time. It consists of all of these complex interrelationships and weaving together of patterns on multiple different levels that is very, very difficult to describe. That's one of the things, I think, that makes it so powerful. You respond to it physically, you respond to it corporeally, but there's also something in it that's absolutely beyond the physical and that is beyond possession. The last thing I'll just say about this point is, I think the fundamental things to be aware of in music are, well, it's a multi-parametric art. We'll talk about that a little bit later. That's a mouthful. But basically, it's an art form that has many different dimensions to it that are all operating on different levels at the same time. There's rhythm, for example, but there's also pitch, there's also melody, there's also tone, there's also timbre, there's also intensity, how loud or soft it is.
- 1.23** But the key things, I believe, are rhythm by way of the feet, and melody by way of the breath. Melody is a proxy for human speech, fundamentally. That's where I believe it

comes from. If you've ever been to a really deadly public lecture where you're just falling asleep, you can't stand it, it's so boring, often what's going on is the person who's speaking doesn't know how to modulate their voice. So they're talking in a monotone all the time like this, and there's no —you know, that kind of thing. It's like, that'll put you to sleep because there's no melodic dimension to it. I don't know how else to describe it. The voice rises, it falls, it has a sense of cadence, it has a sense of conclusion, it has a momentum to it, it has a rhythm. There's a melody to speech. And that's where I believe melody comes from, fundamentally.

- 1.24** So this course is going to focus almost exclusively on Western music. And I'm not making any claim as to the superiority or the greater complexity of Western music. It's simply a reflection of the fact that I am a Westerner. This is my culture. This is what I've been studying my whole life. And so that's going to be my point of emphasis. But there's a couple of things that I think we have to establish a little bit. One of them, and we're going to get into this in detail, is what makes Western music what it is? What makes it different? What makes it distinct from other cultures?
- 1.25** Then the other question is, because I'm going to be talking a lot about what we call classical music. You talk about classical music to people, and very often their eyes will glaze over. They imagine themselves sitting at some interminable concert that's 90 minutes long or even longer, maybe. There's too many instruments, and it's too complicated, and there's no rhythm, and it's boring, and it's expensive, and you have to get there, and you have to dress up, and all this stuff. It's like, who wants to bother?
- 1.26** We don't actually reflect on what classical music is, where this concept came from. The concept of classical music is really a matter of historiography, which is to say that, this concept had to be invented. It stems from a very particular period. It really stems from the 19th century, I would argue. It's a creation of the 19th century in the sense that, before that point, there was just music. There was just the practice of music. There were all kinds of different sorts of music, obviously. There was light music, there was serious music, there were popular styles, there were savant styles. There was music for dancing, music for courts, music for ceremonies, and so on. But we didn't have this concept of classical music as distinct from other forms of music.
- 1.27** So why is that? Why do we now have this idea that there's classical music on the one hand, and then there's, what, popular music, commercial music, dull music, all these other genres on the other side? See, I think these things are all actually closely interrelated. I don't believe that this division is particularly relevant, particularly in the modern age. I have to be a little bit careful making that claim because there's lots of—it's trendy at the moment to talk about pop music in an academic setting, which is—I'm not going to do that. I think that's silly. It can be, it can be very interesting, but that's not what I'm intending to do.
- 1.28** I should also make clear, by the way—I didn't mention this—but I'm actually not an academic. I'm a practical musician. I'm a working musician. I'm somebody who writes

music for a living and who is involved in music on that level, on the level of an artisan or a craftsman, I would say, and an artist, certainly. But I'm not somebody who primarily spends time, what, researching, writing theses, writing dissertations, articles, papers on music. I don't do that. I really am a practical musician, I would say. But nevertheless, I would argue that pop music is an out-branching of the classical tradition. They have quite a lot of things in common.

- 1.29** Actually, when you really get down to it, to what might actually separate these things, I think there's one primary distinction that I would just make right off the bat, which is that classical music, what we call classical music, is fundamentally a written tradition. It's the concept of notation, music notation. This is really key. The way pop music tends to function is that, first of all, it's a type of public property. It belongs to everybody. Pop music is something that anybody can enjoy, anybody can access. You don't need any special skills, particularly to play it. You can learn a few chords on a guitar, you can learn a couple of chords on the piano, and you can play a huge amount of pop music already. You might not do it well. We might not necessarily want to listen to your voice as you try to sing, but it's accessible. It's really accessible, and you don't need to learn to read music for it.
- 1.30** There are extraordinarily gifted pop musicians that can't read music. I find this amazing. Paul McCartney can't read music, and he's the most successful composer in history, really, in terms of just the sheer number of people that listen to his music, certainly the success of his different musical ventures. And he can't read music. So how about that? Being able to read music is not an essential prerequisite for being a musician, even a great musician.
- 1.31** But there is a distinction between musical traditions that are fundamentally written traditions and musical traditions that are, what would you say, passed down from person to person, where someone can show you how to play a few chords. You don't need to be able to—you just need to know where to put your fingers on the fretboard. You can do that. The melodies, you can pick them up pretty intuitively. Occasionally, you'll meet someone who's so tone-deaf that they can't do that. They can't sing a simple melody. That does exist. But think of it as a bell curve. I think most people can do that. Most people can do that.
- 1.32** What does it mean that classical music has a written tradition? Well, it means that we have these documents that we call scores, which allow a composer to fix for an extended period of time the parameters of the piece that he wants the performers to play. He can specify pitches, he can specify rhythms. As music notation developed throughout history, it became more and more precise, more and more specific. It made it so that, even if you lived 2,000 kilometers away from where the piece was written, it didn't matter. Someone could send you the score, you could pick it up, you could read it, you could reproduce that music. That was an absolutely fundamental revolution in the development of this art.

- 1.33** Now, things look a little bit different because, well, if you consider what music notation is, we still think of it primarily as notes written on a piece of staff paper. But there's other things that music notation can be. If you wanted to be a little bit provocative, you could even suggest that newer technologies like MIDI, for example, which is a way for computers and musical instruments to interact with each other, is a form of notation. You can play a sound on a MIDI keyboard, M-I-D-I. It's a keyboard that translates pitches, intensities, durations, and so on into numerical values and then sends that on to a computer. This is technology that is very widely adopted in many different areas of music. That's a form of notation. It's a way of encoding music and then being able to reproduce it somewhere else.
- 1.34** The advent of the player piano also is, in a sense, it's a type of notation. It's a way of fixing a piece of music and being able to reproduce it. You know what they do with player pianos? There's a roll of paper and they perforate it. They put little perforations in it. Then there's this pneumatic contraption that causes that to send an impulse to a hammer, and then the hammer strikes the string. It's amazing. As with lots of things, these definitions are somewhat fluid, I would say. But for the vast bulk of what we consider to be the classical repertory, this distinction does hold clear, I think, which is pieces that are primarily written.
- 1.35** So what does that do? If a piece of music is written down, you can do things that you can't do extemporaneously. It's the same with writing. It's the same with literature. It's like, if the only way to write a novel were to memorize the entire thing and then say it and then have other people memorize it too, and then transmit it to each other that way, not very many novels would have been written. With music, what that allows you to do, if you're writing down a piece of music, is to go more slowly. It's to think very carefully about what you're doing. You can also achieve a degree of complexity and richness that might be difficult otherwise. You can also coordinate all kinds of different parts. You can write a piece for 200 instruments if you want. Write an interesting part for each of them. That's very, very difficult to do if you don't have some a written tradition. Those are a couple of points.
- 1.36** There is a distinction on that level between popular and classical music. But I also want to just make another point about this, which is that, in some circles, popular gets a bad rap, which is completely wrong. I want to talk about that. We tend to confound or to confuse the popular and the commercial. Those are not the same thing. I want to immediately make that distinction clear. Popular music is not equal to commercial music. I don't know why we lump those two categories together. You can think of this as being like a Venn diagram. There's the popular and there's the commercial, and sometimes they overlap. There's a bit of—they meet in the middle, but they don't overlap completely. Popular musical traditions exist all over the world, and they're the life's blood of music in a sense. In fact, classical music really stems from that. It's not

the other way around. Classical music comes out of, to a large degree, popular traditions.

- 1.37** What's commercial music? Well, commercial music is music that's written primarily to generate profit, or at least, if not primarily, then that's a significant part of the incentive. But again, we have to be careful because it's tricky. You can write music for profit and be a profound artist. But I just wanted to start by making that distinction. Classical music, pop music, commercial music, they all have points in common. They all overlap to some degree. Even the most serious, profound, ambitious composer, it's very rare to meet a composer who isn't fond of, is completely allergic to making a bit of profit. Again, these categories do meet.
- 1.38** So I talked about the fact that classical music, or Western music generally, has some aspects that are different from, or at least that have different points of emphasis, from other musical cultures, and we're going to talk about that a little bit today. When we talk about music in the West, I think that there are certain things that have simply been greater points of emphasis, that have been higher priorities. You can see this. We're going to look in a minute at examples of different music cultures from around the world and what makes them distinct. What you'll see is, it's such a complicated phenomenon that, of course, there's lots of different ways that you can approach it. There can be different things that you're focusing on.
- 1.39** Western music has tended to focus on the development of harmony, which is to say notes being played together. Let's talk about that for a second. There's a thing in music called the interval, and an interval is simply the space between two different tones. So you can have a small interval or a large interval. A small interval would be something like this. [music] And a larger interval would be something like this [music] or [music] or [music]. You can have huge intervals like that that span multiple octaves. Or you can just have a semitone, which is the smallest interval you can play on a piano. [music]
- 1.40** So what is harmony? Well, harmony is two or more tones sounding together. This is a melodic interval, two tones played successively, and a harmonic interval is two tones played at the same time. I want you to retain that concept. Melody is a succession of tones [music], and harmony is tones played simultaneously [music]. OK, so when you have harmony, you have multiple tones being played together. Of course, a chord can be very simple, it can be very complex, it can be 2 notes, it can be 12 notes, it can be anything in between. When you have the possibility of different sounds happening at the same time, you have the possibility of polyphony. Polyphony is simply multiple sounds, or more precisely, multiple layers in a piece that are going on at the same time.
- 1.41** Again, in order to push polyphony to a high degree, you either need to have a musical tradition that is absolutely immersive, which is to say, where music is just an absolutely integral part of everyday life. We're going to talk about that in a second. There are cultures that are like that, where everybody's a musician. They work at it all the time, and they're extraordinarily virtuosic at it. You need either something like that, where

you can spend whatever, 10 hours a day memorizing a piece of music, and then you can have these intricate polyphonies, because it's very, very difficult to do. Or else you need some kind of method of transcription or notation that allows you to write down a line or a voice, then write down another voice that accompanies that first voice.

- 1.42** So these are dimensions that I would argue are particular points of emphasis in the Western musical tradition. But before we talk more about that, let's talk about what other cultures look like. So I mentioned a little bit earlier, I used this inelegant word, which is multi-parametric. So music is a multi-parametric art form. What does that mean? What it means fundamentally is that sound has all these different attributes. Music has different attributes to it. You can focus on one of them. You can focus on all of them. You can have one to the exclusion of the others. Or you can combine them in lots of different ways.
- 1.43** So for example, frequency. Well, what's frequency? Frequency is what's actually happening with sound. It's funny because we use metaphors to describe tones. We use metaphors to describe melodies. We say, "Well, that melody is going up. It's going down." We say, "That's a low note. That's a high note." We use spatial metaphors for a phenomenon that has nothing to do with space. What's going on are frequencies, fundamentally. If you play a low note, what we consider to be a low note, then the frequency, the length of the cycles is longer. If you have a high note, then the cycle is much, much closer together. So instead of talking about low notes and high notes, we should probably be talking about slow notes and fast notes, but that's totally counterintuitive. I don't think anyone's going to adopt that anytime soon.
- 1.44** So we have frequency, and frequency is, well, what I just played. It's pitch. It's the fact that I can sit down on a keyboard and say, "This is a C [music]. This is a C sharp [music]. This is a G [music]. This is a D [music]." All of these tones are associated with specific frequencies. For example, the A above middle C [music], you can measure that in hertz. That's 440 cycles in hertz. Any harmonic tone can be expressed that way. So that's pitch, that's frequency. The other is duration. We don't typically think of music in terms of duration. Duration is a kind of, what would you say? Well, it's just not how we listen to music. We think of rhythm. Duration is a slightly different concept.
- 1.45** Rhythm is a relative phenomenon. Rhythm is something like the way a musical figure interacts with a pattern. It's usually not exactly lined up. If you had a music in which there was an underlying meter, an underlying pattern, and then you just stuck robotically to that pattern and never changed it, then the rhythm would be extremely simple and probably not very interesting. But the music that we love, the music that has this quality of liveliness and music that's exciting, is always just pushing a little bit against that pattern, against that grid.
- 1.46** Duration is simply a different way of thinking about it, I suppose. You can talk about it in phenomenological terms. For example, we know that there's a limit point beyond which it becomes very difficult to distinguish between different durations. If I play you a

sound and it lasts for half a second, and then I play you a sound that lasts for 2 seconds, you can clearly tell that one of those sounds is longer than the other without any difficulty. At a certain point, you can't make that distinction anymore. We're not actually really wired for understanding duration on a sophisticated level in the same way that we might be able to appreciate differences in pitch. Humans can distinguish between incredibly refined, minute, different pitches. Duration, not so much.

- 1.47** If I were to play you a sound that was 6 seconds long, and then a sound that was 6.5 seconds long, I guarantee you, none of you would hear the difference. I certainly wouldn't hear the difference. So beyond a certain point, duration starts becoming a matter of—well, you just can't make the distinction anymore. So we're really oriented towards shorter durations, generally speaking. But think of duration as a category that encompasses different aspects of rhythm. Some rhythms are periodic, which is to say that they have a regular pattern of weak and strong beats. One, two, three, three, four, one, two. That's a basic simple march rhythm. That's a periodic rhythm, which is to say that that pattern just repeats itself indefinitely until the piece is over.
- 1.48** But there are other musical phenomena where you don't really hear a grid. You don't hear a pulse. You hear sounds that last for a certain amount of time. There's lots of ambient music that's like that, or certain traditional musical cultures where the sounds are held for a very long time. There's no sense of pulse, really. That might be a meaningful distinction that we can get into a little bit later. Pulsed music versus unpulsed music. That's duration.
- 1.49** Timbre. That's another thing. So timbre is an interesting one. What is timbre exactly? You're probably familiar with this term, but timbre is the quality of a sound. It's what makes it so that a piano sounds different from a clarinet. It's what makes it so that a guitar sounds different from a bass guitar. It's not just the fact that the sound is lower or higher. It's that there's a different quality to the sound. That's where things get a little bit complicated because timbre is actually closely related to frequency, strangely enough. What timbre actually is, and this is where we have to get slightly into the physics of sound, is it's the particular overtone spectrum of a sound.
- 1.50** Now, If I play you a sound on the piano, the C, for example, [music], what you're actually hearing is not just one tone. You're hearing a whole spectrum of sounds. You're hearing a fundamental tone, which is the lowest sound. Then you're hearing all of these overtones, and you're not aware of this. Our perception plays tricks on us. We just hear this composite sound. But harmonic tones, tones that have an exact pitch to them, have overtones, sympathetic vibrations, if you like. And timbre has got to do with that. It's the particular distribution of energy, of intensity, in the sound spectrum.
- 1.51** So if you have a sound that has a lot of energy towards the upper end of the spectrum, it'll sound buzzy. If you have a sound that has a very, what would you say, a relatively, not a very rich spectrum, it'll sound hollow. A flute, for example, sounds a lot more pure in a sense than, let's say, an oboe or an electric guitar, simply because the electric

guitar is going to have way more partials in the sound. It's a more complex, it's a richer sound. That's timbre. I said that it has something to do with frequency, well, because it has to do with the number and the distribution of overtones in that sound.

- 1.52** Finally, intensity is the fourth major parameter of sound that we're going to talk about. Intensity is easier to describe. It's the energy in the sound, how loud it is. Now, all of these parameters are, weirdly enough, interrelated. You can't really separate them out very easily. If I change the intensity of a sound on the piano, I'm changing its timbre. If I play a note on the keyboard, very, very softly, there's going to be less energy in the sound, and I'm not going to be exciting the upper partials as much. The timbre is going to be a lot more, what would you say? Pure is the wrong word. Warm, maybe. It's going to have maybe a slightly darker sound. Hard to describe these things, again, without getting into visual metaphors.
- 1.53** If I played that same note really loudly on the keyboard, it's going to take on a bit of an edge to the sound. It's not just because it's louder, it's because timbre has changed. Let's try this. I'll take a relatively low note. [music] Now, the harder I strike that key, the more energy will go into the string, the more the string will vibrate, and the more the upper partials in the sound will come out [music] versus [music].
- 1.54** It's not the same timbre. You can see this also very clearly with instruments that can change their level of intensity easily, like a clarinet or a saxophone, for example. Ask a saxophonist to play a sound as softly as they can, and then the same note as loudly as they can, and the quality of the sound will change, not just the intensity. So music is a multi-parametric art form. And what that means is, again, different cultures tend to emphasize different things. There's all kinds of reasons for that. I think on one level, you can make a distinction between indoor cultures and outdoor cultures. Cultures where people are, for whatever reason, because the weather sucks, they're indoors all the time. And then cultures where it's warmer, a warmer climate, you're going to be outdoors all the time. You make a totally different music, depending on what the climate is.
- 1.55** My wife's Danish, so I spend a lot of time in Denmark, and yeah, the weather's pretty bad most of the time. It's the same with other parts of the world. Holland, for example. Why do they have all those beautiful paintings, the Dutch old masters, of people in dimly lit rooms? Interiors, it's always interiors, Dutch interiors. Well, because they have to be indoors. It's cold out. And that has consequences on the development of a musical culture.
- 1.56** So here's a few different examples of music from around the world. So we're going to listen to an example of Carnatic music. And Carnatic music is, well, perhaps the way of thinking about it is, it's the classical music of the Indian subcontinent. It's a vast body of compositions. It has a very different point of emphasis. One of the things about Carnatic music that I find fascinating is, in Indian music, of this region, you have this emphasis on melody, and not just any type of melody, but a highly ornamented form of

melody. You don't just have the skeleton, one note after another, but you have all these rich ornamentations and little trills and little subtleties and slight shifts of pitch that are unbelievably sophisticated.

- 1.57** They have this repertory of hundreds and hundreds of different modes and scales that they use, which don't actually correspond with the scales that we use in the West. Different tuning systems, obviously different instruments, and different points of emphasis. There's an extraordinary melodic virtuosity in Indian music that never developed in the same manner in the West. So what we're going to do, we're going to listen to an example of a piece by Thyagaraja, probably murdering the pronunciation. Thyagaraja was a composer of Carnatic music, who lived from 1767 to 1847. And he wrote thousands and thousands of pieces, thousands of songs. In that sense, he's sort of like an Indian Schubert. Schubert also wrote countless songs, hundreds and hundreds of them. And these are largely devotional songs, and I think they're very beautiful. So let's listen to one just to get a sense of what Indian music sounds like.
- 1.58** [musical piece]
- 1.59** OK so if we pull that apart, I think that's really beautiful, actually. It's a really beautiful piece. There are three components to that sound that we just heard. There's a drone. That's the first thing you hear. It's this long-held tone played on a sitar or a similar instrument. That's an example of what I was describing earlier when I was talking about unpulsed music. There's no pulse to that. There's no rhythm. It's just a sound. It's modulated. The sound changes over the course of the piece. It gets brighter, it gets darker, but it's a drone. Then there's a rhythmic layer. There's the percussion. Tabla. Those rhythms are really complex. The masters of tabla playing, Indian drumming, can do things that make my head spin and at a velocity that's just incredible. I mean, these patterns are really, really complicated.
- 1.60** Then there's the melodic layer. The melodic layer, besides the fact that it's obviously, I think it's really quite beautiful, the melodic layer has all of these beautiful, subtle inflections in it. Those are the three components, I would say, to this piece, if I had to pull it apart the way a composer might do. But you'll notice that there's no harmony. There's no chords in this piece. There's no chord changes. The drone lasts for the entire piece from beginning to end. We have this fundamental tone that doesn't change. The melody, the melody is all based on a single scale. It doesn't modulate. We don't move to another key. We don't change chords or anything like that. And then there's this rhythmic layer.
- 1.61** The particular point of emphasis there, I think, is on this melodic invention and also, of course, on the rhythmic virtuosity. Let's listen now to something from another part of the world, from Indonesia. Indonesian music has fascinated composers and musicologists for a very long time at this point. There's a particular reason for that. There are two islands in Indonesia that have an absolutely extraordinary musical

culture. It's not absolutely unique. You can find other musical cultures in that part of the world that have some points in common.

1.62 But they have this type of orchestra in Indonesia called the gamelan. The gamelan is largely made up of percussion instruments, tuned percussion instruments for the most part—metalophones, to use the specific term, which is to say instruments that have metal bars or sheets of metal that are tuned and that can play different scales. They also have skin drums and bamboo flutes and other things like that. But the essence of the gamelan, I would argue, is the tuned percussion aspect.

1.63 This music is found mainly on the islands of Java and Bali. For one reason or another, we tend to associate it, probably incorrectly, with Bali, Balinese gamelan music. We're going to listen to an example of this. This music is also, as I mentioned earlier, a type of musical culture that is unbelievably intricate, and they don't have scores. They don't have a written musical tradition for this, as far as I know. I could be wrong about that, but as far as I understand it, these pieces are memorized, intricately memorized by all the people in a village. Basically, everybody is participating in this to some degree. Here's an example of Indonesian gamelan music.

1.64 [music]

1.65 OK. So, as you will certainly notice, that's a very different musical culture than the music of southern India, with, again, different points of emphasis. Now, one thing they have in common is they have these really incredibly intricate rhythmic patterns, different sorts of rhythms obviously, than what we heard the piece by Thyagaraja. But there's another point of emphasis in Balinese music that I think is really worth pointing out, which is this extraordinary timbre of all of these tuned metallic instruments. It's such a beautiful sound. One of the things that they do in these orchestras is they have different instruments that are tuned very slightly apart from each other. You have one set of gongs, for example, or a tuned metal bars that is tuned just very slightly higher. When you play both of them at the same time in unison, you get what are called beats, rhythmic beats. It creates this kind of shimmer, this sheen to the sound that is just really, really beautiful.

1.66 Now, in the extract that we just heard, we didn't hear the bamboo flute in that, although they do have that. The melodic dimension of that music is quite different from what we heard in the extract of Indian music. The melodies here are these little short figures that are repeated, sometimes in loops and sometimes in more complicated patterns. But basically, they're shorter, they're more discrete than what we had in that piece by Thyagaraja. Again, a different point of emphasis. You have this emergence of an entire orchestra of these instruments that is really unique to the gamelan. But again, to my Western ears, not really harmony, at least not the way that I would think of it. Maybe they think of it that way. I don't know. I'm obviously not Indonesian, but I can't help but hear things from my particular vantage point. So a different point of emphasis.

- 1.67** Timbre is extremely important. Rhythm is important. Melody is a little bit, what would you say, made up of these shorter fragments that are repeated and laid on top of each other and stitched together. But notice that there's no conductor. It's amazing that they can synchronize these complicated rhythms all playing together in unison. It's really extraordinary. OK, so I'm going to play you another example. And this is an example of West African music. So sub-Saharan African music has—this is a vast generalization, by the way, because sub-Saharan Africa is an extremely vast geographical region with many, many different cultures, with different musical cultures. And an ethnomusicologist would be able to expound upon this for hours and hours. I'm not a specialist by any means, but I am very fascinated by the music of this part of the world.
- 1.68** A lot of it is really focused on percussion, sometimes exclusively on percussion. What that means in practical terms is, well, no tuned pitches. You hear percussive sounds. And again, there's a level of elaboration, a level of sophistication, of virtuosity in these rhythms for which we simply do not have an equivalent in Western music. This is what happens if you focus on a particular dimension. You can go incredibly far with it. I would argue that—I talked about the fact that polyphony and harmony are particular aspects that are developed in the West. I would make the argument that in order to have polyphony, in order to have the capacity of superimposing multiple different layers in a composition, multiple different lines, different melodies going on at the same time, you have to simplify to a degree in order for that to function. Because you can't have everything be complex on all levels all the time. It would overload our senses. It would overload our perceptions.
- 1.69** My sense is we've accepted a trade-off to a degree. We've accepted that our rhythms are nowhere near as complex as what you're about to hear for the most part. You can listen to classical and Baroque music, and quite honestly, it's all based on 5 or 6 very simple rhythmic patterns. We've had to accept a degree of simplification on the level of rhythm and pitch in order to develop this concept of polyphony and harmony. Anyway, let's listen to this example of wula drumming. A wula drum, W-U-L-A, is a type of djembe, and a djembe is a skin drum that's tuned high. What you'll notice here is that there's a soloist playing in the middle, and then there's an orchestra of percussionists that surround him. Let's listen to this example of West African percussion music.
- 1.70** [musical piece]
- 1.71** OK, so here's the thing I find really phenomenal about this. I love this music. You notice that they're moving a lot as they play. The dance aspect of it is obviously integrated into the performance aspect. In the example of Indonesian music that we heard, they don't really move very much. It's got a hieratic dimension, which is to say, almost a kind of a, what would you say, a kingly kind of, very poised. Now, there is a whole tradition of dance in Indonesian music, but the dancers tend not to be doing the same thing as the performers. They tend to be a separate performer within the orchestra. In this

example, the dance aspect, the moving around, is an integral part of the performance, and it's wonderful to see.

1.72 So that's a little overview then of different musical cultures. Obviously, that's a small sampling and there's all kinds of different traditions that we could also talk about. We could talk about Korean music, and we could talk about the indigenous music of Australia or all sorts of different parts of the world. Arabic music. But unfortunately, there's only so much time. So I chose those 3 examples as illustrations of the idea that, well, that you can have different points of emphasis in different musical cultures. Let me play you something at the keyboard that illustrates this concept of polyphony. So here's a little—I'm just going to play 10 seconds of a piece by Bach. See if you notice something about this.

1.73 [music]

1.74 You notice that there are two melodies going on simultaneously in that piece. The right hand is doing this.

1.75 [music]

1.76 The left hand is doing this.

1.77 [music]

1.78 If you put them together,

1.79 [music]

1.80 The dialogue. Dialogue going on within one person who has to play both of these parts at the same time. So that's what I would call a different point of emphasis. And so the rhythms of what I just played are actually really simple. It's this repeating rhythmic idea. But it's in the combination of these different parts that you start to see this particular richness, I would say, or this different point of emphasis. So I'm going to stop there for today. And that's already quite a lot, I suppose.

Lecture 2: Musical Evolution

At a glance — Musical evolution across early periods.

- 2.1 We left off last time talking about music of different cultures and what makes them distinct, what makes them different from each other. We're going to shift our attention now to music in the West. What I'm going to try to lay out here is a bit of a chronology of the development of Western music. Now, obviously, you can go quite far back with that. But the problem is the farther you go back in time, the harder it is really to have precise elements that you can describe. And part of the reason for that is there was no music notation that we know of or that we're able to decipher before the Middle Ages.
- 2.2 So we have descriptions, again, of music in ancient Greece, for example. We have depictions of the instruments. We know that they had lyres, and we know that they had aulos and all these different instruments. But we certainly, obviously, don't have recordings. We don't have scores. A lot of it is necessarily going to be conjectural if we try to understand what that music is like, although we do some sense of how it fit into their culture. But what I'm going to talk about now is primarily the development of notation and the development of theoretical concepts in music. This really started to flourish in the Middle Ages.
- 2.3 The difficulty of talking about medieval music is that it covers an extremely broad time span. The Middle Ages is the period between the fall of the Roman Empire and the Renaissance, which covers approximately 500 AD to 1500. So that's 1000 years of music history. Imagine that. Think about everything that's happened musically in the past decade. Imagine what 1000 years is like. So that's a difficult thing to grasp. So obviously, you can divide that down farther into distinct periods, and you can look at particular regions of the world and things that were happening in different parts of Europe and so on.
- 2.4 But it's useful to think of this, I think, as a block of time, because this is really when music notation and music theory was developed. And not only in the Western world, I have to make that clear as well, because there are other musical traditions that have different forms of notation. But there's a particular emphasis on that, once again, in the West, and on finding precision in music notation that allows compositions, music, to be written down and transmitted across great distances.
- 2.5 The first thing you might be wondering is, well, why have music notation in the first place? The earliest ideas we have about a kind of, what would you say, a serious or an art music tradition has to do with music that was found in monasteries. We know that the monks in the Middle Ages used to chant, in unison, and this is called plainchant. What they're really doing is taking religious texts and assigning a shape to them, a melodic shape to them. They would chant these texts in unison. That obviously has a incantory function, a spiritual elevation that can come from that thing. But it's also a

mnemonic aid. This is kind of an interesting experiment. For a lot of people, it's a lot easier to remember the lyrics to a song than it is to memorize a poem. I think part of the reason for that is we associate the melody, the rising and falling of the melody, with the words, and it becomes a lot easier to remember them.

- 2.6 To a degree, the idea of putting a melody to words is a mnemonic aid. It allows you to remember the words more easily, more distinctly. But the earliest attempts of writing this down somewhat took the form of a notation that is going to look very strange, very foreign to people who've never seen this before. And it's not a precise notation. It's not an absolute notation, it's a relative notation. You can see an example of that on the screen. What you've got there is a text. Then above the text, you've got all these markings. These don't really look like notes. There's no staff, there's no lines that these notes are placed upon. These signs, there's a repertory of approximately 20 of these signs, and they're referred to collectively as neumes, N E-U-M-E-S. And this neumatic notation, as it's called, became standard by about the ninth century.
- 2.7 So what do these signs mean exactly? Well, again, it's a mnemonic aid. They don't tell you exactly what tones to sing. They don't tell you if it's a C or a D. And well, those appellations didn't even exist at that time anyway, as far as we know. But they do tell you what to do with your voice. So there are 2 simple symbols in neumatic notation that are very easy to demonstrate. There's a symbol like this—an acutus, it's called. These are extremely old signs. You see these in classic Greek and Roman notation. A sign like this, which means you go up, and a sign like this, which means you go down, and that's called a gravis. So acutus and gravis. With these two simple signs, you can tell a singer to raise their voice or to lower it. There are others as well, lots of other symbols in this notation.
- 2.8 So again, it's a relative notation. It's not an absolutely precise one, but it was enough if you were in a monastery and you wanted to memorize it, say, a corpus of compositions or of melodic lines, then you could do that more easily with the neumes, because even if you didn't necessarily know what the exact note was, it would remind you of the contour of the melody. That worked pretty well, provided that you had a lot of people living in one place, and you weren't actually attempting to have the piece be played anywhere else. Because if you were to take this neumatic notation and show it to a monk 500 kilometers away, or somehow get it to them, if they'd never heard the melody before, they would have had a hard time deciphering it just from reading the notes.
- 2.9 So what starts to happen is, you start to have these political entities that get larger and larger, and the need to communicate across vast distances. That starts to increase as well. There becomes a necessity. It's like, well, maybe you'd like to have a musical culture that can span a much greater geographical area. When that's the case, you need a more precise form of notation. So there's a kind of necessity in that sense. If you want to have a, some kind of a centralized musical culture, if you want to have musical

works and be able to transmit them to other people, and you can't just do it on a one-to-one basis—this person tells it to this other person and so on—then you need a more precise form of notation.

2.10 We're going to talk about how that occurred. But first, I want to give you an example of what plainchant sounds like. The first thing to know about plainchant is that there's no rhythm, there's no meter, as far as we know. There is no provision in this notation for writing down rhythms at all. These are just purely directional signs. There's no repetition either. But what you do have is a melody that's chanted in unison. It's always by all men or all women, but you don't have mixed choirs. They didn't have that. Here's an example of plainchant called *Viderunt Omnes*.

2.11 [music]

2.12 OK, so that's an example. Plainchant. You notice, again, no harmony, really. The rhythms are the rhythms of speech, the rhythms of the contour of the text, but they're not yet these articulated rhythms that we think of when we think of composition. So no harmony, just a single melody chanted by monks in unison. And that was the beginning of Western musical composition. You notice that it sounds really good in a cathedral. That's another interesting point here, that pieces of music and different musical traditions exist within not just this abstract sense of, well, there's singing, there's instruments and so on, maybe there's dancing. Music is performed in a space, and the space has acoustic properties, and that really changes things.

2.13 I've been inside Notre-Dame Cathedral in Paris. It's a huge building, unbelievably resonant. If you play a sound or just snap your fingers in Notre-Dame Cathedral, it'll resonate for 10 seconds. That has an influence on the music you do. If you're in a small dead room that doesn't have a lot of reverb, then you can play things that, if you were to play them in a cathedral, would just sound like a wash. You wouldn't hear any of the details. So different spaces for different types of music and the different acoustics that they create, I think that's also an important point.

2.14 That was an example of plainchant. The other point I want to make about medieval music is the fact that they didn't have the scales that we have today. Scales are a particular ordering of tones that we use to construct melodies, harmonies, whatever it might be. They used a very different approach to constructing these lines. The term for that is modes —modes as distinct from the fixed 7-note scales, the diatonic scales that we use in virtually all areas of music. It doesn't matter if it's pop music or classical music or whatever, we have these 7-note scales that form the basis for our musical culture.

2.15 In the Middle Ages, it's a little bit different. These modes consist of a particular patterning of tones or whole tones and semitones. Remember earlier today, I talked about the fact that the smallest interval you can play on a keyboard is between a white note and the black note above it. If I move from C to C sharp, [music], I can't play a smaller interval than that on the keyboard. That's called a semitone. You can think of

that as the basic, what would you say, the smallest unit that we have in most types of music in the West. Now, if I play two adjacent white keys on the keyboard from C to D, [music], that gives me a whole tone. So remember that distinction, semitone [music] and whole tone [music].

- 2.16** So the medieval modes consist of an alternation of whole tones and semitones. That's where these scales are derived from. Actually, it's the same with the diatonic scales, but they're just patterned differently. The thing here is that we look at a contemporary piano keyboard, we see all these, we see this beautiful, symmetrical arrangement of tones, and they're all really nicely tuned, and it sounds good and all of that. But we think of this as being inevitable, or this is just the way things are. But what I want to impress upon you is the fact that this is actually—there's a high degree of contingency in the way that these things developed.
- 2.17** Our keyboard layout could have been completely different than the way it is. The scales that we use and the tuning system that we used could have been completely different than what it is. Well, for proof of that, all you have to do is look at other cultures. Remember the gamelan music that we heard used a different scale and a different tuning system than what we use. But that's how it developed. In the music of the Middle Ages, what you have are these modes. And there was a theorist who's very, very important in the history of Western music called Guido, Guido of Arezzo, who, around the year 1000, comes up with two ideas. Now, I doubt that this was a solitary genius and nobody else had thought of these things. Probably he either codified it or he took it farther, or it's usually not that cut-and-dried. But he's credited with two things.
- 2.18** One of them is the development of what's called the hexachord. The hexachord is a 6-note scale. This hexachord is the embryo of the scale that we use today, the major and minor scales. The hexachord consists of 6 tones that have syllables associated with them. These syllables will probably be familiar to you if you've ever heard The Sound of Music, "Doe, a deer, a female deer," etc. My daughter's really into that song right now, so I'm hearing it quite often. That's where these syllables come from, about the year 1000.
- 2.19** The tones are ut, re, mi, fa, sol, la. Six different syllables. They correspond to this—ut [music], re [music], mi [music], fa [music], sol [music], la [music]. OK, so you might not be familiar with the ut part of it because, in more recent times, we tend to say "do." Do, re, mi, fa, sol, la. But the origins of these syllables are interesting. There's a couple of dimensions to this. One of them is that Guido based these syllables on a poem in Latin, and it's a kind of acrostic. He took the first syllables of different lines of the poem, and that's where these—that's where these syllables come from.
- 2.20** The other interesting property of these syllables is that you have all five vowels —ut, re, mi, fa, sol, la—with only one repeated. Fa and la both have the A sound. So you have the five vowels and different consonants attached to them so that you can actually pronounce these as you're singing or as you're learning to sing. So that had to be

invented. That had to be invented by a theorist. Music has so many different dimensions to it, but music can be for entertainment, it can be for purposes of relaxation or consolation also. If you're having a hard time, a lot of people turn to music.

- 2.21 But there's also a scientific, mathematical component to music. Certainly, the Greeks and a lot of medieval theorists were very focused on that dimension. They were trying to understand this phenomenon, trying to understand how they could theorize it, how they could express it mathematically, and how they could take it farther. This idea of having a hexachord was extraordinarily revolutionary because suddenly you could fix the tones. Remember, we looked at neumatic notation and, well, there are no pitches indicated. There's just rising and falling. But now the notes have names, and that ends up being an important distinction.
- 2.22 The other thing that Guido invented was the four-line staff. So four-line staff is no longer universal by any means. Before that point, at some point in history, I don't know exactly when, someone had the idea of putting a line down on paper, just a horizontal line, so that you could at least have a fixed point relative to which you could say this is going up or this is going down. And that turned into a two-line staff. Guido had the idea that you could have a four-line staff, and the four-line staff allows you to fix notes on spaces and lines.
- 2.23 These are two concepts, the four-line staff and this hexachord, that had extremely far-reaching consequences for the development of music in the West. Nowadays, we use a five-line staff, but there was a time where this was not standard at all. We had different numbers of lines. We had four-line staves, five lines, six lines. Six-line staves are actually not uncommon in the early part of the 17th century. Again, with all of these things, it had to be developed. It wasn't some inevitable force of nature that resulted in the scales, the notation system, and the tones. It did have to be developed.
- 2.24 A couple of things about the hexachord that I'd like to talk about. The first thing is that it's symmetrical. Remember, I said that, in these scales, you have a pattern of whole tones and semitones. In the Middle Ages, theorists were very preoccupied with the idea of this ideal, Platonic, mathematically beautiful, perfect music. This hexachord is symmetrical in the sense that it has two whole tones, a semitone in the middle, and then two more whole tones [music]. That's two whole tones [music], a semitone [music], two whole tones.
- 2.25 I've been reading this treatise on music that was written in the 13th century called *Summa Musicae*. There's a wonderful description in this book about this hexachord. He says that "it is proper that soft materials be contained within hard materials." What that means is that they considered semitones to be a soft material, to be a soft interval. They considered the whole tone [music] to be a hard interval. You can get a sense of where this is going. It becomes abstract very quickly. This idea that intervals have different qualities and they can be arranged in different patterns. I just love that idea

about how the semitone has to be in the middle so that it can be enclosed by these other harder intervals. So that's the hexachord. That's Guido of Arezzo. And there's one thing that I haven't quite touched yet. I'll just take a second to explain. But it also points to the fact that it really was complicated to work all of this out theoretically.

2.26 You remember from the song that it doesn't stop at fa, sol, la, right? There's a seventh tone, which is ti or C. And the seventh tone, [music], well, this had theorists tied up in knots for centuries, weirdly enough. The reason is, it disrupts the pattern. If we had two whole tones in a row, a semitone, two more whole tones—if you add the C, [music], you have three whole tones in a row. That's not so good. If you have a B flat, [music], then you preserve the pattern. But then there's another problem associated with that, which is that it doesn't [music] resolve upwards in quite a satisfying way [music]. You can do that, but [music] it has more of a sense of pull to it, more of a sense of resolution when you have that semitone [music].

2.27 The seventh tone had an ambiguous status throughout the Middle Ages, in the sense that it could be either a soft interval or a hard interval, either a semitone or a whole tone. From that concept comes the contemporary French terms for flat and natural. In music, a tone can be natural, it can be sharp, which is to say that it's raised by a semitone, or it can be flat, which is to say that it's lowered by a semitone. You can have a natural tone, a sharp sound, or a flat tone. In French, the terms are *bémol*, which means flat. And what that literally means is a soft B. So the soft B is a semitone [music] above the A [music]. And *bécarre*, which is the term for a natural note, means literally a square B or a hard B [music].

2.28 So you know those little symbols that you've probably seen, the flat sign looks like a lowercase B. That's where it's from. It's the seventh tone that has this ambiguous status. It can be flat, it can be natural, it can be soft, it can be square, to use the French terminology. Hundreds of years for theorists to work these things out and try to invent these scales in a way that could be transmitted across distances and that could result in the creation of a musical repertory. Now, you might wonder how all of this was received and how people actually went about reading music when this was a comparatively new technology at the time. The thing is, there were actually multiple different ways of doing that. Think of music notation as a mnemonic. It's a set of symbols that have an agreed-upon meaning that people in different parts of the world can see and understand what it is they're meant to do.

2.29 In the Middle Ages, there was a teaching device, a pedagogical device called the Guidonian Hand. That refers again to our friend Guido. What that means is, if you had singers that couldn't read music, you could use your hand. The teachers would do this. Each of the joints and each of the tips on the thumb and the fingers were associated with specific tones. You have two joints on each finger and a tip on each finger, one joint on the thumb and the tip of the thumb, and that gives you 19. You could go, "Ut, re, mi," and continue on the inside of the index finger, "fa, sol," etc. You could span

about three octaves on your hand by doing that. A teacher could hold their hand up and point to a joint on their hand, and the singers would know what to sing. Isn't that amazing? Just find that remarkable.

- 2.30** I mentioned that there was a long period in which music notation was not standardized. We had different types of staves, we had different names for tones, and so on. By about 1600, which is really getting into a completely different era, getting outside of the Middle Ages by this point, we have the five-line staff that becomes relatively standard. OK, so I mentioned the fact that in plainchant, in Gregorian chant, there's no rhythmic notation. In a lot of early music notation, we don't have that. There's no real way of specifying how long you should stay on a particular tone. So medieval composers, medieval scribes, musicians had something called rhythmic modes. Remember, we were talking about modes and scales and as pertains to pitch, but there were also modes that pertained to rhythm.
- 2.31** Basically, if you knew what the mode was, then you could know what rhythm to sing a particular song in. These modes will be familiar to you if you've studied poetry because they're the same as the metric feet that we have in poetry. The terms are trochee, which is a long-short rhythm, and that sounds like dom ba dom ba dom ba dom. There's a long value and a short value. It's sort of similar to the way scales are organized in the sense that, in a scale, you have semitones and whole tones, some kind of alternation of them. In rhythmic modes, you have an alternation of long and short pulses, dum ba dum, or different accents placed on different pulses.
- 2.32** Then you have iambs. You might have heard the term iambic with respect to poetry—iambic monometer, iambic pentameter, iambic hexameter, and so on. Iambic means short-long, da dum, da dum, da dum, da dum. Then there are dactyls. Dactylic. Dactylic hexameter and so on. Dactyls are long-short-short, yum da da, dum da da, dum da da, dum. This is where dance rhythms come from. Dance rhythms are just a regular periodic alternation, long and short. Obviously, there's more to it than that. There's lots of subtleties that come into play. But fundamentally, long and short. Then we have the anapest, which is short-long-long. There's the spondee, which is two equal long tones. Then there's the tribrach, which is basically a triplet.
- 2.33** So most of these rhythmic modes were not used in the Middle Ages. There were a few that were a lot more common than others. The trochee, for example, the long-short rhythm, was extremely common. But basically, if you knew the rhythmic mode, then you could figure out what the rhythm was for the whole piece. So that's how rhythmic notation got started. Now, that system works well, but it's got some disadvantages. One of them is, well, you can't very easily change the rhythm in the middle of a piece. The other thing is, if you want to have multiple different voices, if you want to have some kind of polyphony, if you want to have independent voices in a composition, you need to find a way to synchronize them so that they can all fit

together. You don't necessarily have everybody singing the same identical rhythm all at the same time.

2.34 And this innovation took place in the late 13th century, and it's got a name. Mensural notation. And mensural notation is, again, a concept that's, I believe, particular to the West. The idea of mensural notation is that you start with a large beat unit, so a long tone, and then you divide it by halves. So for example, you start with a whole note, which lasts for four beats. Then you could divide a whole note in half and you get two half notes. Divide a half note in half, you get quarter notes, and then eighth notes, 16th notes, 32nd notes, 64th notes, 128th notes, etc. This is the central conception of rhythmic notation as it developed in the West, is that it's divisive. You start with a large beat unit and you divide it by an exact proportion each time and you get these smaller and smaller beat units.

2.35 Now, again, if that's what you're accustomed to, you might think that's the most obvious thing in the world, but it's not. The idea that you would start with a large beat unit and divide it—well, you could do it the opposite way. You could start with a small beat unit and add to it, and then you get additive rhythms. That actually has profound consequences for the type of music that you create. Eastern European folk music, for example, which is another topic of interest of mine, is extremely interesting in that regard. If you listen to traditional Turkish music, Bulgarian music, Macedonian music, music of the Balkans, music of Hungary, they don't have this large beat unit divided in half. What they have are asymmetric beats, long beats and short beats, that are only slightly different from each other.

2.36 The most characteristic rhythm—this is common in Turkey, but also other parts of that region of the world—is the aksak rhythm. The aksak literally means a limping rhythm. It's because you have unequal beats. In a lot of our music, in the West, we have an equal regular pulsation. One, two, three, four. Or it might be in threes—one, two, three, one, two, three—but you notice that the beats are equal. And in additive rhythms, you can have a short beat and a long beat. Your beat could be [music], etc. What you're doing is you're taking a really small beat unit and adding to it, making it a little bit longer.

2.37 This is the thing that for Western musicians, it's hard for us actually to get our head around that because we're used to counting. We're used to thinking in terms of exact multiples of beats. But in traditional Hungarian music, they don't think that way at all. There's not an exact relationship between the long and the short beat. It's a matter of feeling. It's a matter of pulse. Well, it's tricky. It's tricky to master that if you're not from there. That's divisive versus additive rhythms.

2.38 I'd like to give you an example of what happened in music after the advent of mensural notation, divisive notation, where you have different signs and symbols that mean different things, that have different values associated with them. OK, what that means is, and this is really a profound thing, suddenly, you can have rhythmic independence in

a composition. You can have one layer that has a particular rhythmic continuity to it. You can add another voice on top where the rhythms are not the same, and you can synchronize them exactly. Think of that as a profound technological change. Think of notation as a kind of technology. It drastically changed the sorts of music that could be written. Suddenly, composers were intoxicated with the possibilities of this new notation. You could write things of a complexity, of a richness, of a rhythmic variety that had never been possible before.

- 2.39** And as with anything, whenever there's a new technology that arises, there's a kind of—it tends to be a kind of jumping on the bandwagon, a kind of exaggeration that happens. You're really excited that this possibility is here and you want to see how far it can go. There's a manuscript that is preserved in a castle in France, in a town called Chantilly. This manuscript is called the Codex Chantilly, the Chantilly Manuscript. And this is a corpus of pieces written in the late 14th century by various Franco-Flemish composers, mainly from northern France and Flanders. Collectively, these works are known as the *Ars Subtilior*, which means the more subtle art. This is where you see this extraordinary efflorescence of richness, of complexity, of polyphony in Western art.
- 2.40** These pieces are so difficult to sing, it's almost unbelievable. They're really, really, really tricky, very, very intricate. It's not surprising that, as with lots of things in music history, there was a kind of counterreaction after that because you can't really sustain a level of extreme complexity all the time. It's always this tendency. It's like a pendulum swing. You develop a new concept and idea, things get more complex, and the idea gets absorbed, and then go back down to something more moderate afterwards. But the music of the late 14th century in this part of the world, as incarnated in the Codex Chantilly, is extraordinary, some of my favorite repertory.
- 2.41** I'm going to play you a composition from this period, which is called *Fumeux fume par fumée*. I'm going to translate what that means in a second. This was written by a composer. We don't exactly know what their name was. It's attributed to a composer called Solage. It's part of a sub-genre, this might amuse you, of songs about smoking, of which there were apparently quite a few at this time. The words are, "A smoker smolders smokily in smoky speculation. Thus, he steeps his thoughts in smoke. A smoker smolders smokily, for it suits him well to smoke until he gets his way. A smoker smolders smokily in smoky speculation." I love that. So you can tell there's obviously some—they're having fun with the words and all of that.
- 2.42** You'll also notice that this is a piece of secular music. It doesn't have a particular religious function. It's talking about smoking. So let's have it. I'll play *Fumeux fume par fumée*, from the late 14th century.
- 2.43** [music]
- 2.44** OK, I just think that's so remarkable. So hopefully the difference between that and the 10th century, 11th century Gregorian chant is clear. Rather than just one line chanted

in unison, it's a trio. You have three different lines. They're all rhythmically independent from each other, and they're all synchronized. And so that obviously resulted in a whole raft of new possibilities for composers. So some composers took this unbelievably far in terms of complexity. In fact, you really have to wait until the 20th century to find a musical repertory that goes quite this far in terms of technical complexity.

2.45 I'd like to play another example, which is a piece by the French composer from the late 14th century, Guillaume de Machaut. This piece is called "Ma fin est mon commencement." My end is my beginning. This is a pretty remarkable thing. This piece is a palindrome. It's the same backwards and forwards. There were all kinds of remarkable bits of technical ingenuity in construction that go into making pieces like this. Composers at this time, because of the possibilities of music notation, because of the fact that you could slow down, you could write things that were extremely intricate, pushed this very, very far. You have pieces in which you have multiple different rhythmic layers going on at different speeds simultaneously, and they're looping, but they're looping at different times and just really unbelievably complex things. Here's "Ma fin est mon commencement," my end is my beginning, by Guillaume de Machaut.

2.46 [music]

2.47 That's a little sample of Guillaume de Machaut, 14th century French polyphony. As my mother-in-law might say, it's not exactly a toe-tapper, but it's got something else. There's a degree of elaboration in this music that is extraordinary. There's a kind of hidden symbolism in it, too, which is really interesting. If you think about this palindrome idea, you have a line that's the retrograde of another line, where there's one that's going forward and one that's going backwards at the same time. That's not obvious when you're listening to it unless you know the piece really well. But think of that as a kind of endless cycle of rebirth, endless cycle of new beginnings. The piece doesn't have a particular directionality to it. It's not moving forward from A to B. It's a loop. It's a circle. It just keeps going and going and going.

2.48 A lot of this music, a lot of the music of this era, you have these different rhetorical devices or different forms of symbolism that, while they might not be obvious to us today, provide another dimension to the music. The Fumeux fume that we heard earlier by Solage, the song about smoking, is another example of that. You have these really sinuous, twisting, turning lines, like smoke spiraling upwards into space and making all of these interesting shapes. There's a degree of figuration in that as well.

2.49 The second part of this lecture is going to have to do with the parameters of music, the different elements that go into making music what it is. We've talked about some of those things already, but I'm going to go into a little bit more depth this time around. Remember, we were talking about frequency, duration, timbre, intensity. I want to expand upon these concepts slightly. The first thing is frequency. Remember, we were

talking about the fact that a harmonic sound, a tone, has a particular frequency associated with it. 335 hertz, 400 hertz, 442 hertz. And that holds true for melodic sounds, for sounds that are tones or tones that you can sing, but it doesn't hold true for every aspect of music. The percussion music that we heard from West Africa, for example, does not use harmonic sounds. It uses percussive sounds.

2.50 The way you can think about this, it's sort of like the difference between vowels and consonants in speech. Vowels are sustained sounds that typically have a pitch associated with them, even if you're not consciously singing or thinking of a tone. But if I hold a vowel, a, or u, or a, or i, there's going to be a pitch to what I'm saying. Consonants, however—k, b, t, these sorts of sounds—are not harmonic sounds. They're closer to, to use a technical term that isn't often thought of as a technical term, they're noises. We think of this distinction between harmonic sounds and noises. The distinction is simply to do with the fact that in harmonic sound, there's a periodic vibration that is constant. Then in addition to the fundamental tone, typically, you also have these upper partials that exist in an exact multiple to the fundamental sound.

2.51 In noises, you don't have that. Noises are much more complex as acoustic phenomena than harmonic tones. A noise might consist of every frequency in the entire audible spectrum from about 40 hertz to maybe 16,000 cycles for most people. So for example, I don't have a snare drum with me, but if I had a snare drum or a cymbal or a bass drum or something like that and played them, you would hear sounds where the attack, the initial onset of the sound, has an enormous amount of energy in it, and the sound rapidly dissipates. It has a rapid decay afterwards.

2.52 But if you were to analyze that sound, look at it on a sonogram, you would see that there's energy in almost every frequency zone in the audible spectrum. That's not the case with harmonic sounds. So that's the first distinction, is you can have frequencies, but the frequencies might not necessarily be—well, they might not be periodic, they might not be harmonic tones. We also have noise. Well, noise is an absolutely indispensable component of music. It's present in virtually every type of music you can imagine. For example, when I strike a key on the piano, [music], you're not used to thinking of sound this way, probably, but in the attack, that initial 3 or 4 milliseconds, that's actually a noise. If you were to cut that off and just listen to that very short duration, you wouldn't hear a tone. You would hear noise. You would hear the sound of the hammer striking the string. And as the sound progresses, it gets softer and purer and less rich in partials.

2.53 Sound inevitably has these two components to it. A frequency component. Well, not inevitably, because not all sounds are harmonic. But musical sounds anyway, will typically have some admixture of frequency and noise. If you want to know what music sounds like when there's no noise at all, well, synthesizers can do that. You can listen to a sine tone, for example, which is just a pure sound wave produced by an oscillator.

There's no noise in the attack. It's just a pure sound. But it also doesn't have much character, and you can only produce these sorts of sounds electronically.

2.54 As for duration, well, that's a concept that we've talked about. I want to expand slightly on that. We've talked a little bit about rhythm and how rhythm was developed in Western music, how it required, to a degree, the advent of mensural notation in order to be able to write these things down and to be able to fix them and have multiple different levels going on in a piece at the same time. Well, rhythm is a really complicated concept if you actually try to understand how it all fits together. Because in rhythm, you have meter. That's the first thing. What meter is, is a regular pattern of weak and strong beats.

2.55 And so, in a march rhythm, for example, which I demonstrated earlier, you have a downbeat, which is the first beat of the bar, and the downbeat is an accented beat. So it goes one, two, three, four, one, and the one is the strongest. It's the beat that's the most accented. Three is a slightly weaker accent. There's an accent on three, and then two and four are the weakest beats. Then again, that accented downbeat. So meter is really the organization of accented and unaccented pulses.

2.56 But that's not quite rhythm yet because, as I mentioned earlier, if you just stuck literally to that all the time, it would be the most foursquare and boring music you could possibly imagine. You only have to listen to—pick any random pop song, any branch of popular music. Listen to ragtime, listen to the Delta blues, listen to Dixieland, listen to Thriller. The rhythms are extremely engaging because there's this pattern disruption. There's the grid of the meter, the pulse, that lies beneath everything. But then there's another level on top of that, which is the disruption of that pattern, which is where musicians are never exactly on the beat. They push it, they pull it, they're ahead of the beat, they're behind the beat, and they're constantly sort of playing around with it. That's what makes the music so lively. That's what makes it so exciting.

2.57 I would say in music, generally, you have these two things going on at the same time. There's what you might consider to be a structural level, which is the level of the basic pattern that we're hearing. Then there's an informal level that is more spontaneous, that is determined in the heat of the moment by the performer. Those are all the little micro-decisions that a performer is going to make when they're actually playing a piece. They're never going to just be like a rigid metronome unless they don't know how to play. But a good musician knows how to play with that, and they know how far they can push it and pull it before it breaks.

2.58 That's the other thing. Ragtime and swing music are a good example of that because in swing, you push the beat so far that it almost becomes completely irrelevant, but not quite. That's where this wonderful tension comes from. You're always ahead of the beat, behind the beat. But there's that tension because you know what the underlying pattern is and you're pushing it. That really makes for very exciting rhythms.

Lecture 3: Musical Parameters

At a glance — Musical parameters, including timbre.

- 3.1** We talked a little bit also about timbre. We talked about that with respect to Balinese music, about the way that they tune their instruments so as to create this beautiful sheen, this beautiful quality in the sound. I want to play an example of a clarinet playing some low tones, just as an illustration of what I'm talking about. What you're going to hear is a musician playing a low D, a low E flat, and then a low D on a clarinet. You'll see that the timbre of that, which I'll also play on the piano, is very different. Also, as he increases his intensity, the timbre is going to change. The sound will become more brilliant, it'll become darker, but it won't be the same as he changes the volume. So here's an example of some low clarinet sounds.
- 3.2** [music]
- 3.3** So if I play those same tones on the piano [music], I'm playing the exact same pitches as what you just heard on the clarinet. But the timbre is different. If I put the soft pedal down on the keyboard [music], it can create a whole other type of sound. Performers, skilled performers, sensitive performers, do this all the time. You can control not just the intensity of sounds, not just the rhythm, not just the pitch, but you can also control the color, timbre, by focusing on particular, what would you say, frequency components in the sound, making the sound brighter, making it darker.
- 3.4** Nowhere is that more in evidence, perhaps, than in jazz, where if you listen to John Coltrane playing the tenor saxophone, I mean, what he can do with that instrument is absolutely astonishing. There's this colossal range of different timbres, different tones, different colors he's able to pull out of the instrument. It's not just, not just the notes you're playing, it's the different timbres that you're able to pull out of your instrument.
- 3.5** So there's another dimension here, which probably is not something that we would intuitively think of as being a particularly important part of music, and that's space. Space is the fact that sounds, as we saw with the example of a plainchant, sound propagates itself in space, and it has different acoustic characteristics depending on the size of the room we're in. But in addition to that, space can be made to be a parameter of a musical composition. A simple example of that, if anyone's ever heard a 5.1 surround sound system, if you've ever been in the movies, then you'll be familiar with this. The sound can come from in front of you, behind, left, right, and it can move around in the space.
- 3.6** That's obviously something that wasn't really possible to do, at least not in recorded music, until relatively recently. That's a whole new frontier. It may well be that space will become another equivalent dimension of sound of equal importance to the others. Who knows? These parameters are not necessarily fixed once and for all. I can say that in different musical cultures, again, in the different areas of music history, there are

different points of emphasis given to these different things. For example, timbre. If we take baroque music as an example, the period of Bach or Handel, timbre was important. It had its role, but it wasn't nearly as important as it is now. You could take a composition by Bach and play it on a piano and you can play it on an organ. You can play it on a harpsichord, you can play it on a quartet of saxophones, you can play it on a synthesizer, on 100 flutes, it doesn't matter. You won't destroy the music when you do that. The music will still make some degree of sense. It'll be coherent.

3.7 But if you were to take a contemporary pop song and re-orchestrate it for any of those different instruments, you will not just be slightly modifying the sound, you'll be actually changing the meaning of the song. That's something that becomes increasingly important after the late 19th century. Timbre itself, the physical material of sound, becomes absolutely capital. When you think of bands like Pink Floyd and things like that, they're considered to be innovative bands. Why were they innovative? A lot of it has to do with the sound, the particular production techniques they were using, the instruments they were using, the way they were combining instruments in layers to create entirely new sounds. The Beatles did that, too. That's when timbre becomes this really fundamental aspect of the way we listen.

3.8 But with all of these things, there's a dynamic interaction between these different parameters, and they do change over time. I mentioned at the beginning that music is fundamentally a kind of sublimated, abstracted movement, a movement that you can achieve in this abstract realm of music, but that you can't necessarily experience in everyday life. So we're going to expand on that concept a little bit. In order to understand the concept of movement in music, there's two sides to this. There's obviously physical embodied movement. So if I'm dancing, if there's a rhythm, then that's a kind of literal physical movement because I actually have to move my feet. I have to move my body in order to be synchronized with the music.

3.9 But there's another form of movement in music that is more abstract, but that is just as meaningful. And that is the concept of harmonic motion. The way I would describe this is that, because we have scales, because we have an ordering of tones, because we can say, "This is the first note of the scale. This is the second note of the scale," the tones are organized in a hierarchy. What that means is, you have the possibility of music being in a key, and the key is a central tone to which all of the others refer.

3.10 If we say that this symphony is in B flat or this symphony is in D major or this violin concerto is in E minor or whatever, those refer to keys. If you're in E minor, that means that your piece is largely articulated around the E minor scale, and the tone E is going to be a central tone in that composition. This is a very central concept in the music of the West. It doesn't matter what the genre is. You'll see this concept of key or tonality. It's extremely widespread, and it's deeply ingrained in our conception of music.

3.11 Let me give you an example of what I mean by harmonic movement. If I'm in C major, then I'm using this scale [music]. And this tone [music] is, for want of a better term, the

central tone. It's the tone to which all the others refer. So if I'm in C, I can play this C major chord [music]. And as long as I stay on that chord, I have a kind of automatic coherence, a kind of automatic stability that comes out of that. Why? Well, because it doesn't need to go anywhere. It's static. It's just sitting there [music]. I can just keep repeating these tones over and over again and just stay in C major [music]. I could just stay in C major forever. It'll be boring. You probably wouldn't want to listen to it for very long, but I can do that.

- 3.12** But in this tonal music, in music that's written in a key, you can move to other chords [music]. You can move around from one chord to the next. When you do that, a very interesting thing happens. You have a dialectic or a dualism between things that are stable, such as the main key of the piece, or the tonic as it's referred, the main tone, the first tone in the scale [music], and then different scale degrees, as they're called, because, remember the do, re, mi, fa, sol, la, si, and so on? Each of those tones is a different scale degree. You can assign them numbers, too, which is what a lot of theorists do. One, two, three, four, five, six, seven.
- 3.13** So the C is the first tone of the scale, so it's referred to as the first scale degree [music]. D is the second scale degree [music]. E is the third [music], F is the fourth [music], and so on. Now, if you're staying on the first scale degree for an extended period of time, then, like I said, you have this stability, you have this stasis. If I move from one to four [music], then suddenly I've disturbed that equilibrium. I've moved from somewhere stable to somewhere unstable.
- 3.14** You know how there's this trope about how pop songs consist of three chords? You've probably heard that. Well, the three chords are one, four, five. If we're in C major, it would be C [music], F [music], G [music], and then back to C [music]. That pattern, we've all heard that pattern thousands and thousands of times. Why is that pattern so common? The one represents a point of stability [music]. The four represents an intermediate zone of stability [music]. And the five represents instability. It's the polar opposite [music]. If I play that chord [music] within this system, within this framework, you want to hear it go back to [music]. You want to hear that resolution at the end.
- 3.15** Baked into these scales, baked into the fabric of music written in a key, is the idea that you can have stability and instability as opposing forces. That brings us to the concepts of consonance and dissonance. So these are terms that are not widely understood by the general public. If you say, "Oh, that music sounds dissonant," a lot of what you'll immediately think of is something that's really hard on the ears, something that's unpleasant, it's harsh, it sounds ugly. That's not what these terms mean, technically. Consonance and dissonance are another way of considering the idea of stability and instability. A consonance is something that is stable, and a dissonance is something that creates a disturbance that wants to move.
- 3.16** Within the framework of tonality—again, music written in a key—there's a dialectic between those two things. You don't want the music to be stable all the time because it

would be boring and there'd be nowhere for it to go. You don't want the music to be completely unstable all the time because there'd be nothing to hang on to. There'd be no patterns, there'd be no ability to recognize what you're hearing. So what you have is this ingenious balancing act, this dialectic between consonance and dissonance, between stability and instability.

- 3.17** In an acoustic sense, we can actually describe this technically. So Pythagoras, around 500 BC, is credited with discovering or developing the concept of the harmonic ratios. This is where it gets very slightly mathematical. But he discovered that if you take a string and suspend it between two points, and then you stop the string at exact ratios along its length, you can obtain harmonic tones. For example, if I have a string, and let's say it's tuned to C [music], and I stop the string, put my finger on the string exactly at the midpoint—I'm cutting the string in half so that there's two equal sides to the string—I obtain an octave [music].
- 3.18** And there's all these different harmonic ratios that allow you to then obtain the fifth [music], fourth [music], third [music], all these different intervals. There's a scientific and physical basis for this, which is that the simpler ratios we tend to hear as consonant and as stable. So one to two, for example. As you start getting into extremely complex ratios, 41 to 43, for example, then you get tones that are more difficult to relate to each other acoustically in terms of the ear. It's just fascinating. The Greeks had this understanding that music was not just an auditory phenomenon, but it had a grounding in mathematics as well.
- 3.19** If I play intervals that are very distant from each other in terms of this succession of harmonic tones, then you obtain things that sound unstable to the ear, or at the very least, are difficult to relate to one another. So for example, if I were to play a C, and then a C sharp, three octaves higher [music], then you can tell that that's a more complex relationship. That's a more complex ratio. It's a more complex sound than, for example, a perfect fifth [music] or an octave [music]. These are very simple consonances. And they're simple consonances because, well, mathematically, they're based on extremely simple ratios. So there's a relationship between all of these things.
- 3.20** To a degree, consonance and dissonance are relative phenomena. So even though there is a physical basis for this, in reality, the scales that we have, the instruments that we have, are not based around mathematically pure intervals. They can't be. The reason for that is, we can create instruments that have exactly in tune octaves, exactly in tune fifths. But the moment we do that, the thirds, the fourths, and the seconds are out of tune. It's a problem in acoustics that's known as the Pythagorean Comma, which is the fact that if you were to try to create a scale that would consist only of exactly mathematically perfect consonances, exact ratios between the notes, your octave would be too big. The tones would not be in tune with themselves, and you'd have a problem.

- 3.21** The result of that is that we've had to find compromises that allow us to have scales of equal numbers of tones, either seven tones or, in the case of the contemporary keyboard, 12 tones, 12 chromatic tones, C, C sharp, D, etc. In order to do that and have the notes be approximately in tune with each other, there's a compromise, which is that almost none of these intervals are exactly in tune. So again, there's this naturalistic fallacy that the way that the keyboard is now, or the way that we sing tones now, is inevitable, that it's perfect, but it's not the case. There are compromises that allow us to have these sorts of scales.
- 3.22** The ancient Greeks knew this. They knew that you would run into problems very, very quickly if you tried to base your scales and base your music on these abstract mathematical principles. So there's the physical reality of sound and acoustics. Then, well, there's the cultural layer that goes on top of that. That is not exactly consistent with that, let's say. So that's what I mean when I say that consonance and dissonance are relative phenomena to a degree. To another degree, they're contextual. They're located within the work of art itself. If you're listening to a piece of music that has, let's say, a more complex harmonic fabric to it, maybe it's not using a lot of simple chords, maybe it's using very complex chords, or maybe the harmonic structure of it is very complicated, then you're listening to something that, relative to a very simple piece with two chords, is going to sound dissonant.
- 3.23** But when you're in the world of that piece, it becomes a relative thing. Then you start hearing a different scale of consonant sounds versus dissonant sounds. We 'll talk about that a little bit more in the latter part of the series when we get into music of the 19th and 20th centuries. The point of all this is that some degree of differentiation between stability and instability is important in music. It seems to be so that you can have oriented movement, so that you can write melodies that have a direction to them or some kind of consistency, some kind of pattern in the music that is audible.
- 3.24** But there's many ways of doing that. And as we move through music history, you'll see that the methods, the range of possibilities open to composers for how do they create these polarities, how do they create this feeling of movement, expands continuously. So you can think of this simple progression that I just played, this one, four, five [music], as a very simple description of this duality, this polarity of tension versus relaxation and movement versus stasis [music]. If I were to do this [music], that would be a surprise. Not expecting that chord. Why is it surprising? Well, because there 's this expectation built into these scales. It's very, very interesting. You can surprise your listener. You can do the thing that's not expected, which opens up a whole realm of expressive possibilities.
- 3.25** We've talked about what you might call the genetic substructure of music, the way that our scales are formed, the way that tones are formed. But there are other dimensions of music that I want to talk about a little bit today. So one of them is the concept of segmentation or form. This is something that may operate on a more or less

unconscious level when you listen to a piece of music. You might not be thinking about it, but usually there's some measure of articulation in a piece of music—articulation in the sense of a finger having joints.

- 3.26** What that means is, it's not just one undifferentiated thing, but a piece of music can be broken down into sections. To give you a simple example, you might have a little introduction. Then in a song, often you'll have this, and then maybe a first verse, and then a chorus, and then a second verse, and then another chorus, and then maybe an outro or some kind of concluding section. Well, that's a form of articulation. It 's a form of segmentation. That's what gives rise, among other things, to concepts of musical form, which have to do with the way the piece is patterned and the way that the different sections relate to each other. In order for that to have any kind of meaning to the listener, there has to be some notion, on some level, of repetition.
- 3.27** A lot of forms are described in terms of a sequence of letters, ABA form, AB form, ABACADA, etc. What does that mean? Well, it simply means that the A, every time it returns, is a repeat of the same idea or a variation on the same idea. The intervening sections present you with contrasting material. One of the most elementary forms that we have in music is ABA form. You have a first opening section, contrasting middle section, and then a return to the opening section at the end. Very, very common. This is the prototypical form of a lot of popular music also. It's the verse, chorus, verse idea. It's like you have a verse which usually tells a story, and then a chorus which is more declamatory in nature, and then you go back to the story again afterwards.
- 3.28** This alternation of the same and the different is extremely ancient in music. Ancient Greek poetry is arranged that way very often. You have these different formal segmentations, different ways of patterning a piece of music, where you have elements that repeat and elements that provide variety. You want an interplay between those things. You don 't want just repetition in a piece of music. It would lead to monotony. You don't want just change all the time because that would drastically increase the cognitive demand of listening to the piece.
- 3.29** So some kind of balance, and obviously that will take different forms in different composers. You have pieces that are extremely repetitive, and you have pieces that are extremely non-repetitive, and everything in between. That says something about style. It says something about what sort of sensation the composer is trying to create. OK, so the thing about form and formal segmentation, what I just talked about, is that fundamentally it unfolds over time. In order to appreciate a form and a piece of music, you have to listen to it in sequence. You have to say, "OK, well, there's our introduction, then there's a different section, then there's something else that's different, then it returns to the thing we heard earlier." All of that unfolds over time.
- 3.30** Structure is a different thing. Form and structure are not the same thing. Pieces of music have a structure to them, but the structure is something that exists outside of time. Form exists within time. Structure is the network of abstract relationships that

hold a piece together. Something like the scales that I just played, some of these tensions that exist within the tones. You could think of those as structural elements. They're things that hold the thing together, that provide coherence, but they're abstract. They don't exist in the dimension of time.

- 3.31** In a lot of classical pieces, in fact in some of the things we just listened to, there's a structural layer that's often unbelievably complex, but that you can't necessarily hear because it doesn't unfold in time in the way that we're used to. You remember the Guillaume de Machaut piece, "My end is my beginning"? Remember, there's a voice in there that's going backwards while the other is going forwards. That's a kind of a structural element. It does unfold over time, but it's very, very difficult to perceive it unless you know what's actually going on.
- 3.32** OK, so next we come to the concept of the phrase. The phrase, again, we talked about melody and how melody has its origins in speech. Phrase has to do with the breath. This is a central concept in poetry. It's a central concept in speech, obviously, and it's also extremely important in music. In poetry, you have a division into stanzas and line breaks. There's a point where the line ends and you start a new line. That is related to the breath. How many words you're going to say in a single breath before you take a breath and start again.
- 3.33** Music is typically organized into phrases of different lengths. That is another form of segmentation in music. We talked about form, about how you can have sections of a piece that might repeat or that might relate to each other. Phrase is an even smaller segmentation of that. So for example, here's a little excerpt from a Mozart piano sonata [music]. What you heard there is a melody. It seems to have a coherence, a beginning and an end [music]. But you can easily segment that into four little sub-phrases [music]. See how it's very articulated. It's, in fact, not just a continuous stream of notes.
- 3.34** There are four phrases in that [music]. Take a breath [music]. It's four phrases. It's related to the breath, once again. Of course, you link them up. When you're a performer, you try to make it so that we don't have the sense that the music is starting over again with every phrase. You try to link them up into larger units that have a musical sense and a feeling of flow. But you can break these things down into farther and farther, finer and finer distinctions.
- 3.35** Next comes texture. So if I talk about texture with respect to music, what does that mean? I wonder if that says anything to you. What would texture mean? We think of texture as being the way a sweater has texture or something like that, or the texture of a floor. But what do you imagine texture might mean in music? It's the overall global characteristics of a piece of music, not on the level of detail, not on the level of this note plus that note plus this rhythm plus that rhythm, but what you might think of as global characteristics. In the same sense that a physical object can be rough or smooth, the texture of a piece of music has something to do with, well, how is it laid out?

- 3.36** It can be, for example, a series of chords with no particular melodic dimension [music]. The texture of what I just played is a chorale-like texture, or a harmonic texture, which is to say that when I play three or four tones at the same time, you hear them as chords, as a block. You're not listening to individual tones, and you're not really listening to a melody. You're listening to a series of chords. That's a type of texture. Another type of texture would be the Mozart example that I played earlier, where there's a melody in the right hand or a phrase in the right hand, and the left hand is playing a repetitive figure. So there's a division of labor, so to speak, between the two hands.
- 3.37** The left hand is playing this [music]. That's a figure. That's that repeating rhythmic motif that provides a kind of continuity to the piece. It provides a rhythmic dimension. That's a little repetitive rhythmic figure. And then the right hand [music], you can see that the two hands are not doing the same thing. The left hand is providing essentially an accompaniment that supports what the right hand is doing. The right hand is playing a melody. We had a harmonic texture, which is a succession of chords. We had a melody plus accompaniment type of texture. Of course, you can—endless variations on that [music]. You can, it's a very common approach to playing the keyboard or lots of other instruments, which is where you have melody in the right hand and some kind of chordal accompaniment in the left. That's called an accompanied melody.
- 3.38** Another type of texture is a contrapuntal texture. Contrapuntal refers to counterpoint. What counterpoint is, is the art of playing multiple lines simultaneously, multiple distinct voices. Remember the example that I played of that in the previous lecture? [music] That's a different type of texture. We don't have a succession of chords. We don't have an accompaniment. We have two equal lines of equal degrees of complexity going on simultaneously. So, once again, texture of a piece, you can have a piece of music that's consisting entirely of rhythms. You can have a piece that consists of one line on top of another line on top of another line. You can have a piece that consists of chords. You could have a monophonic texture where you've just got one line, like in Gregorian chant, where everybody's chanting in unison.
- 3.39** So that's the level of texture. Again, when I talk about the texture of a piece, it doesn't tell you anything about the piece, except those global characteristics. So I'm going to play you an example. We're coming to the end of this lecture, but I want to play you an example of a piece from the French Baroque, a composer called Jean -Philippe Rameau, R-A-M-E-A-U. Rameau was a near exact contemporary of Bach. He was born in 1683 and died in 1764. This is going to just be a little illustration of a couple of things. The concept of texture —how do you lay out the different elements in your piece of music? Is it chordal? Is it contrapuntal? Is it polyphonic? Is it melodic? Does it have an accompaniment, etc.?
- 3.40** Then also the concept of form that we talked about earlier, because this piece by Rameau has a very distinct form. So let me play it and then talk a little bit about the

piece. This piece is from a collection of pieces for solo harpsichord, and it's called Les Tendres Plaintes, which translates to tender complaints. So it's got this kind of a certain sadness to it, I suppose.

3.41 [musical piece]

3.42 I think that's a really beautiful piece from the Baroque. One of the things I love about this piece is it's so richly ornamented. I talked about the fact that you can have structural elements in a piece, and then you can have another layer to it that's more spontaneous or more improvisational. The ornaments in this piece consist of things like trills [music] and little, what are called mordants, little shakes [music]. That's what gives a lot of this music its character, the fact that you have this constant ornamentation, which makes it, I think, have a very beautiful surface quality. If you were to strip the ornamentation away and just play it in a very simple, straightforward manner, then you would get this [music]. It's nice, but it's not quite as rich. It's not quite as beautiful.

3.43 So there's that aspect of it. There's a great deal of ornamentation in this music, and that's very characteristic of the Baroque. We're going to get into that in more detail when we talk about Bach very shortly. The other thing I just want to briefly say about this piece is the importance of formal segmentation, the fact that there's quite a lot of repetition in it, in fact. What might sound at first glance as a continuous outpouring of music in fact is highly structured and highly repetitive. So this opening theme [music], we hear that multiple times over the course of the piece.

3.44 In between repetitions of that initial theme, we have contrasting episodes. The contrasting episodes are in other keys, and they have slightly different material. It's not an enormous amount of contrast, but nevertheless, there's a distinct division of this piece into sections. After this initial statement of that melody, we have [music], etc. What's happening there is we're actually moving towards a new key. It's very subtle, but the main tone of this passage changes. We move from something a little bit dark and a minor key towards something slightly brighter. And these slight changes in the harmony keep happening periodically in the piece and provide us with the possibility of contrast and a sense of movement and a sense of, well, when the main melody returns, you have a sense of coherence and return and something that's familiar.

3.45 So there's this, even if it's very subtle, even if we're not necessarily accustomed to hearing music this way, there 's this constant push and pull between the familiar and the unfamiliar, between stability and instability. In the second contrasting episode, we move to F major [music], etc. That passage sounds a little bit brighter, maybe possibly a little bit happier than the [music]. Why? Well, because we've moved from D minor into F major. We've changed, in fact, the harmonic texture of the piece.

3.46 So that's just a few concepts maybe to encourage you to go a little bit farther with your listening. Notice phenomena of repetition, notice phenomena of sections returning, and see where that takes you. You can practice it. There 's an art to listening. That's a

grand tour of some of the concepts we're going to be talking about. I'll stop there for now. But we've talked about the way scales are formed. We've talked a little bit about some historical notions of when music theory developed in the West. We've talked about different musical cultures. We've talked about formal segmentation, talked about ornamentation, and different types of musical textures. We're going to stop there for now, take a break, and then we're going to talk about a single composer who had an immense impact on the development of music in the West.

3.47 So we'll see you in a few minutes.

Lecture 4: Bach's Mastery

At a glance — Bach's mastery.

- 4.1 So up to now, we've looked at historical developments, and we've looked at the works of a number of individual different composers. But I want to focus on a single composer this time. And in fact, that's the only time I'm going to do this in this entire series of lectures. For me, and not just for me, for many musicians, many historians, many musicologists, many listeners, there's one name that seems to sit at the summit of the pyramid, and that's the name of Johann Sebastian Bach.
- 4.2 That's not simply an assertion I'm making. This seems to be the case that what Bach gives us is something that no other musician has ever managed to do to the same degree. There are other composers who are as gifted. There are other composers who have created individual works that are as powerful, as meaningful, as the works of Bach. But Bach represents the summit somehow of this musical enterprise. All subsequent musicians, to some degree—this might sound like hyperbole, but it really isn't —owe something to Bach.
- 4.3 It's not that easy to put your finger on exactly why that is. Exactly, is there one dimension to it that makes it more remarkable? Is it the melodies are particularly beautiful? Is it the harmonies? What is it exactly? Well, there's a lot of different dimensions to it, and we're going to talk about that today. I think part of it is that what Bach gives us, perhaps to a greater degree than other musicians, is a remarkably sophisticated and complete worldview. Music can express all of these different things. It's an extraordinary thing. In Bach, there's, of course, the whole esthetic side. There's the composition side. There's the beauty of the pieces, but there's more than that. There's the pedagogical side.
- 4.4 Bach wrote an immense number of pedagogical works that are intended to that either demonstrate what you can do in a given direction technically, or that are intended to help younger people in learning how to play music. There's a whole corpus of works that Bach created that are explicitly pedagogical in their orientation. But he also gives us theology. Bach represents a remarkably sophisticated vision of theology and how it can coexist with music in the musical domain.
- 4.5 There's organology. Bach was an incredibly sophisticated person when it came to understanding instrument building, instrument manufacture, the different parameters that go into building harpsichords and oboes and lutes and all these different instruments. He understood all of that, and he was following very closely the developments in the building of instruments and making recommendations to builders. He had a handle on the craftsmanship side, the actual physical building of instruments.
- 4.6 There's the esthetic side. There's the idea of producing compositions that have a beauty, a depth to them, that is in a richness that is really unparalleled. There's the

scientific aspect to it. He understood music theory extremely profoundly, perhaps more profoundly than any other musician since. Bach is always trying to take the musical possibilities of his time and take them farther than anybody else had ever managed to do. He pushes music to its absolute limits, almost to the breaking point. How far can you go in a given direction? How rich, how complex, how perfect can you make a musical composition? There's a science to that. There's a science to counterpoint. There's a science to combining different lines together.

- 4.7 Then there's the theoretical dimension. There's the performance dimension. He was a supremely gifted improviser, supremely gifted organist. He was extremely famous in his lifetime for being one of the great organists of his day. He was able to do all of these things and do all of them at the highest level. It's something that for contemporary musicians is almost a source of puzzlement. It's very hard to understand how a single person could have done that, in addition, obviously, to leaving us this body of works that is virtually unparalleled in its breadth, its depth, and its quality. It actually causes a measure of disbelief. It doesn't seem humanly possible.
- 4.8 When you understand the mechanics of his production, what he had to actually physically do in order to write all of these pieces, in addition to the fact that he had 20 children, then you really start to wonder what exactly is going on with this man? At one point in Bach's life, he undertook the composition of not just one, but several cycles of cantatas. Cantatas for every occasion of the liturgical calendar. So we're talking about 50 cantatas, 50 complete works of music, that are usually around 20, 25 minutes long. He would write one of those every week. Can you imagine? And some of these cantatas are, even today, absolute hallmarks of the Western musical canon.
- 4.9 He was having to crank out 20 to 25 minutes of music every week, in addition to teaching, in addition to his family life, in addition to performing, in addition to his many and various duties as an employee. It's astonishing. It's simply astonishing. I don't have the capacity to understand how he did it. Theorists and historians, biographers also really struggled to understand this. But there it is. It's a fact.
- 4.10 One of the other things that I think makes the music of Bach so remarkable is that, in many respects, he's a very pragmatic musician. He's very concerned with the function of music in a society. He's very concerned with the function of an individual piece. In other words, who's it for? Who's going to be playing it? In what circumstances? How can I make a piece that will respond to those requirements in the best possible way, the most optimal manner? He's not torturing instrumentalists. He's not writing things that are extraordinarily difficult to play. I mean, OK, if you want to play Bach well, it's difficult. It is very challenging. There's no question. But there's something about it. It fits really well under the fingers. It's impeccably conceived for the instruments. It's impeccably conceived for human voices, human hands. Everything just seems to work remarkably well.

- 4.11 Not only that, but he's able to present you with an incredibly thorough picture of what it's possible to do on a cello or what it's possible to do on a lute or a keyboard instrument, and present you with this wealth, this richness of possibilities, that is really quite astounding. So he's pragmatic in that sense. He knows how to write music for people, and he knows how to write music for specific occasions. Music for dancing, music for worship, music for edification, music for learning how to play a keyboard instrument. He can do all those things.
- 4.12 In addition to that, though, in addition to this pragmatic dimension, it's some of the most abstract music ever written. What I mean by that is, there's something about it that resists description. Of course, you can analyze a piece of music. You can say, "Here's what key it's in. Here's what the motifs are. Here are the melodic ideas. Here's the texture. Here's how the harmonic structure works," and all that sort of thing. But in Bach, there's a dimension that just goes so far beyond that, that you almost can't believe it. There's this constant concept in Bach of patterns upon patterns upon patterns that is of a complexity that is just simply mind-boggling.
- 4.13 He seems to have had the capacity to hear a fragment, a little melody, a little motif, and instantly grasp all of the different ways that you could combine that with other melodies. It's just amazing. These are very, very, very difficult things to do. But he could take a little phrase, a little motif, and imagine, "Well, here's what would happen if I flipped it backwards upon itself and had them both going at the same time." Or, "Here's what would happen if I played it with itself at the same time, but one of them is moving at half the speed." Or, "Here's what would happen if I took these two completely unrelated themes and put them together and then made a little canon or a little fugue out of that." He could just do that. He had this extraordinary combinatorial ability, the ability to hear any musical idea and instantly imagine all of these possible ways of treating it.
- 4.14 Even the most unlikely candidates, the musical material that you would think has very little potential or there's nothing interesting in it, he could find something in it that would be remarkable. So pragmatism, high degree of abstraction, a desire to push his art to the limits, a very typically Germanic trait also of thoroughness, extreme preparation, wanting to be very organized and very, what would you say? Well, you can see that in a lot of his projects. He doesn't just write a piece. He'll write a suite of 12 pieces or a suite of 24, 48 pieces that present this incredibly thorough vision of what it's possible to do in a particular form or a particular genre. He leaves no stone unturned, no possibility unexplored. So that's Bach.
- 4.15 Now, historically speaking, you might think, well, how does such a figure emerge? Where does this come from? There's a couple of things. One of them is that Bach actually comes from a family of musicians, a very, very extensive family of musicians, living in middle Germany. In terms of Bach's immediate family members, you can count approximately 30 members of the Bach family that were all professional musicians. All

of his, certainly his parents, his siblings, his cousins, his uncles, they were all musicians. Now, you see this to a degree with other figures of the Baroque and classical eras where there's a kind of, there'll be a family of musicians. It's the case with the Haydns, it's the case with the Mozarts, it's the case to a degree with Beethoven as well. You can see there's a family proclivity, let's say, for music.

- 4.16 But the fact that there are 30 Bachs, and not just that, but he also had many children who went on to become extremely well-known composers in their own right, tells you something. He's coming out of this milieu that is supercharged with musical genius. I don't know how else to describe it. In addition to that, he had the good fortune to be born during a time of relative stability in this region of middle Germany that he lived in. He didn't really move around too much. That was his area. He didn't travel that much. He lived in three or four different towns and cities, all within the same very small vicinity. His life, as far as we can tell—we don't have a huge number of biographical elements—but his life seems to have been relatively uneventful. Well, you would hope that it was, given the fact that he was writing this absolute colossal amount of music all the time. There had to be some measure of stability in order for that to be possible.
- 4.17 So Bach, in many respects, is the greatest master of the Baroque era, the composer whose achievements just stand above all others, I would argue. There are other composers who are extremely interesting, but I don't think there are any that have a similar breadth to what Bach was able to achieve. Paradoxically, however, in many respects, his music has a lot more in common with the music of the Renaissance than the music of the Baroque era. I'm going to explain what I mean about that a little bit.
- 4.18 We haven't talked much about Renaissance music. The Renaissance was a comparatively short period in comparison with the 1000-year history of the Middle Ages. The Renaissance, about 150 years of music, approximately, and one of the things that characterizes it, we've seen a little bit about this in late medieval music, and to a degree, these things dovetail. I mean, this division of music into periods of medieval music, Renaissance music, classical music, and so on, is to a degree, these are simplifications, because it's not as though you wake up one morning and it's like, "Right, the Renaissance has started." It's not that simple.
- 4.19 But what you start to see in Renaissance music is this flourishing of counterpoint and this increased codification also of the musical language, which is to say that, whereas in the medieval era, you see a lot of different competing theories, a lot of different competing concepts of how music can be written. You see multiple different scales, multiple different modes, different ways of writing rhythms and so on. By the Renaissance, things are starting to stabilize a little bit. Music theory starts to be a little bit more widespread. It starts to be shared to a wider degree amongst different parts of Europe. There's this emphasis on a linear, melodic counterpoint.
- 4.20 What happens in the early Baroque with composers, for example, like Claudio Monteverdi, is you start to see the emergence of a music that is no longer predicated

entirely on counterpoint, on individual lines stacked up on top of each other, but rather a music that has a simpler texture where you've got a primary melody and an accompaniment. The accompaniment is often harmonic or chordal in nature. Remember, we talked about these different types of texture. Well, these things do change over time. You don't really see textures like that in the music of the Renaissance, but you do see it in the Baroque.

- 4.21** So what that carries with it is a great deal of simplification. If there's one line where you can say, "That's the primary line, and the others are subordinate to it, the others are simply supporting the main line," then that gives you a degree of simplification over the textures that we had in the Renaissance, where there might be three lines, four lines, or in some cases, 40 individual distinct lines. None of them is necessarily the main line. They're all equally important.
- 4.22** That's what starts to happen in the Baroque. You get this simplification and you get this idea of the basso continuo, or the thorough bass. This is a concept that I'll just briefly touch upon here. It's important because it has ramifications for the music the composer is writing in the Baroque. What that is, is a system of notation whereby, instead of having to write out all these individual lines —three different voices, four different voices—you just have to write a bass line. And above the bass line, you write numbers. Seven, four, three, six, four, etc. There's this whole language of numbers. And what those numbers tell you is, well, the performer on the basis of those numbers knows what chord to play above that bass tone and in what position.
- 4.23** And so that was a kind of simplification, a kind of shorthand that allowed composers to write music with a simpler texture, to not have to write out all the parts necessarily, and also to allow performers to a degree to improvise over these bass lines. In a lot of Baroque repertory, you'll see just a bass with the numbers appearing above the bass that tell you what chords to play. Then there'll be a melodic line. The melodic line will be a skeleton. It's sort of like in the Rameau piece that I played earlier where there's all this ornamentation. A lot of these pieces, they won't actually show you the ornamentation. The performer has to do that themselves. They have to figure out a tasteful way of interpreting the melody.
- 4.24** So you'll get a kind of skeleton. You'll get a harmonic outline. You get this simple melody, and the performer will have to, to a degree, improvise on that. That's very characteristic of a lot of Baroque music. I think that's why a lot of jazz fans love Baroque music, incidentally, because there's a degree of improvisation that goes into it. Not everything is scrupulously written out. There's a great deal left up to the performer to work out. But see, Bach doesn't really fit into that picture. So although he's a creature of the Baroque era, indisputably, what he's really doing is furthering the project of the musical Renaissance. The idea of counterpoint is absolutely central in everything he does.

- 4.25 Bach is unable to simply write a melody. When he writes a melody, it's a melody that is immediately conceived with a countermelody, and then that he can immediately then do all kinds of things to. He's always thinking in terms of layers. He's always thinking in terms of two voices, three voices, four voices. What's weird about that is, even when he's writing a solo unaccompanied piece for a monophonic instrument, for example, a flute or an oboe, an instrument that can only play one note at a time, even there, there's a latent polyphony, where the voice, the melodic line, will be jumping around from the bass to the treble all the time, hopping back and forth, which creates a virtual polyphony. Even when he's only writing one line, he's actually thinking in terms of multiple lines all the time. This is just absolutely intrinsic to how he conceives of music.
- 4.26 What that means is, the music of Bach has a kind of fractal quality to it. You know how nature is fractal? It's like, you can look at nature from different levels of magnification and see the same patterns. You look at a forest from a distance of 5 miles and you'll see a pattern to it. If you get a little bit closer and look at individual trees, you'll see other patterns. You'll see patterns in the bark. If you look really close, you'll see patterns in the individual leaves. Then you can keep zooming in and seeing new levels of magnification, new levels of patterns.
- 4.27 The music of Bach is like that as well. There's this depth to it that, I don't know what to say. I actually lack words. It's like contemplating the achievements of Shakespeare or Leonardo da Vinci. It's hard to grasp the scale of genius. Let's talk a little bit about some specific attributes of Bach's music. We talked a little bit about the textures that you get in the Baroque, how you tend to have bass line with some chordal accompaniment that's determined by the performer. You get a melodic line on top of that, and you get this clear division of the functions of these different layers in the piece of music.
- 4.28 On the screen here, I've got an example of figured bass notation, so you can see what that looks like. If I play a C on my keyboard [music], there's lots of different chords that I could conceivably play along with that C. The figured bass notation tells me what to play above it. I could play this [music], or I could play this [music], or I could play this [music]. Or I could play this [music]. So the simple fact of having the bass line doesn't tell me everything I need to know. I need to know what chords to play on top of that bass.
- 4.29 The thing is, it's actually not that different from jazz notation or from tablature notation that guitarists are familiar with, where it doesn't tell you everything, but it tells you what to do, approximately. Then there are multiple different ways that you can realize a figured bass. You can actually play the part. So this is a new predominance in the Baroque of accompanied melodies, harmonies plus chords. Another way to think about that is that, in terms of pitch -space of music, the melodic dimension of music, there are two dimensions. You might want to think about it this way. There's a vertical dimension and there's a horizontal dimension.

- 4.30 Horizontal is everything that unfolds successively over time. The horizontal dimension is melody. It's like this [music]. That's a horizontal dimension. It's something that unfolds note by note by note by note. The vertical dimension are sounds that we hear simultaneously, chords [music]. So that creates a bit of a division, you might say, in musical textures between the horizontal and the vertical. What Bach does, and this is really quite extraordinary, is he manages to seamlessly integrate both of those things. So you don't just have, as you might have in the music of the Renaissance, a succession of independent voices. And you don't just have an accompanied melody or a succession of chords. You have both of those things at the same time in Bach.
- 4.31 So there's both this exquisite attention to detail where every tone, every voice, within a texture is exquisitely crafted and can be a stand-alone melody. If you were to take just one line out of one of Bach's complicated polyphonic works like the St. Matthew Passion or any of the Brandenburg Concertos, you could just take third trumpet part or just the oboe part and listen to that on its own, and it would be a completely coherent melody. But he combines that so that, when you add all the voices together, you have a coherent vertical harmony that's very, very clear. But then you also have this horizontal dimension that's extremely clear.
- 4.32 Let's hear a couple of examples of music that represent, alternatively, this horizontal or this vertical dimension. The first is an example from Vivaldi. This is just a short excerpt from Vivaldi's Mandolin Concerto in C. Vivaldi was a contemporary of Bach, an Italian composer who wrote hundreds of concertos, extremely beautiful music. And you'll see that in Vivaldi, there's this clear division between accompaniment, chords, and melody. Here's the Mandolin Concerto in C by Vivaldi.
- 4.33 [musical piece]
- 4.34 OK, so what that example illustrates is the fact that there are very, very clear layers in that piece. There's this sort of repetitive taca, taca, taca, taca, taca, taca rhythm that you hear. There are chords being played by instruments that are realizing the figured bass in this piece, the basso continuo, as I described earlier. Then there's a soloist, which is the mandolin, playing this beautiful line on top of that. But there's a clear division of labor. The solo part is the thing that your ear is naturally drawn to, and the other dimensions in the texture are there to support what the soloist is doing, effectively. Now, if we listen to an example of Renaissance music, you'll see the difference.
- 4.35 [musical piece]
- 4.36 That was a brief excerpt from a motet by Palestrina. You can see the difference here. There's one texture in which you can't easily separate out the different lines. They're completely woven together. In the Vivaldi example, where there's obviously a rhythmic or an accompanimental figure and then a melodic aspect on top of that. Weirdly enough, the Vivaldi has a lot more in common with the Beatles than it has in common

with the Palestrina. A typical pop song will also have some of melodic dimension, usually the voice part. Then there will be a continuous rhythmic figure that will be provided by the drums or the bass or whatever it is. So that kind of relative simplification of texture is something that's very characteristic of the Baroque.

- 4.37 Now we're going to look at some examples of Bach's music and see what he actually does with this. So one of the themes in this series of lectures so far is that music is, of course, tied up with the social and political and technological conditions of its time. Inextricably, it's very, very difficult to separate out these things. Bach comes at a time, as I mentioned earlier, of stability. But he also was very, very influenced by two things. One of them was the Lutheran Reformation. Now, the Reformation happened 200 years before he came onto the scene, when Luther nailed his 95 theses to the door of a church and ushered in the era of Protestantism. Bach's music is completely tied up with that idea, and I'll explain how.
- 4.38 One of the central concepts that you see in Lutheranism, in Protestantism, generally, is the idea of taking the scripture closer to people by presenting it, first of all, in vernacular language. Instead of having to read Latin, you can read the vulgar tongue, the speech of everyday people, which would have been German at this time. So that's the first thing. The second thing is he wanted to encourage active participation in the liturgy. So rather than simply having a priest deliver a sermon of some kind, you would have the entire congregation singing together.
- 4.39 And in order to make that happen, Luther took it upon himself, in collaboration with composers of his day, this is in the early 1500s, to compose a series of hymnals which take poems, either of his own devising or other poets, and put them to music. And these songs are then set for 3 or 4 voices, the idea being that every member of a congregation can sing together. If you have a high voice, you sing the top part. If you have a low voice, you can sing the lower part. Then you have the inner voices as well for people who have more of a medium range to their voice.
- 4.40 This led to the development of what's called either the hymnal, these collections of melodies based on religious texts, or of the chorale. The idea of the chorale is very simple. These are typically vocal compositions written in 4 parts with a seamless blend of all the voices, and all the voices are of equal importance. Again, that's that sort of Renaissance idea. None of the voices is the predominant voice. To a degree, the top voice, I suppose, is the one that draws your ear the most because it's the highest one. It's the soprano. But the other voices are all of equal importance to creating this beautiful texture.
- 4.41 In 1524, Luther publishes the first collection of chorales. It's called the Geistliches Gesangbüchlein, which means the little spiritual hymn book. What's remarkable about this is that the melodies that you see in this book, 200 years later, are still a central part of German spiritual life. I find that absolutely remarkable. These are compositions. They had to be written by somebody. They had this central position in that culture for 10

generations. That's something that's very difficult, I think, from our present vantage point, to grasp—that a culture could be that stable and that consistent, that across 10 generations, everybody still knows the same tunes.

4.42 We'll make fun of the pop music of the '70s as though it were another epoch altogether and make fun of disco and things like that. It's like, well, that's only 50 years ago. Imagine what would happen if we had to wait 200 years? What would remain of the popular music of our time in 200 years? We don't have that vantage point. And what do we know about the pop music of the 1820s, for example? Technology has distorted our relationship with all of these things, of course, because now we have music recording and you can go on Spotify and hear anything, absolutely anything.

4.43 But nevertheless, in a time where there was no recording, there were no LPs, there were no cassettes, there were no playlists, there was nothing of that sort, that you could have these hymns that would persist for 200 years, and that by the time Bach comes along, he can take these very familiar things and weave them into his compositions, which he did very, very, very regularly in his passions, his oratorios, his cantatas, all of these different genres that he was working in.

4.44 I find that absolutely remarkable. He could take a piece like Ein Feste Burg ist unser Gott, Our Mighty Fortress Is Our God, which is one of the pieces that appears in this first hymnal by Martin Luther. He weaves that into one of his compositions where it's just one strand. Then there are all these voices going on, commenting upon it and ornamenting it and all these other things, and create a 20-minute piece out of that. In order for that to be possible, you have to have this culture that's really focused and centralized in a way. I don't actually know if that would be possible today. But you do see that in Bach.

4.45 So I'm going to play you some examples of chorales that have their origins in the endeavors of Martin Luther, and I'll show you what Bach does with them. So these are songs, vocal compositions, that would be played, that would be sung by a congregation together. And the first one I'm going to play is called Ach wie flüchtig, ach wie nichtig, which means —this is a real optimistic little piece—it means, "oh, how fleeting, oh, how worthless are the lives of men." Right. OK, so now what Bach would do, typically, is he would just take the melody, the melody that Luther had invented or some other composer, and then he would add 3 more parts to it. So he would harmonize it, so to speak. Here's the top line. Here's the melody that Bach would have been working with that he didn't write himself, but that was well over 100 years old at the point that he wrote this piece [music].

4.46 Ach wie flüchtig, ach wie nichtig. How fleeting, how worthless. OK. Don't worry, it has a happy ending.

4.47 [musical piece]

- 4.48 Each of the tones in that piece is a line that belongs to a particular voice, soprano, alto, tenor, and bass, the 4 main voice categories. I can separate any of them out and just play it on its own. Here's the bass [music]. Perfectly coherent, beautiful little melody. That's just the bass line. Here's the top voice again, the soprano [music]. Here's the alto [music]. Here's the tenor part [music]. It's 4 melodies being played at the same time, and the result is this.
- 4.49 [musical piece]
- 4.50 So even today, the discipline of writing chorales, in other words, taking a given melody, a top line, and having to add the other parts, constitutes a rite of passage for composers. When I was a student, I had to do this hundreds and hundreds of times. The thing that writing chorals does that is such a fantastic discipline is it forces you to understand the interaction between the vertical and the horizontal, writing multiple different melodies together that create this cohesive, coherent harmony. You also have to be able to do that at a desk, away from a keyboard, so that you can hear the sounds inwardly and add the additional parts. Then when you sit down, it still makes sense. It still works. It's an extraordinary discipline.
- 4.51 It's not an absolutely universal thing, but I would certainly recommend any composer who wants to sharpen their skills to try to write their own chorales. You can just take the top line of any of these melodies that Bach wrote and add the other three parts. Then you can compare it to what he did, and then you can feel really small because, of course, yeah.
- 4.52 Let me play you one more. So some of these are quite dizzying in their elaboration. This one is called Allein zu dir, Herr Jesu Christ. And the words are, "On you alone, Lord Jesus Christ, my hope on Earth depends. I know that you are my Comforter. There is no comfort elsewhere. From the beginning is nothing chosen. No man has been born on Earth that can help me out of my distress. I call to you in whom I have my trust." Now, all of these, obviously, are religious texts. Bach weaves this expressive component into all of these pieces. They're relatable, they're human, they deal with emotion, they deal with everyday problems that people have, struggles that people have. He weaves those into these amazingly expressive compositions. Here's what the opening phrase looks like, just the melody.
- 4.53 [musical piece]
- 4.54 There's about 400 of these, and it's an inexhaustible corpus of exquisite music. What these are, these are not piano compositions. These are intended to be sung. They're intended to be sung, obviously, in a congregation. But they also form a part of many of Bach's compositions, large-scale compositions, oratorios, passions, cantatas, and so on. What will typically happen in one of these compositions is you'll have multiple movements. Some of them will be solos for a soprano or an alto to sing. Some of them will be what are called recitatives, which is where there's a story that's briefly told,

usually by a tenor or a baritone voice, with a very spare accompaniment. Then there are choruses and chorales, and then chorales will often be placed at the end of the composition as a kind of musical summary of the themes of the work as a whole.

- 4.55 Those are a few examples of these beautiful chorales that Bach wrote. And this is just a tiny drop in the bucket of what he achieved as a composer. So let's look at a couple of other things. We're going to come to a close fairly soon. Now, I mentioned the fact that Bach is always thinking polyphonically. He's always thinking in terms of multiple lines, no matter what he's writing for. I want to give you an example of how that actually works in a composition. One of the sets of pedagogical pieces that Bach wrote is a set of Inventions and Sinfonias. What these are are little short keyboard pieces. The inventions are written in 2 parts and the sinfonias are written in 3 parts, which is to say there are 3 lines going on simultaneously in the texture of these compositions.
- 4.56 They're short little pieces, and they were intended for beginners to play. He wanted to write just a set of keyboard pieces that somebody wanting to learn to play contrapuntal music for the first time could sit down and play, ideally without too much difficulty. The irony, of course, is that these pieces are ridiculously sophisticated to a degree that —he just couldn't write anything simple. He was criticized for that, by the way, during his lifetime. When he died, there was a remark by a contemporary of his who said that "Bach's work suffers from an excess of artfulness." It's too much. It's just too much. It's too rich. It's too much going on.
- 4.57 He was completely out of phase with his time. Everyone moving towards these simpler sorts of textures, more melodic, more easy to listen to, easy to play. What he does is, as he gets older, he really digs into this. He starts writing music that's more and more arcane, more and more abstract, more and more contrapuntal. Even a simple little piece, supposedly, like this keyboard piece you'll see, is actually quite sophisticated. Here's the 8th Invention. It's a piece in F minor by Bach.
- 4.58 [musical piece]
- 4.59 So that's Bach writing something simple. The thing that's so ingenious about this composition is there's almost nothing in it. If you actually boil it down to its component parts, you'll quickly see that there's a great deal of structural redundancy in it. Repetition, in other words. The motifs that he's working with are really simple [music]. What seems like a continuous melody is actually, you can break it down into a series of little descending phrases [music]. It's just 5 descending notes [music]. That's the skeleton that this piece is built out of.
- 4.60 The left-hand part, which is a distinct part, is a little different [music]. It starts with this large ascending interval [music]. It's a separate melody, but again, the components of these strands and the texture are actually quite simple. What's not so simple is the way he combines them. This entire piece, and actually the entire corpus of inventions, all of these little keyboard pieces are written using a rather sophisticated technique called

invertible counterpoint. What invertible counterpoint means is that you write lines that you can combine together where you have one voice in the left hand and another voice in the right hand. If you flip the hands around, it still works. I can have a melody in the right hand and maybe a bass line in the left hand. Then once I'm done doing that, I flip the parts around. The top voice goes in the left hand and the lower voice goes in the right hand.

4.61 That's what he's doing in all of these pieces. It's dizzying because you have to be able to imagine combinations that work even when you flip them upside down. It's absolutely crazy. I'll just play a brief illustration of what I mean. Remember, we had this motif in the right hand [music]. All right. And in the left hand, we had [music]. So you put them together, it sounds like this [music]. OK, so what happens when you flip them around? Well, it sounds like this [music]. In the right hand, we have [music]. That's the bass line that we had previously [music].

4.62 There's this constant, incredibly virtuosic play of materials being swapped back and forth between the hands, being played backwards and forwards and upside down and who knows what else. In order for that to work, you can't just invent a little motif, a little melody. You have to simultaneously invent the other voices that go along with it and make it so that they work together coherently. That's the faculty that Bach had that makes me think he could have been a mathematician of genius just to be able to intuitively and instantly see these patterns and put them together. It's really quite astounding.

4.63 The last thing I'm going to talk about very briefly with respect to Bach are the cantatas. So a cantata, just for a definition, is, as the name implies, it's a sung composition. Cantatas in Bach's time were essentially religious compositions, for the most part, written in multiple movements. And lots of composers had written cantatas before Bach. He didn't invent the form, but you can argue that he pushed it farther than anybody else had done. He wrote, well, we have approximately 250 surviving cantatas that he wrote. Remember, these are 20- to 25-minute-long compositions for large groupings of instruments, soloists, choruses, and they're big pieces.

4.64 What I find astounding about these cantatas is—just set aside the question of the flourishing musical genius and the overflow of incredible musical ideas, one after the other. But remember, I said at the outset that Bach is a pragmatist. He's always thinking in terms of what a piece is for, who's going to play it, in what circumstances, what's it going to do for them? He's not just writing light music, music for dancing, or music for entertainment or whatever. It goes beyond that. In these cantatas, what he typically does is he'll set out a problem in the first half. Remember, these are intended to be played in a church.

4.65 The first half of the piece, typically, especially in the earlier cantatas, will present a problem of the order of something that everybody has to contend with. "You're a mortal. You're going to eventually die." OK, that's a problem. Maybe that problem can

be treated musically. Then the second half of the composition will present you with a solution to the problem, a way to face the problem, a way to take solace, a way to cope with it somehow or another. So this I find really quite remarkable. There's a kind of, what would you say, an almost advice-giving dimension to these pieces, where they teach you how to live. They teach you how you can live productively and optimally.

4.66 So one example that I'd like to play is from a cantata called *Gottes Zeit ist die Allerbeste Zeit*, which is a composition that Bach wrote very early in his life. We don't know the precise circumstances surrounding the composition of this piece, but it seems to have been composed around the year 1707, which, to put this into perspective, means that he would have been 22 years old when he wrote this piece. It's one of his great achievements. What the title translates to, the time is, the time of God is the best time of all. That's a slightly difficult title to interpret. But it seems to have been a funeral cantata that was written for the occasion of the death of somebody who was a notable person in the city that Bach was living in at the time, maybe a mayor or something like that or a political figure. We don't exactly know. So it's a funeral cantata.

4.67 I'm going to play you 2 short movements from this piece. The first is the opening sonatina. It's an instrumental composition for string instruments and two recorders. We'll start with that, and then I'm going to play you a very short 1-minute movement called *Bestelle Dein Haus*. We'll start with the sonatina. The sonatina, with its very slow-moving pensive dimension, I think is a very fitting opening to a funeral cantata.

4.68 [musical piece]

4.69 That's not bad for a 22-year-old, I would say. Now, I'll just close with a quotation from another part of the cantata. There's a movement called *Bestelle Dein Haus*. So this might sound a little familiar. The words of this section of the cantata are "*Bestelle dein Haus, denn du wirst sterben und nicht lebendig bleiben.*" And what that means is, "Put your house in order, for you will die and not survive." Another cheery message. The idea being, you're not going to be around for very long, so you better take care of your affairs while you're here. And what I take that to mean is something —it's a variation on clean your room, right? Yeah. We've heard that before. So, *Bestelle Dein Haus*, put your house in order. So this is just a little 1-minute extract from the cantata.

4.70 [musical piece]

4.71 OK, so what I've tried to demonstrate is this cantata is divided up into stern warnings, and then in the second half, a more hopeful message. A lot of the compositions are like that, a lot of these cantatas. So I hope that I've demonstrated something, just an inkling of the breadth of what this musician accomplished in his 65 years of existence. It's really quite amazing. So we've just scratched the surface, but there's more to come, and I look forward to continuing tomorrow.

Lecture 5: Shaping Classical Music

At a glance — Shaping classical music.

- 5.1** OK, so it's very good to see everyone again. I'm really excited about today's series of lectures. This is, I think, going to get really quite interesting. We're going to talk today about 3 different historic periods. And once again, I said this yesterday, but I really have to stress that these different historic periods that we tend to divide music history into, they do correspond approximately to major shifts in style and emphasis in Western music. But you always have to take these things with a grain of salt.
- 5.2** So for example, today we're going to talk about the high classical style. And the dividing line, once again, between that and the Baroque, and then the Romantic period that follows, is not really completely clear. But by and large, this is considered to be the style that emerged following the death of Bach in the late 1700s, and that continued approximately on to the death of Beethoven, the death of Schubert, which places us around 1827, 1828. So take that with a grain of salt, but that's just to give you a bit of a bracket to understand roughly what chronological period we're in.
- 5.3** So we talked yesterday about the fact that we tend to refer to classical music as a general term to cover everything that is serious music that's not popular or commercial music. What exactly do I mean when I talk about the high classical style as opposed to other periods in music history? It's a very specific concept. The high classical style refers to a number of things that start to emerge and take form following the Baroque. For example, there's this idea of Enlightenment values that start to seep into music. Enlightenment values being humanism, for one thing. So the human subject is placed at the center of the artistic enterprise. So this is music that speaks directly to human emotions.
- 5.4** The other thing is that it achieves another level, I would say, of simplicity. Some refer to the classical style as a galant style, which simply means that there's something very approachable, very beautiful about it. It's very melodic. You also have an increasing simplicity of texture. Remember, we talked about different types of musical textures yesterday. We talked about contrapuntal textures, harmonic textures. We talked about monophonic textures, such as are found in Gregorian chant. What seems to happen in the classical style is that melody becomes an absolute priority for many composers.
- 5.5** So whereas in the earlier polyphonic styles and the music of the Renaissance, the music of the Baroque, you see this emphasis on multiple different lines superimposed, you see this emphasis on complex polyphonic textures in which you can't easily say what is the primary line—they're all equally important, and they create this beautiful woven tapestry of lines. In the classical style, what you see is the emergence of a music where the melody is front and center. It's by far the most important element in the texture of

the piece. And then, of course, in addition to that, you have various accompanimental figures, such as what we looked at yesterday.

- 5.6 I played you a brief excerpt of a Mozart piano sonata where you have this in the left hand [music], etc. These are accompanimental figures. In other words, they're repetitive figures where there's a [music] shape that repeats. It provides, well, it's essentially a broken chord. If I take this figure [music], I can play it as a succession of tones like that with that pattern, or I can play it as a chord with all the tones at the same time [music]. Then I can have a melody of some kind in the right hand. So these sorts of textures become a lot more common, where you typically have some kind of accompanimental figure in the left hand, or in the orchestra, or in the string quartet, or whatever it might be, and then a clear, distinct melodic idea in the upper voice.
- 5.7 In the classical style, the high classical style, what you see is this emergence of a clear division between the melodic layer and the harmonic or accompanimental substrate of the composition. You see this emphasis on human emotion, the human at the subject, at the center, of the artistic endeavor. Rationalism, clarity, a kind of emphasis on symmetrical dispositions also in these compositions. There's a kind of incredible clarity to the way that the pieces are outlined, which is to say that you can very clearly distinguish between sections. These pieces are highly articulated in the sense that you can tell we're on a first theme and then we're going to modulate to maybe another key, then there will be another distinct theme. And the music is animated by all of these different contrasting characters.
- 5.8 That's another concept that is absolutely central to the high classical style, is the idea of a greater amount of contrast in the music, fundamentally. Whereas Renaissance and Baroque music tend to be relatively continuous, that is to say, we have a texture that begins and continues, and there's not necessarily an enormous amount of contrast within a single piece. I gave you an example of that yesterday when I played the piece by Rameau, the *Les Tendres Plaintes*, where you have this very, very richly ornamented texture [music]. And the whole piece basically has that texture. There are some contrasts in the sense that we modulate, we go to different keys, but they are keys that are closely related to the beginning of the piece. We don't really have really gigantic changes in texture in the piece.
- 5.9 But in the classical style, what you start to see is the rise of dialectical forms, forms that are extremely rich in contrast, forms that are extremely dynamic. And so what that means is, we move away from something that is highly unified and synthesized towards something that wants to put contrast and opposition in the center stage. So this is something that becomes more and more a prominent feature of music as the 19th century wears on. You might ask where exactly that concept comes from. Well, I think there's a couple of things. On one level, it seems to be almost a fundamental property of tonality itself. We talked about this a little bit yesterday, how in tonal music, music

that is written in a key, there's a central tone. There's a tone against which all the others refer.

- 5.10** And because of that, you have the possibility of having phenomena of stability and phenomena of instability. You can establish a home, a home center, a home chord, if you like, and then you can move away from that chord, and then you can go from a stable place to an unstable place. With that, automatically comes this idea of a tension between the same and the different, between the stable and the unstable. What the high classical style does is it dramatizes that even further. It makes that the entire basis of the whole musical project. It extends it not just to harmonies, not just to the idea that you can move away from somewhere familiar and reach out into increasingly exotic areas, harmonically speaking, but also on the level of the melody, also on the level of the theme, so that you can have multiple different themes in a piece that have completely contrasting characters.
- 5.11** While this is not an absolutely indispensable dimension of classical music, because there are composers who will tend to predicate their dialectical contrast, if you like, more on the level of harmony, but there are many composers, like Mozart, for example, who, when they're composing, they'll invent a profusion of different themes. And the different themes will have really sharply contrasting characters. You might have a theme that's very light, very lyrical, lilting, like the theme that I just played. And then you might have a theme that's much more assertive, could be even militaristic in its overtones, or it could be much more aggressive, or whatever it might be, or playful or comical. But you have these different themes that bring to mind different characters.
- 5.12** And then in the composition, these characters are brought together, and they're brought into a dialogue. This is something that we'll expand a little bit later on, but it gives rise to a concept that is absolutely fundamental to the high classical style, which is the concept of the sonata and sonata form. But there are other things that tie into this development of the high classical style that we need to talk about. So what I'm always attempting to do in these lectures is to take specific musical examples and show that they don't simply arise out of nothing. They arise out of changing conditions for musicians, changing conditions for audiences. And what starts to happen in the 18th century, in particular, is a dramatic, dramatic change in the status of music.
- 5.13** So up to this point, throughout the Middle Ages, the Renaissance, the Baroque styles, composers by and large are the employees of courts, of nobility, of the church. And the vast bulk of what they're producing is heard in these sorts of venues—in courts, in monasteries, in churches, in cathedrals, and so on. Not absolutely exclusively, but by and large, that's where their musical production is concentrated. And well, what happens during the Enlightenment? Well, you start to see a massive clash between the middle class and the aristocracy. You start to see an emergence of a much larger and a much better -off middle class. Of course, that had dramatic consequences for musicians as well.

- 5.14** With the expansion of the middle class, you start to see a larger population of people, first of all, that has better living conditions, and secondly, that is interested in the arts. And so what starts to happen, in addition to that, is you see an increasing commercialization of music. It becomes a lot more of a popular concern in a sense. So in order to hear a piece of music composed, you no longer have to be a member of a court, you no longer have to be in a cathedral. You can go to new secular venues and hear pieces of music being performed. Now, this is not an absolutely new concept because already in the early 1600s, the Italian composer Claudio Monteverdi was writing operas that were, in fact, performed rather in public venues, in public halls. But that remained a relatively uncommon thing, whereas in the classical era, this starts to become generalized, starts to become much more common.
- 5.15** There's a few things that are happening here. Composers have new opportunities. They have a vastly larger audience suddenly, and they also have new commercial opportunities. What they do is they respond to that demand. Suddenly, their audience is changing and it has changing interests, and composers were responding to that. This audience starts to become extremely interested in individual artistic expression and novelty. That might sound a little bit familiar because essentially, those are the values that are still predominant today in terms of the audience for music. But it wasn't obvious at that time. It's something that had to arise, in fact, out of changing conditions for audiences. You had to have enough people with the necessary conditions and the necessary access to music for that even to be a possibility.
- 5.16** You see this enormous expansion in the demand for music during the classical era. Absolutely countless sonatas, string quartets, symphonies—approximately 15,000 symphonies were composed in Europe during the second half of the 18th century. The repertory that we might be familiar with today is just the tip of the iceberg. There are European libraries filled to the brim with manuscripts that no one has time to look at, and we still don't know what's in all of these collections. Not everything is cataloged. It's absolutely astounding how much was produced during this era.
- 5.17** Those are a few key concepts that I would love for you to retain. The first thing is an increase in clarity in the texture of music, an increased emphasis on melody, and not the long, winding, sinuous, ornamented melodies of the Baroque. Now, this is a different type of melody. These melodies are a lot more straightforward. They're shorter, and they tend to be grouped into symmetrical phrases. And they have a clear beginning, middle, and end. Whereas in some of the Baroque pieces that we listened to yesterday, there's a much more continuous outpouring, outflowing of melody, and it's not so articulated. So clarity, rationality, humanism, Enlightenment values, and also combined with this dramatically expanded new audience for music.
- 5.18** I'm going to play you an example of an emblematic piece from the high classical style, which is Beethoven's First Symphony. This is a piece written in 1795. This is 4 years after the death of Mozart. Mozart, unfortunately, died very young. This is a

comparatively early piece from Beethoven, but it shows you all of the main attributes of this music. You will see right away that it starts with a rather dramatic opening gesture. It's very expansive, it's very expressive. Well, let's listen to it and we'll see what you think.

5.19 [musical piece]

5.20 In that example, you can see that, first of all, there's a new medium, which is the orchestra. The orchestra was not an absolutely new thing at the time. We certainly had orchestras during the Baroque, but what starts to happen is the orchestra starts to become a stable entity with a fixed order of instruments in it. The classical orchestra consists of, first of all, a range of different colors in the woodwind section. That's one of the things that makes the symphony so distinctive. It's what makes it sound the way it does. You have pairs of oboe, flutes, clarinets, and bassoons. Then you typically have horns, trumpets, and a string section.

5.21 The orchestra is this new, powerful, expressive medium that allows composers to experiment with lots of different colors, different mixtures of timbres within the orchestra, and also to have a greater dynamic range, which is to say, you can have all kinds of interesting effects where the music oscillates between very quiet passages and very loud passages. And again, that might seem like a relatively banal observation, but in the Baroque, first of all, a lot of the instruments of the Baroque era couldn't actually technically do that. On an organ, for example, you can have effects of a change of volume, a change of dynamics, by adding stops, so adding additional pipes to what you're playing. But you can't directly influence the sound with an increase in velocity on the key. You can't do that on an organ.

5.22 It's the same with the harpsichord. There are various ways that you can create an illusion that the music is getting louder or getting more intense. But fundamentally, you can't really alter the dynamics very much simply by pressing a key a different way. So Baroque instruments, in many instances, have either a very reduced dynamic range or they're softer or they're made for smaller concert halls. Then in the classical era, what you start to see is, not only are pieces being played in these gigantic new secular venues, in other words, concert halls built all over Europe for the express purpose of listening to music, and not in some kind of religious or ritualistic or ceremonial function, but just in order to have an esthetic experience. That was an entirely new conception.

5.23 When we think about this, this is still familiar to us today, this concept that you might go to a concert and sit down in a chair and listen to people playing on stage. You can pay to go and hear them play, but you don't have to commission the pieces yourself. You don't to actually go and commission the composers or the songwriters or whoever it is to write you a song. You just have to buy a ticket and attend the concert, and you sit down and you listen, and when the concert's over, you get up and leave. Now, this seems completely normal to us, but in fact, if you look at the broader history of music,

that's an extremely strange thing to do. Music never functions like that, if you look at it in a very broad historical sense. Typically, music is for dancing, it's for a ritual of some kind, it's for a religious ceremony, and you're either participating directly through dancing or through playing yourself, or else you're taking some kind of an active role, typically, or you're attending a ceremony.

- 5.24** The idea that you would sit in a hall quietly, not move, not be dancing, not be doing anything else, having an esthetic experience listening to a piece of music is really a creation of the classical era. So you start to see music that, because we're no longer actually physically moving to the music, we're not actually dancing to it anymore, we're not actively engaged in some kind of a ceremony, the music, interestingly, has to become somewhat less abstract and more dynamic and more exciting, I think, to compensate for the fact that you're actually just sitting there in a chair. It has to dynamize this concept of motion, of movement, through all kinds of new devices.
- 5.25** One of them, as I mentioned earlier, was this increase in contrast in the music. Another is the concept of dynamics, and not just any dynamics. I mean, in Baroque pieces, you do have forte and piano markings occasionally, which is to say, a section of a piece that's meant to be played loud, or forte, and a section of a piece that's meant to be played quietly, or piano. And you can alternate between the two. A lot of Baroque concertos are like this. It's sort of like Nirvana songs, how they have a loud section and then a quiet section. That's it. There's nothing in between. A lot of Baroque music sort of functions that way.
- 5.26** And what happens in the classical era is that you have the concept of the crescendo, the idea that you can start very softly and gradually get louder and louder and louder. Then you also have the decrescendo. So dynamics or the volume, the intensity of the sound, becomes a whole new dynamic parameter in the music, and it makes it much more vivid, much more 3-dimensional, and much more exciting to listen to. So this is music that's hyper-charged with dynamism on every level and contrast and a dialectical concept.
- 5.27** We talked about the fact that the classical era reflects this new era of humanism. How does this dialectical idea fit into that? Well, see, I think this is really extremely interesting. The dialectical idea is really old. This is the Socratic principle of a dialogue between two intelligent people who might have different points of view. And what do you? Well, you have a discussion. Person A thinks one thing, person B thinks something else. They have a dialogue, and then the differences are hashed out. And at the end, potentially, both people come to a, what would you say? Not necessarily an agreement, but they might shift their position as a result of having had the discussion. This is, again, a very humanistic concept that you would do that.
- 5.28** And this is actually reflected in classical music, in the sense that you set up an opposition between either two harmonic regions. Typically, it's this polarity, as we mentioned, between the tonic and the dominant. So that's this [music]. That's a

dominant chord or a 5 chord [music]. And then this is the 1 chord [music]. A lot of classical pieces end like this [music]. It's this fundamental polarity in classical music between the place of stability, where you start from [music], and then the place that you move towards [music].

- 5.29** So this dialectical idea is related to the Socratic principle. You take two things that are different, that are opposing characters, let's say, or opposing regions harmonically in a composition, and you patch them out. And what you end up with at the end is a synthesis. So this is really the fundamental principle of this music. But I'll save the details for a little bit later when we get into the nuts and bolts of how the sonata form actually functions. Right. So along with these new opportunities for composers, the new social conditions for music, the new secular concert venues, we have quite a range of new commercial opportunities for composers. And along with that comes the rise of the independent artist.
- 5.30** And that is actually a very big deal because it means that artists are no longer exclusively writing for kings and princes and dukes and the church and whoever else it might be. The artist is suddenly an emancipated figure, able to choose what they write, able to choose the direction that their music is going to take, to a more or less great degree, depending on the circumstances of the individual composer. But we know that Mozart, for example, especially in his later years, was essentially a freelancer, writing for whoever wanted to commission a piece from him. It could be a private individual. It could be any kind of patron, effectively, or on occasion, writing pieces really just for his own pleasure. That is also a new concept.
- 5.31** And along with that came the development of printed music. So we talked about the rise of music notation. We talked about Guido of Arezzo and how it took centuries for music notation to become relatively standardized and widespread throughout the European continent. What starts to happen is that, along with the development of the printing press by Gutenberg, well, logically enough, some musicians start thinking, well, maybe we could make movable type for music scores as well and actually print music. And this starts to become a possibility around the year 1500. It's at that point still a very uncommon thing to actually print a piece of music and be able to distribute it or be able to sell it commercially. But by the time of the classical style, by the time Haydn is getting to be an old man, it becomes a fairly widespread technology.
- 5.32** It allows composers not only to make multiple copies of their compositions and have them be sold in different parts of Europe, but in addition to that, they can actually be paid more than once for what they're writing. And that was never the case before. So if you were a composer employed by King Frederick of Potsdam, for example, you would write a piece and you would be paid in ducats or gold coins or whatever currency they had at that time, and that would be that. You would write your piece, you would get your commission, and then you would go home. Suddenly, with the advent of music publishing, of printed paper music, a composer can be paid more than once. If the

piece is a popular success, then so much the better. So that also brings with it new incentives for composers to write pieces that are going to be popular, that are going to sit well with this new audience.

- 5.33** And so, whereas during Bach's lifetime, only 4 of his compositions were ever published. Now, think about that for a second. This is a guy who wrote hundreds of cantatas, passions, oratorios, hundreds of pieces of keyboard music. None of this was available during his lifetime. You couldn't actually buy a copy of these things. You couldn't play them unless you had a copy of the manuscript. If you wanted a copy of the manuscript, you had to copy it out yourself by hand, the entire thing. There were only 4 pieces of his that were published during his lifetime. These repertoires were indescribably fragile. You have one copy of this piece in many cases.
- 5.34** We know that Bach's estate was divided up after his death, and bits of his collections of manuscripts were parceled out to his various children. One of his children, unfortunately, was a drunk and a gambler, and literally gambled away these scores of many of Bach's cantatas, which are lost forever. It's terrible, but that does start to change a little bit in the classical era, when you have the possibility of music printing, music publishing, a wider dissemination of works. So by the time Haydn, once again, in his mature years, he can secure contracts with music publishers. He can receive royalties. He can benefit from the sales of his pieces, and that resulted in a lot more security and a lot more freedom for composers. With that, you start to see a greater variety of musical styles also that is emerging in order to respond to this new expanded audience.
- 5.35** By the time we get into the late 19th century, this is just absolutely widespread, and music becomes an enormous industry, which it still is today. Of course, the forms that it takes are constantly changing. But this is really the beginning of that whole epoch. So while we have this image of classical music as being predominantly serious and predominantly a high art form, in reality, the high classical style is extremely interesting because it presents us with a kind of amalgamation of two things that you might think as being not particularly compatible. On the one hand, you have this erudite style. You have this style of, this savant style that involves a high degree of mastery, a high degree of technical accomplishment, and a kind of nobility, let's say, a reaching for something transcendental and serious.
- 5.36** But there's also a popular dimension to this music. During the classical era, this concept of a division between, let's say, vernacular or popular styles and serious or savant styles, isn't so clear-cut, not at all. In fact, the basis for the classical style is the popular. We have elements that are drawn from popular European dances. As I mentioned a little bit yesterday, if you boil it down, if you look at all the range of rhythms that you get in classical music, it's very limited. You have jigs, marches, sarabandes. There's maybe 6 or 7, at the most, fundamental rhythmic patterns that you see in this music, and they're all pretty much derived from dances. And popular dances.

- 5.37** Then in addition to that, you have this simpler melodic style. There's a kind of braiding together in the high classical era of the popular and the savant. Nowhere is that more the case than in Mozart, and particularly in Mozart's operas. It's worth stressing that this is not a normal thing. If you look, again, in a broader historical sense at the development of Western art music, you see again and again this polarity between the popular and the savant styles. In Mozart, suddenly these two things come together in a brilliant flash. In his operas, such as *The Marriage of Figaro*, you have the two seamlessly coexisting. You have a popular style that's immediately accessible that anyone can listen to and enjoy. But you also have the highest degree of compositional sophistication.
- 5.38** You might think, "Well, why doesn't every composer do that? Why doesn't everybody reach for this incredible, elevating, noble sophistication of expression and this popular accessibility?" Well, the answer is, you have to be historically lucky to a degree. You can't just will it. You can't just decide to do that. Even Beethoven was not able to achieve that to the same degree as Mozart was. But in some of these operas, it's a brilliant flash. You could argue that this never really occurred again to the same degree, where you have this amalgam of the popular, the accessible, and the sophisticated. That's one of the things that I think makes Mozart extraordinarily fascinating as a composer.
- 5.39** You have this coming together of light music, which is written in a simpler, more accessible style. That can take the form of dances. It can take the form of music to listen to during dinner, music that's essentially made for entertainment, and it tends to have a more superficial character compared to the music that was being written during the time of Bach. And then also this wonderful sophistication of expression.
- 5.40** So now we're going to talk specifically about some forms and some genres that start to emerge during the classical era. And I'm going to give you an example. So we're going to talk about sonata form a little bit. Sonata form is something I've alluded to a couple of times today, but it's really an absolutely central concept. Sonata form absolutely dominated Western music for 200 years, between the death of Bach, 1750, and approximately 1950. Now, again, take that with a grain of salt. It's not as though January 1st, 1951, we're done writing sonatas. It doesn't work that way. But just as a kind of convenient historical bracket, you can say that this mode of thought was absolutely dominant in the European art music world for 200 years.
- 5.41** So what exactly is a sonata? You've probably heard this term before, but what does it mean? Well, a sonata is two things. It's a genre, and it's also a form. When I say it's a genre, what I mean is, it's a body of works typically written for solo instruments. And the term sonata comes from the word "suonare," which is an Italian word simply meaning "to sound." The reason it has that name is that these are exclusively instrumental compositions, so there are no voices. Sonatas are written for instruments. It might be a sonata for piano or a sonata for cello and piano or a sonata for violin. But

whatever it is, it's an instrumental composition for either a solo instrument or a very small grouping of instruments.

- 5.42** So it's a form in that sense. Sorry. It's a genre in that sense, but it's also a form. What sonata form is, is the development, the extension, and the dramatization of this dialectical principle in music, which becomes absolutely generalized during this era. We talked yesterday about Bach. Remember, I played you this excerpt from a Bach keyboard piece [music]. There again, there's not much contrast. Everything is very closely braided together. It's not a particularly dramatic form of expression either. But what happens in the, in the, in sonata form is that you get this generalization of the dialectical principle. So whereas in Bach, you tend to have this highly unified texture where everything comes together, everything is braided together very tightly, in the classical style, it's all founded upon oppositions, changes, dynamics, contrasts.
- 5.43** And this takes the form of a very particular approach to writing music. So I'll explain what happens. Typically in a sonata, and of course, there are lots of outliers or lots of exceptions to this, but you can think of sonata form as a kind of process, a musical process that plays out over the course of a piece. It typically takes on a 3-part dimension, which is to say, you'll have an expository section or an exposition, a development section, and then a recapitulation at the end. So what does all this mean? In an exposition, what you're doing is you're laying out the main ideas that the piece is going to be dealing with. And by main ideas, I mean the basic tonality of the piece.
- 5.44** You'll establish clearly what the key is. You'll establish clearly what the second key is because typically you modulate from the tonic to the dominant or some other tonal region. So you have an opposition between two harmonic centers, two key centers in the piece. And in many instances, you also have opposing themes. And that's the exposition. So idea A, a little modulatory passage or a transition, and then idea B. So you start out with an opposition. The development is the section of a sonata in which you take these different ideas and chop them up and put them into a dialogue with each other, and you modulate all over the place. You have all kinds of different chords and different harmonic regions that the piece goes through, some of them extremely distant from the starting point.
- 5.45** So you can also think of this as a dramatization and a complexification of the idea of instability. You start with something stable, you move towards something unstable, and then comes the recapitulation, the third section of a sonata form. And the recapitulation is the happy ending of a Socratic dialogue. It's when you have these two points of view, they're hashed out, we have a discussion, and then there's a synthesis of these two ideas, the two equal partners, the two equal intelligent people listening to each other, come to some kind of a synthesis of their points of view. And that's what happens in a sonata.
- 5.46** So the opposition that was initially laid out in the piece between two harmonic regions, two different thematic ideas, whatever it might be, two different characters, is in a

certain sense resolved. Those oppositions are synthesized, resolved, and we hear everything in the same unified light. It's a really quite beautiful concept, and it's perhaps not surprising that it was absolutely dominant in instrumental music for so long. So that's a brief overview of the sonata. It's both a form, a 3-part form, typically, and it's also a genre. It's a new type of highly dramatic, highly contrast-filled instrumental music. So I'd like to play you an example of a sonata. This is a string quartet by Wolfgang Amadeus Mozart in C Major.

5.47 [musical piece]

5.48 OK, so that's a typical emblematic piece of the high classical style. I want to just play a couple of examples from what we just heard, just to illustrate this point a little bit better. So in this string quartet that we just heard, there were several different themes, in fact, that contrast with one another. The first thing that you hear in this piece is this theme [music]. OK, it could not be simpler. This is a very, very simple, completely symmetrical phrase. It spans exactly one octave. It rises and then it goes right back down again. Absolutely symmetrical. We start here [music], we gradually climb up to here [music], and then we go back down to here again [music]. Let's listen to it again [music]. Perfectly symmetrical. It could be a children's song. It's so simple.

5.49 So you see this increased clarity, this simplicity, this fact that these are very easy melodies to sing, also. And this division of phrases into little discrete symmetrical units. We have an opening phrase, which is sort of like a question [music]. And then a response to the question [music]. And then it continues from there. So very short phrases, discrete, very clear character. And that establishes the tonality of the piece. It tells us we're in C major [music]. Without a shadow of a doubt, this is the key that we're in. This establishes a stability. It establishes a place from which we can then go on and have adventures. We can visit different keys, different harmonic regions, and establish contrasts and oppositions within the piece.

5.50 And what in fact does happen is we have a second theme, which appears here [music]. Which is quite different from the first theme, and also it's in a different key. This second theme is in G major [music]. Remember, the first theme was in C [music]. Second theme is in G [music]. So even something as simple as that allows the composer, in this case, Mozart, to create an opposition. And so what happens is, we hear this exposition twice. By the time he gets to the end, by the time he's modulated to G, we've heard this second group of ideas, the second theme group. Then he repeats the exposition.

5.51 This is the nice thing about classical music because there's a lot of structural redundancy in it. You hear things multiple times. You hear themes again and again and again. You hear the same rhythmic patterns again and again and again. There's just so much repetition in it. And the entire exposition is repeated identically a second time, which gives you a chance to become more familiar with the melodies, more familiar with the themes. By the time we get into the development section, you've heard them 5 or 6 times already. So if you're paying attention, you can hear these different themes,

you can remember them easily, and when they return later on in the piece, you can recognize them and follow their trajectory, follow their development.

- 5.52** So we talked about the fact that the sonata is a new concept in classical music. It's a new genre, it's a new form. There are a whole wealth of different forms that are developed during the classical era. The symphony, for example. And the symphony is taking really the same concept as the sonata, but transposing it to the orchestra. And what you have is typically a large public-oriented composition. By public-oriented, I mean it's something to be played in a large concert hall for hundreds or even thousands of people. You can think of this as a public form of expression as opposed to the string quartet or the piano sonata, which would typically be things to be played in a small room amongst friends or even just alone.
- 5.53** These are pieces that an amateur musician could play by themselves. A solo piano piece, for example. You can buy the sheet music, you can take it home. If you can play the piano, then you can play the piece. Or if you have some friends, you can put together a string quartet and play pieces in your home. But the symphony and the opera, on the other hand, are vast public-oriented compositions involving large numbers of instruments on stage and typically played in much larger concert halls.
- 5.54** What I love about the symphony is—this is again a very classical concept—the symphony attempts to give you a complete picture of the human being. It gives you everything. It gives you a sonata first movement in sonata-allegro form, which represents the thinking side of things, the intellectual, the analytical faculties. You have a slow movement, which is the beating heart of the piece. It's the lyrical dimension of the piece. It's the piece where the emotions come to the fore. There is typically a scherzo or minuetto, which is a rapid movement for dancing, typically in a humorous or light-hearted style. So that's the movement that, well, that's for the feet. You have the slow movement for the heart, the fast movement for the feet, and the sonata-allegro movement for the mind.
- 5.55** And then there's a fourth movement at the end, which is another allegro movement that typically ends things on a high note. So you get this very complete picture of the human faculty of expression and emotion and all of our different capacities. I think that's a wonderful concept. This idea of writing pieces in multiple movements, where it no longer has to be a set of disconnected little dance movements, as was the case typically in the Baroque. Suddenly, you have these symphonies that are like just a complete picture, a complete picture of the human being.
- 5.56** In addition with that, there are a couple of other genres that we can talk about. There's the lied, for example, L-I-E-D. The plural form is lieder. This is a type of German art song that was cultivated, particularly during the 19th century, by composers such as Franz Schubert, who wrote hundreds and hundreds of these. These are typically songs set to poems by German poets. Extremely expressive pieces, typically very short on the order of 2, 3, or 4 minutes at the most. You can think of this as the origins of the popular song

form that we have today. All of these different forms are developing, being created, being invented by composers, and reaching increasingly large audiences.

5.57 I'm going to close this lecture off with a performance of a piece by Beethoven. This is a very, very short little composition. What I think it shows is that even in a miniature piece, a piece that lasts just about 2 minutes, Beethoven is able to create all kinds of different textures, all kinds of different contrasts, and present you with an extremely rich and dynamic music classical experience, even in the span of what he terms a bagatelle. A bagatelle is a small, insignificant piece, something that you might dash off. That's what the name implies. But as you will see, it's not so insignificant. It's actually, I think, a very beautiful piece. So this is from Beethoven's Bagatelles Opus 119.

5.58 [musical piece]

5.59 OK. This piece is something that I just absolutely love because he can't sit still for even 2 seconds. He always has to come up with some kind of new texture. He has to take the music somewhere surprising. He has to do something unexpected. So even in the span of an extremely short little composition, we just have this little microcosm, this little world, that opens up. So you can have enormous contrast, variety, and drama, even in a drop of water in a classical style.

Lecture 6: Romantic Transition

At a glance — The Romantic transition.

- 6.1 We've been talking about classical ideals of balance, of symmetry, of clarity, humanism, Enlightenment values. You might wonder where that all leads. After the death of Beethoven in 1827, you start to see composers very preoccupied with expanding the range of what you can do in music, expanding it quite considerably. Beethoven is an interesting figure in this sense because he really bridges the gap between these two fundamentally different attitudes, the classical style and the Romantic attitude.
- 6.2 Beethoven is difficult to classify for that reason because his earlier pieces are clearly indebted to Haydn. They exemplify this clarity, this transparency of texture, this measured aspect. In his later pieces, Beethoven becomes obsessed with the idea of taking any kind of formal constraint of some kind and pushing it to the breaking point, almost to the point where the musical language stops functioning. He wants to push his material as far as it will possibly go. He wants to impose his strong will, his strong personality, on the music material. It's a gigantic exacerbation of individual subjectivity.
- 6.3 Where that starts to lead is things like the Ninth Symphony. The Ninth Symphony of Beethoven is an absolutely colossal composition. It's a vastly lengthy piece that not only includes a symphony orchestra, but it also includes soloists and a choir and all kinds of new instruments that had not appeared in the orchestra up to that point—instruments such as the trombone and the piccolo and the contrabassoon and all these new instruments that just drastically expanded the range of colors, the range of sounds, that you could have in an orchestra.
- 6.4 But the idea that, in the middle of a symphony, you would have a soloist stand up and start singing, and then you would have a chorus begin to entone this famous Ode to Joy that we have in the finale of the night—what that did, the effects of that innovation of Beethoven's, where suddenly the form just collapses in a sense, it's like anything can now happen—and that had such profound and far-reaching consequences. Every composer after the Ninth Symphony of Beethoven was then left with a challenge. What do you do after that? What do you do after a piece of music that completely throws open the doors? All bets are off. You no longer know what Beethoven is going to do in his late pieces.
- 6.5 It's not just the case in the symphonies. It's the case also in his late string quartets, where suddenly you have this intensely metaphysical reflection upon the nature of music itself. The music becomes increasingly inward, increasingly esoteric, increasingly strange in its developments, increasingly unpredictable as well. The old classical forms are starting to burst at the seams. They can no longer convey, they can no longer contain the turbulence, the violence, the complexity of Beethoven's thought. While he's still nominally operating within classical forms, they're bursting at the seams at this

point. The composers that followed the generation after Beethoven had to respond to this challenge because suddenly it's no longer enough just to write a 20-minute string quartet. Suddenly, the composer is tasked with establishing a new expressive world, with expressing their own individual subjectivity, with a new degree of power, a new degree of force that had not been seen before.

- 6.6 So one of the things that starts to happen in the era after Beethoven is that music, in many respects, becomes increasingly distant from anything that you could consider to be a pragmatic functional dimension. Composers are no longer responding directly to the requirements of their audience or their patrons. They're no longer necessarily even writing music for today. Beethoven considered himself to be writing for tomorrow, for the future. He was often asked by performers, by audience members, "I don't understand this piece." And he'd say, "It doesn't matter. You'll understand it later on. If you don't understand it now, maybe your children will understand it." This is a new shift in this artistic attitude.
- 6.7 The composer is no longer simply responding to the demands of their time. The composer has become, in a sense, a kind of visionary, someone who is going to be sensitive to things that ordinary people cannot see or hear or comprehend. And this new figure of the composer as visionary becomes almost like a kind of expectation. But in so doing, again, the composer becomes detached. The music becomes detached from everyday life. It's no longer music for a church service. It's no longer music for dance. It's no longer music necessarily even for entertainment. It's a kind of elevated activity of the spirit.
- 6.8 And in many respects, that's the era that we're still in. We're still in this kind of late, late, late, late, late Romantic ethos, even today. Of course, other things have happened since then. We've had modernism, we've had the postwar era. We're going to talk about all those things tomorrow. But in many respects, the Romantic attitude has become deeply ingrained culturally and societally in terms of how we think about the figure of the artist. In earlier epochs, the artist was an artisan. What does an artisan do? An artisan learns a collection of tools and techniques, knows how to do them, is the absolute master of these techniques, and can produce things of high quality.
- 6.9 And fundamentally, the role of the artisan is to learn from their predecessors and to continue a tradition, continue a way of—it can be anything. It can be making shoes, it can be making stained glass windows, it can be composing music. The idea of the artist, per se, as an emancipated figure, as somebody who's not merely continuing a tradition or a craft, but who's, in fact, in a certain sense, inventing the future—this is a concept that begins to emerge in the Romantic era. So along with that, as I mentioned, comes the idea of music as a kind of abstracted activity of the spirit. And if you don't understand it, then in a certain sense, the artist might say, "Well, maybe later you will or maybe you won't. But I still have to say what I have to say." The artist as somebody who's obsessed with visions, things that only they can imagine.

- 6.10** So along with that comes a new emphasis on mysticism, the concept of the sublime, as opposed to empiricism and rationalism. Art becomes about things that we cannot necessarily apprehend with our senses, that we cannot necessarily understand, but that promise to open up entire new vistas of experiences to listeners, to anybody who would be willing to pay attention. And along with that comes the rise of individualized personal idioms. This is starting to fray at the edges of what historians will sometimes refer to as the Common Practice period in music. What do we mean by the Common Practice period? Again, it's a matter of historiography. These are terms that are invented after the fact by musicologists attempting to make sense of the past and divide it into sections.
- 6.11** But broadly speaking, the Common Practice period refers to a period of music history where, by and large, there was a certain form of consensus about how music notation worked, how music theory worked. These things were relatively stable and consistent from country to country. If you're a musician living in Madrid, or Vienna, or Paris, or Berlin, or Amsterdam, or London, you share basically a common technical framework, a common musical language. You'll see approximately the same forms, approximately the same genres, the same understanding of harmony, the same sorts of technical constructions.
- 6.12** Of course, there are regional cultural differences. You see different points of emphasis. Opera was hugely developed in Italy, for example. Comic opera or operetta was hugely developed in France. Germany, in the Romantic era, tended to focus more on instrumental music, and so on and so forth. There are different points of affinity of emphasis in these different cultures. But by and large, there's a shared musical language. But with the rise of the subjective experience of the composer, of individual personal expression, the emphasis on the individual as opposed to the collective, you start to see the emergence of styles that are highly idiosyncratic. It's no longer a given that if you go to attend a concert and there's a premiere, that you'll instantly understand the piece or that the genre will be familiar to you, or that the form will be familiar to you.
- 6.13** Instead, you see composers breaking out of these things increasingly, attempting to go farther, push forward into new domains, new types of experience. In order for this to be a reality, you need to have, first of all, again, a continued emancipation of the middle class. You need to have people that have leisure time available, that are independent to a certain degree. This, of course, accelerates dramatically with the Industrial Revolution. After that point, the living conditions for the average person in Europe are dramatically, dramatically improved. And along with that comes more leisure time, more ability to spend time on the arts, more ability to attend public concerts as secular events as well. So all of these different forces go into creating the Romantic era, the Romantic style.

- 6.14** I want to play you an instance of a piece that I think exemplifies the Romantic spirit. This is a piece by the Hungarian composer, Franz Liszt. Now, I've played a bunch of examples at the keyboard up to now. I'm actually not a pianist. I play the piano well enough in order to demonstrate the odd piece. And of course, it helps me in my work as a composer, but I'm not a performer. What you'll see here, well, this is where we come to the limits of what I can demonstrate. I can't play this. Quite frankly, most people can't play this either. That's another thing that happens in the Romantic era. You start to see the rise of this, again, the figure of this genius, solitary figure who stands completely outside of society. He's no longer a servant of the nobility, no longer a person plying a craft, but rather somebody who is completely apart, who can do things that nobody else can do.
- 6.15** With that, you see the rise of this almost demonic extreme of virtuosity. And along with that, performers who are able to reach new heights of difficulty, of complexity, of technical accomplishment. And that results in an enormous expansion of the range of music. So let's listen to an example of a composition by Franz Liszt. This is from his *Étude Transcendentale*, the *Transcendental Études*, so named because their difficulty is transcendent. So these are pieces that are accessible only to a very small elite number of pianists. So here's the *Étude Transcendentale*.
- 6.16** [musical piece]
- 6.17** So, Franz Liszt, everybody. Just a simple little piece. So this is really a very new type of expression. It really takes off where Beethoven left off and goes even farther. And then that leaves us with a problem because, well, again, where do you go from there? Music has become so complex, so virtuosic, that it's accessible only to a very small number of performers. Liszt was able to play that, by the way. He was an absolutely extraordinary virtuoso player. We had this whole new class of genius performers and composers, like Paganini, for example, who is popularly thought to be possessed by the devil. There is no other way to explain his abilities on the violin. He wrote a series of caprices that even today are absolute benchmarks of the violin repertory. They're the summit of difficulty in terms of violin performance.
- 6.18** You have the *Étude Transcendentale* of Liszt. You have the piano music of Chopin also, which is similarly demanding. There's this whole range of figures that start to emerge. Now we come to the image of this composer as detached, separate from society, standing apart, able to do things that nobody else can do. Some of these composers have an increasingly difficult time, in the sense that it's not obvious what the market is for this sort of music. A lot of these pieces would be performed in private salons, in small gatherings of people, patrons of the arts, lovers of the arts. But that also tends to increasingly isolate the figure of the artist.
- 6.19** I want to talk about a development in a parallel art form, which is poetry. We had a question earlier about poetry, so I'm going to get back to that a little bit. These artistic developments are not isolated to music, obviously. You can see them in all the arts. You

can see them in painting and poetry and literature, theater. They don't necessarily happen concurrently. For example, the Baroque style in painting emerged quite a lot earlier than it did in music. But nevertheless, you see these different shifting attitudes. There's a quote from the English Romantic poet, John Keats, that I'd like to, I'd like to share here. Keats had a concept that he called negative capability. I think this explains a lot of the Romantic and, ultimately, the modern attitude towards the arts. And this concept stems from a letter that Keats wrote where he's describing what he holds to be one of his cardinal values in art. He calls it negative capability. I'll explain what he means, and I'll read the quote.

- 6.20** Keats says, "I had not a dispute, but a disquisition with Dilke on various subjects. Several things dovetailed in my mind, and at once it struck me. What quality went to form a man of achievement, especially in literature, and which Shakespeare possessed so enormously—I mean, negative capability. That is, when man is capable of being in uncertainties, mysteries, doubts, without any irritable reaching after fact and reason. Coleridge, for example, would let go by a fine, isolated verisimilitude caught from the Penetrallium of mystery, from being incapable of remaining content with half-knowledge. This continued through volumes would perhaps take us no further than this, that with a great poet, the sense of beauty overcomes every other consideration, or rather obliterates all other consideration."
- 6.21** So what Keats is laying out here is the idea that if you're overly focused on reason, on logic, on certitudes, then you're not going to go very far anymore as an artist. That's no longer enough. You need to be comfortable, or rather, comfortable with being uncomfortable in the presence of uncertainty. Art is no longer predicated on certainty. It's predicated upon mystery, ambiguity, doubts. And he's criticizing Coleridge for being overly concerned with that and passing by things that cannot be explained. This is, again, a fundamental shift in attitude. As you get into the Romantics, you have to accept mystery. You have to accept things that bypass reason and logic. We're getting into the domain, once again, of the mystic.
- 6.22** And that simply continues over the course of the later 19th century. So this, in a sense, is the essence of the Romantic project. It's, on the one hand, the rise of the individual subjectivity elevated to an absolutely predominant quality. And it's also this increasing searching for things that cannot be explained, things that cannot be understood. Now, all of this is taking place against the backdrop of a profound shift in life in the Western industrialized world, the Industrial Revolution. By this point, music becomes not only a significant industry, financially speaking, commercially speaking, but it also takes on a new public and popular dimension that, in a sense, is lacking from the other arts.
- 6.23** For example, if you're interested in painting or sculpture, you can maybe go to a salon show or exhibition or something like that and see some contemporary paintings, but you certainly can't buy them. They're too expensive. Music, on the other hand, becomes accessible in principle to absolutely everybody. Anybody who can buy a cheap

ticket to a concert can experience the piece and have the same experience as anybody else. In a sense, this took music out of the domain, the exclusive domain of the courts and so on, and made it a matter of public concern. There are more and more concert halls being built, and not just concert halls, but entire organizations, festivals, a series of concerts intended for very large numbers of people, thousands and thousands of people.

6.24 With that comes a number of developments. One of them is that, because music is being played in larger and larger concert halls and the audience is bigger, well, you can no longer have the same instruments as before because they weren't loud enough. That's simple. For example, the piano goes from being, well, in its prototypical form, first, you have organs and clavichords and harpsichords. Then those are all wooden-framed instruments. The clavichord and the harpsichord are wooden-framed. There's a limit to how much string tension you can have on a wooden-frame instrument. In other words, the range of the instrument can only be so great. You can't have extremely low bass notes, can't have extremely high treble notes. The strings can only be a certain length. As you have more strings on a piano, and as the strings get longer, the soundboard is under more and more tension.

6.25 So there needed to be a fundamental shift in the way these instruments were being built. In order to play something like that Liszt piece, you certainly can't do that on a wooden-frame piano. You need something like this, a steel frame construction. And when you have that, you can have seven-octave range on the keyboard or even greater. You can have extremely high notes, as we heard in the Liszt piece [music], all the way down to notes that are so low that you can even barely hear the pitches [music]. Piano becomes almost a surrogate for the orchestra by that point. There's virtually no frequency that the piano can't produce. Seven octaves, that's a gigantic range. So the piano becomes this enormous thing, costing vast sums of money, requiring years for its fabrication by an entire gigantic team of dedicated artisans.

6.26 And not just the piano, every instrument of the orchestra is profoundly changed during the 19th century. The woodwind instruments go from being essentially hollowed out pieces of wood with holes on them to being enormously complex mechanical devices with keywork. And what the keywork on a clarinet or a flute allows you to do is to close holes that you can't directly cover with your fingers. So you can have a hole on this end of the flute, press a key, then a lever activates a pad, which then closes the hole at the other end of the instrument. You could never do that unless you had gigantic hands. But suddenly you can do that. What does that allow you to do? Well, you can have completely chromatic instruments that can play any tone, any combination of tones with great fluidity, and it's all perfectly in tune. That was not possible during the Baroque or even a good part of the classical era. The instruments were simply too limited. There was just a physical limitation to the ergonomics of the hand.

- 6.27** Suddenly, keys become a standard thing on woodwind instruments. There's a race amongst European instrument builders to produce ever finer, ever more spectacularly perfected flutes, oboe, clarinets, bassoons. Brass instruments get valves so that they can play in every key. They can play a huge range of tones, which was not possible before. Baroque trumpets and horns don't have valves. They're simply brass pipes. You can play different partials by altering the pressure of your breath and your lip pressure on the instrument. But you cannot play all the chromatic pitches on these instruments. Suddenly with valves, that becomes a possibility.
- 6.28** In addition to that, entire new families of instruments are being developed for all these new circumstances in which music is being played. We needed new instruments for the orchestra so that you could have an orchestra of 100 people and fill a huge hall and everybody can hear it all the way to the back of the hall. We also have military instruments, outdoor instruments, instruments for marching bands, and then all kinds of instruments intended for amateurs, such as the upright piano, the smaller pianos that you can fit in your home. This becomes a gigantic industry.
- 6.29** The saxophones are invented around the middle of the 19th century by Adolphe Sax, a Belgian music instrument builder. What's so funny about the saxophone is, for Sax, it was actually intended to be a marching band instrument. It was intended to be an instrument that was loud enough to be played outdoors and that could replace the woodwinds or replace the brass or represent an amalgam of both in one instrument. They're both a brass instrument and a woodwind instrument at the same time. But he also invented the saxophone as potentially another addition to the orchestra, another section for the orchestra, and invented not just one, but eight different saxophones, the soprano, alto, tenor, bass, baritone, all these different instruments, and provided them in multiple different keys.
- 6.30** There's just this explosion of inventivity in instrument building that just completely takes over Europe. In Germany, Wilhelm Heckel, invents a perfected bassoon, and he also develops the contrabassoon, which is this gigantic woodwind instrument. It's taller than I am that can play the lowest notes on the piano. You really have to hear these sounds to believe them. It's really quite extraordinary to hear that played on a woodwind instrument. All kinds of new instruments are developed. The piccolo becomes more standard, the contrabass clarinet, the English horn, all of these instruments, and a whole wealth of brass instruments, the tuba, and so on. So you see this dramatic expansion in the range of instruments that are available, the range of colors that are available to composers, and just the sheer variety of sounds, and orchestral music starts to reach its apex during this period.
- 6.31** What composer would not dream of having an orchestra of 120 instruments at their disposal, capable of playing any conceivable frequency in any key? You can modulate all you want, you can play any rapid figuration, you can do anything you want, really. It's just the limits of your imagination are the only constraint. So the Romantics seized

upon this possibility. Just as Liszt did with his piano music, you start to see composers like the Frenchman Hector Berlioz with his *Symphonie Fantastique*, which is a vastly ambitious composition. Or again, the symphonies of Gustave Mahler.

6.32 Mahler is a kind of poster child for the excesses of the late Romantic and early modern era, writing these pieces that are so extravagant that they just completely burst open the conventional limitations of music. Perhaps the emblematic example of that would be his famous *Symphony of a Thousand*. This is a composition that he claimed to have written in a matter of weeks during a summer holiday in Austria and later orchestrated. It's a piece lasting well over an hour. It's called the *Symphony of a Thousand*. It doesn't literally require 1000 performers, but it certainly requires several hundred. Seeing a performance of this piece, I can tell you, is really quite spectacular. An enormous orchestra, multiple choirs, 8 soloists. You need not just a regular concert hall. You need a gigantic stage in order to put this piece on.

6.33 Mahler is not the only one writing pieces like this. You also have the *Gurre-Lieder*, which is an early piece by the Austrian composer, Arnold Schoenberg. You have pieces like the Berlioz *Requiem*, which requires an entire battery of timpani and percussion. I think there's something like 16 timpani required to play this piece. Just instrumental forces that you can't even imagine. Of course, these make an absolutely colossal sound, something that has to be experienced live. Let me play you a brief excerpt from the *Symphony of a Thousand* by Mahler.

6.34 [musical piece]

6.35 That gives you a small taste. I mean, you have to hear these pieces in a concert hall, of course. There's no replacing that experience. But composers were not writing tons of string quartets during this period, little pieces to be played in private gatherings. They're writing these colossal, gigantic compositions, require enormous amounts of rehearsal in order to play. That really brings us, I would say, towards the limits of the Romantic style. We're edging into modernism at this point. But there's a couple of other things that go along with this idea. One of them is the emergence of the concept of classical music as distinct from other forms of music. That is really, I mentioned this yesterday a little bit, but I'll expand upon it today. That is a creation of the 19th century.

6.36 How did that come about? We talked about the fact that music printing, music publication, becomes more and more widespread. With that, you can write a piece and have it played in multiple different concerts in different countries around the world. You can write a piece and have hundreds or even thousands of people purchase a copy and play it at home. This is all happening against the backdrop of the rise of European nationalism, which is to say that, rather than having Germany, for example, consist of a very large number of independent little microstates, little duchies and principalities and tiny little entities that have their own laws and their own politics and their own leaders,

you start to see the emergence of unified European states that start to have a concept of nationhood.

6.37 This doesn't just happen in Western Europe. It happens in Eastern Europe as well, the Balkan countries, for example. And along with that comes another problem, which is creating a national culture. If the culture is no longer defined by city states, by very small entities, but rather by vast geographic areas, then there starts to become a bit of a political problem, which is how do you have a coherent entity on a national level? One of the things that starts to happen is the emergence of the artist as a national hero, the artist as exemplifying this or that particular culture —the German artist, the French artist, the Italian artist. These different nations, these cultures, become conscious of the fact that they have these figures in their past who can serve as emblematic figures. Beethoven, Goethe, Hector Berlioz in France. Then you see this emergence of the artist as not just exemplifying a particular style or a particular attitude, but in a sense representing their country.

6.38 So all of these things are happening against the backdrop of this rise of personal subjectivity and an increase in these different styles. Then we also see the emergence of this concept of classical music. So what that means is, the great figures of the past start to be venerated, start to be turned into heroes. Beethoven is no longer a spectacularly successful German musician from northern Germany. He becomes a hero. He becomes enshrined, enthroned. He becomes one of the pillars of the canon of great musical works. In order to have a canon, in order to have a repertory, there needs to be some degree of consistency, some degree of permanence of the works. You can't just have a piece be played once and then put on a shelf somewhere in a library. You have to have a "repertory", in quotation marks, which is to say, a corpus of works that are held by some degree of consensus to be important pieces. Then these pieces are played not just once, not just twice, not just a few times on the occasion of a premiere, but potentially hundreds, thousands of times.

6.39 And then what starts to happen there is, the music of the past starts to encroach upon the music of the present. We see the emergence of a museum culture, culture in which there's a division. On the one hand, there are living artists, artists who are creating new things, who necessarily, in the shadow of Beethoven, are having to develop new ideas, are having to always go farther, always search farther into the outer regions of what we can do in art. This comes into conflict with this new audience that is interested in, "Well, what are the greatest pieces? What are the important pieces? What are the canonic pieces?" And they want to hear those pieces again and again. They want to go and hear Mozart. They want to go hear Beethoven. By this point, Beethoven had been dead for 50 or 60 years, but he's been canonized.

6.40 This creates a conflict because suddenly, artists are no longer being judged against their peers. They're being judged by the accomplishments of the past as well. That puts artists in a new and quite awkward position because suddenly, you're competing with

Bach, you're competing with Beethoven. You're not just responding to the changing tastes and interests of audiences. And that leads to a drastically new situation. So what you start to see are artists becoming more and more isolated in a sense, not necessarily in a harmonious relationship with their audiences, because the audiences have their pick. They can go hear the great classic works of the past, which are increasingly being edited and published in authoritative editions and widely distributed all over Europe.

- 6.41** There's a society that was devoted for years and years to nothing else than collecting every possible work of Bach that could be found and editing them and publishing them and making them widely available. Hundreds and hundreds of pieces being published. That's the emergence of this repertory. So the artist is suddenly in a bit of a beleaguered position. And what you see is the emergence of an artist that is really accepting that to some degree and saying, "Well, I'm going to do what I have to do, even if my audience doesn't understand it, even if it's very different from what they're accustomed to, even if it's very different from the music of the past. I have to move forward."
- 6.42** You start to see the emergence of not just an audience, but multiple parallel audiences that are interested in different things. There's going to be an audience, and it's always a fairly small subset of the general population that is interested in new things, who's seeking novelty, seeking things that they haven't heard before. Then there's always going to be another percentage of the audience that is going to stick with things that they already enjoy, that they already know about, that are universally acknowledged as being great masterpieces. Then in addition to that, with the rise of the Industrial Revolution, the possibilities of music publishing, and eventually also of music recording, you see this tremendous flowering of popular and light music as well.
- 6.43** Suddenly, composers are in this awkward position. It's like, how can you possibly compete with all of that, with this enormous expansion of the range of music? Well, increasingly, as we saw in the Liszt piece and the Mahler, we see these pieces that are explicitly intended to provide a kind of elevation of the spirit and that don't have a particular function anymore. They're esthetic experiences, they're experiences that are new, that are novel, that are extreme in many cases. And that's their wheelhouse. That's where composers are situated in this era.
- 6.44** On the one hand, there's this dramatic expansion of the size of the audience and the interests of this audience. But at the same time, you see the artist as being a somewhat more marginal figure in some instances. We're going to stop there for now as regards the Romantic era. I'd like to play you one more piece that I think illustrates this Romantic attitude quite well, because there's one other component to this. Because you have all of these composers who are expanding outwards, farther and farther, trying to find new forms of expression, trying to break apart the conventions of the language of their time, the forces that had kept music coherent and unified, and that

had allowed for the rise of the high classical style in the Common Practice period, are starting to fray at the edges.

- 6.45** One of the most fundamental aspects of that is tonality —well, simply having a clear point of stability in a composition. Harmony starts to become increasingly complex, increasingly ambiguous, increasingly chromatic, which means to say that it's departing from the seven tones of the diatonic scale, do, re, mi, fa, so, le, si, do, that we talked about yesterday. You start to see pieces that are just wandering all over the place chromatically, that no longer have a strong sense of direction, a strong sense of teleology, a strong sense of resolution, where in keeping with the Keatsian idea of negative capability, to listen to these pieces, you have to accept that you don't know where you are, that you don't know where the piece is heading, that you don't know what's going to happen next.
- 6.46** Let me play you a short piece by Chopin, a Polish composer. This is a prelude in E minor.
- 6.47** [musical piece]
- 6.48** That piece, from a harmonic standpoint, I think is very interesting because you can't predict where it's going to go next. The patterns in it are so asymmetrical, so strange, so unpredictable, that often you simply don't know where you are. Not that it necessarily matters because there's a rhythmic regularity to it that makes it rather hypnotic. But if we look at the actual chords in this piece, we see that he's constantly moving chromatically, which is to say that he'll play a chord and then he'll alter one tone. He'll take it down a semitone. For example [music], top note goes down to [music], where the bottom note goes down [music].
- 6.49** That's known as chromatic voice leading. The reason it's called chromatic voice leading is that each individual tone in these chords is being altered by the smallest interval, semitone, each time [music]. If you take all of these chords and put them together, you get a descending chromatic scale, which just sounds like this [music]. You get every semitone in the chromatic scale. If we look at the melody in the right hand [music], etc., it's all chromatic [music]. Every chromatic semitone in this little melody.
- 6.50** What that means is the music has an increasingly weak sense of direction to it. It's increasingly focused on color. It's increasingly focused on local effects that make it harder and harder for composers, really, to structure their pieces. The thing that had held music together and oriented it was always this polarity between tonic and dominant. That's still present here. But again, it's fraying at the edges. Where that leads next is quite interesting and quite surprising and extraordinarily contentious also. We'll talk about that in the next lecture.

Lecture 7: Redefining Music

At a glance — Redefining music.

- 7.1 We've been talking about the dramatic changes that start to occur in the Romantic era, the separation of the artist from their society. They start to become an increasingly solitary, even isolated figure. We've talked about the changing social conditions brought about by the advent of the Industrial Revolution, by the widespread availability of music, the advent of music printing, the expansion of the audience, the development of concert halls and specialized venues for music becoming more and more common.
- 7.2 As we get into the last decade of the 19th century, beginning of the 20th century, things at this point start to accelerate dramatically. Change is no longer linear, it's exponential. And of course, this had consequences for music. So we're going to talk about that a little bit today, how we can understand this period. In the Chopin Étude that I played earlier, I noted that the harmony had become increasingly chromatic, increasingly complex also, and difficult to predict in its orientations. But the harmony is still ultimately based on triads, which is to say the basic element of tonal harmony, which is to say, chords built upon stacked thirds.
- 7.3 So you have a root [music], you have a root, a third above that [music], and then a minor third above that [music]. That gives you what's called the triad, which is the basic building block of tonal harmony [music]. These simple chords have the virtue of all having basically the same construction. It's a third and then another third on top. And of course, you can extend that. You can add another third, and then you get a dominant seventh chord [music]. We can add another third on top of the seventh, and you get a dominant ninth chord [music], etc., but all of these chords, all the way up to 11th chords and 13th chords [music], these are all extensions of triads, fundamentally. The harmony is all built in the same manner.
- 7.4 In the modern era, that starts to change. We start to have chords that are no longer built on thirds, chords in which the dissonances that composers are writing are no longer resolved in the same manner as before. In the tonal framework, if you have a chord such as this [music], and this is a chord based on a root, a fourth, and a fifth [music]. See, there's a protocol in tonality for taking dissonances like this, which are relatively mild dissonances, and resolving them, turning them into consonances. I can take this F and resolve it downwards to an E [music]. Then we can say that the dissonance is being resolved within this context [music]. Or, alternatively [music], this is a mildly dissonant chord [music]. Or even [music].
- 7.5 We have a framework for both consonance and dissonance within tonality. And dissonant chords, well, there's a protocol for resolving them. So we make a distinction between those two things. But what begins to happen is composers are starting to get more and more interested in the autonomous qualities of these dissonant chords and

less interested in resolving them, rather exploring them as sonorities. If you take chains of these dissonant chords, or these more complex chords, let's say, and don't resolve them in the conventional manner, then the syntax of tonality eventually starts to break down. It's no longer functioning. Your piece might be technically in C major or in D minor or E flat major or whatever, but it's no longer being oriented fundamentally by the polarity of a tonic and a dominant, a one-chord and a five-chord.

7.6 Those sorts of notions become less and less prevalent. They no longer structure and orient our perception. With that comes both extraordinary opportunities for composers to expand the range of sounds, the range of harmonies that music can contain, but also a lot of technical problems. Because, well, if you no longer use the structuring device that had held music together for several hundred years, then how do you write a piece? How do you keep it from falling apart? How do you make it so that the piece has some degree of coherence? How do you structure your thought? How do you write sections in a piece? How do you orient the movement in the harmony? It's a very, very complicated problem.

7.7 That's the technical backdrop to this, and we'll get into the details of that in a minute. But that's one force that's starting to emerge in music, is this increasing ambiguity, or the negative capability of Keats, once again, this concept that, well, we no longer have certitudes. We no longer know that we're going to start in this key and end up in this key. We're going to use these types of chords, and we're going to use this grammar to structure our pieces. That's no longer a given anymore. Instead, we have all of these personal solutions to technical and esthetic problems in music. We have all these different styles emerging, which in some cases are completely incompatible with each other. They might be radically opposed.

7.8 For example, in Germany, in the latter half of the 19th century, you have the Wagners of opera, which are these spectacular, grandiose compositions, operas lasting 4 hours or more for gigantic orchestras and Wagner writing all of these libretti himself, these enormous folkloristic stories that he's developing on the stage. There's a certain grandiosity to that conception of music. In France, you have composers living around the same time or slightly later who are viewing this and thinking, "Well, let's do the opposite of that because that's too German. We need to have French music." This is, again, this nationalistic conception. It's like, well, "We need to have Norwegian music, and we need to have Danish music, and we need to have French music, and German music, and Dutch music, and Italian music." And there's all these different nationalistic concepts that go along with that.

7.9 You have composers like Claude Debussy or Erik Satie writing music that is consciously attempting to do something peculiarly French, do something that's the opposite of what Wagner is doing. Satie writing these tiny little, ironic, humorous little pieces, 2- or 3-minute-long piano pieces. Debussy, writing music that is predicated on this absolutely gorgeous harmony. But in both cases, moving in the opposite direction. So these

different styles that are, in a certain sense, not really compatible with each other anymore. And this extreme individualization of music, this becomes greatly accelerated in the last decade of the 19th century.

- 7.10** So we've talked about how music is no longer predicated on certitudes. We've talked about how there's a shifting technical substratum for composers, no longer using the same tools, no longer having a common language. The relationship between composers and their audiences starts to reach what might be considered an all-time low at this period. Remember that audiences are getting more and more fickle because they have more to choose from. They have not only the increasingly diverse range of things that can be experienced from living composers, but also all the great music of the past. This leads to a situation where some composers are entering into an almost conflictual relationship with their audience or withdrawing from it entirely, from a fear of not being understood, being rejected.
- 7.11** There were many concerts within this era, before the First World War, that resulted in extraordinary scandals, where composers are writing pieces that are going so far out on a limb that the audience simply cannot comprehend it, and reacting, in some cases, quite violently to that. There are many instances of these sorts of scandals. Perhaps the most famous one is the scandal that arose after the premiere of the *The Rite of Spring* by Igor Stravinsky in 1913. But it's not an isolated event. In Vienna, in the aftermath of Gustave Mahler and Richard Strauss, these Romantic figures that wrote these colossal symphonies and tone poems, there's another generation that starts to emerge after them, a new generation of Viennese composers that is growing up in the era of late Romanticism, Freud, the analysis of dreams, this drastically new world that's starting to emerge.
- 7.12** These composers have a fundamentally new attitude towards their art. Perhaps the figurehead of this new group of composers is the Viennese composer, Arnold Schoenberg. Schoenberg is a fascinating figure in musical history because he is indisputably a product of this lineage of essentially Germanic musicians. You can see this as an unbroken line of major figures and innovators from Schütz down to Bach, down to Schubert and Schumann and Wagner and Brahms and onwards. And Schoenberg comes out of that. He's an immensely gifted musician. This is clear from when he's extremely young. Already when he's a teenager, he's writing absolutely remarkable pieces. And he sees himself as being in that lineage. Again, this is this consciousness of a nationalistic lineage. This is a relatively new concept at that point.
- 7.13** He's writing pieces that fit pretty nicely within the framework of late Romanticism in the early part of his career. They have some eccentric features to them, which sometimes resulted in problems with audiences and performers. But for the most part, it's music that is comprehensible within the framework that was familiar at that time. But something starts to happen. Schoenberg is a deeply thoughtful person, and he recognizes, as did many other musicians at the time, that there's a crisis starting to

occur in the very fabric of music itself. Because the attraction force of the tonic and the dominant is getting weaker and weaker, because the triadic grammar of harmony is breaking down, because composers are working on forms that are less and less predictable, because every thing that up to that point had structured music and given it a sense of orientation is starting to get weaker, he's perturbed by this because he's deeply in love with the Austro -German musical tradition.

- 7.14** He considers it to be his personal inheritance. He's part of this lineage of composers, and he wants to see this lineage continue. But he can't accept this contradiction between the fact that these structural principles are still basically operating in music, but they're fraying at the seams. It's making it harder and harder and harder to write music. And Schoenberg takes this final step into the void, into the unknown, and he writes pieces where there's no longer a key. There's no longer a scale in the conventional sense of the term. He steps away from that. He steps away from the last vestiges of things that provided a coherent grammar for music, much as other artists were doing in other fields.
- 7.15** For example, in painting, a figure like Kandinsky starts to make paintings that are no longer figurative, paintings that are amalgamations of colors and shapes. Paul Klee, also a Swiss artist, is doing the same thing, experimenting with different media, different ways of making pictures, and not necessarily all that focused on representation. And this gradual collapse of a shared musical language and this rise of technical speculation and composers trying to figure out how they can respond to this, while simultaneously excited by the possibilities, dramatically expanded possibilities, expressive possibilities, technical possibilities of music, are having to deal with this. We're going to hear some examples of where that led to and what situation arose out of it in a minute. But a couple of other things I want to add to this.
- 7.16** Remember, we talked about the changes that occurred when music publishing started to become more widespread. Well, it turns out that was nothing in comparison to recording technology. Recording technology starts to get going. It starts to become a thing, really, in the very last part of the 19th century. We, in fact, have recordings of Brahms, for example, speaking. Brahms, a kind of typical figure of musical Romanticism. Well, we have recordings of his voice. That wasn't the case before. We start to have photographs and recordings of composers at this point. You have composers able to make recordings, not of the greatest technical quality, obviously, but of themselves playing the keyboard or recordings of orchestra pieces. Gradually, this turns into a viable commercial enterprise. Recordings can be duplicated, they can be sold. What that means is a few things.
- 7.17** Up to that point, if you wanted to play a new piano composition, you had to learn how to play the piano. You had to have a piano in your house. That was a standard part of the education of an average middle-class European in the 19th century, was you could probably find your way around on a keyboard. You could probably sight-read a piece of

piano music. With the advent of the mechanical reproduction of sound, that's no longer necessary. You don't need to be able to play the piano. You can buy the recording, and a machine will play it for you in your house. And as this period wears on, that becomes more and more and more and more common. So again, drastically greater variety available to audiences. You can just buy a recording now and play anything at home.

7.18 And not only that, you can buy a recording of a classical piece, a string quartet or a piano sonata or whatever it is, and listen to it 300 times at home if you want. That wasn't possible before. You have to remember that a piece like Mahler's Eighth Symphony or the Bach St. Matthew Passion, these were not pieces that you'd have the opportunity to hear 300 times. It was a one-time deal, putting on a concert like this. Then maybe, maybe 40 years later, it will be played again. Suddenly, that changes. You can get a recording of pretty much any piece, play it 100 times, analyze it, get really into the details of the sound. That starts to change people's relationship to art in a very interesting way.

7.19 For some, this is a disturbing development because along with recording technology comes the prospect of an artificial relationship to art. For example, I can go into a recording studio and I can make a recording of a piano piece that I can't play particularly well, but I can record it a little bit at a time. I can record 3 seconds of it and then stop and do an edit, paste it together, get a reasonably convincing-sounding recording of it. You can cheat, in other words. Performers can make themselves sound like they have absolutely flawless technique, which then creates a kind of feedback loop. Very interesting. Because audiences are now accustomed to hearing these very polished recordings. Then when they go to a concert, they expect to hear the same thing. They expect to hear perfection. They don't want to hear mistakes. That causes performers to have to dramatically increase their level of technical skill to produce these flawless, polished performances every time.

7.20 This is an era where our whole relationship to the art of music is changing. It's no longer necessarily a collective enterprise. It's no longer necessarily even something where the human subject is at the center. Rather, it becomes a form of mass entertainment, in a sense. Glenn Gould, the Canadian pianist, my compatriot, wrote a very perceptive essay about this. It was called *The Prospects of Recording*. He took a very different view on recording technology. Gould was intensely phobic when it came to performing, when it came to audiences. He didn't like that at all. He was one of the first famous classical musicians to stop doing it altogether, just no longer perform in front of audiences. Instead, he decided he would focus on making recordings, just as the Beatles did actually after what was it, 1965 or 66, when they decided to stop touring. It was just too difficult at that point.

7.21 Gould had the same kind of reflection. "I'm going to make recordings." Our experiences of music are going to be increasingly mediated by technology. Rather than resisting that or being afraid of it, he embraces it. He starts making these spectacularly beautiful

and successful recordings that have a vast audience. This is a complicated background against which composers had to somehow continue to find a way forward. We've talked about the fact that tonality stops providing this basic polarity. It stops providing an orientation for our perceptions. In a sense, you could say, "Well, that's an unfortunate thing because, well, what do you do? It's like we're lost. We're stepping out into the void."

- 7.22** On another level, what it did was, it was a tremendous stimulus to the creativity of musicians, not just in the classical world, but also in jazz and many other genres of music. We now have far more ways of orienting a listener's perception than was the case in the past. You can do it through rhythm, for example. You can do it through timbre. You can do it through extremely creative use of orchestration. You can do it through multiple different ways. You can have entirely new harmonic vocabularies. And that results in an increasing diversification of the stylistic and technical possibilities available to musicians.
- 7.23** I want to play you a few pieces now that give us a little idea of this era, what it was like. The first piece is a piece by Claude Debussy called *Prélude à l'après-midi d'un faune*, the *Prélude to the Afternoon of a Faun*. This is an orchestral composition that was written inspired by a poem by Stéphane Mallarmé, a French poet. While I certainly believe this is an exquisitely beautiful composition, you'll hear right away that it's based upon a totally different approach to writing music. There's a couple of things that I just want to point out before we hear it. The first is that sound itself, timbre, the quality of sound, by this point, is now as important as any other dimension of music.
- 7.24** This piece starts with a flute solo. The flute is playing in front of the whole orchestra, and then the orchestra gradually enters. The choice of the flute is not a completely neutral choice. It's not without consequences. It's a reference to the god Pan, and it has all these associations with nature and all these poetic associations. Debussy is using instrumental color as an integral component in the meaning of the piece. If you were to take this *Prelude to the Afternoon of a Fawn* and re-orchestrate it or write it for piano instead, you would be severely changing the meaning of the piece by stripping away the color. It's now an absolutely essential component of the music.
- 7.25** Now, the second thing about this piece that I want to point out is this cultivation of a permanent state of ambiguity. It starts with this chord, this melody, sorry [music]. When you hear a melody like that, it's impossible to know what key the composer is referring to. Partly because we have every chromatic tone, that's to say, all the white keys and all the black keys between C sharp and G [music]. And that melody spans an interval called the tritone, which is this [music]. The tritone, historically, is considered a particularly dissonant and unstable interval. Debussy uses it to structure his melody.
- 7.26** We don't know what key we're in. It could be C sharp minor, it could be E major, it could be A major, it could be F sharp minor. We don't know. We don't find out, really, until the orchestra enters. And even then, we're still in this world of ambiguity. So you

have to accept this floating, dreamlike, ambiguous state when listening to this melodic fragment [music]. So let's listen to the piece.

7.27 [musical piece]

7.28 That was an excerpt from the first couple of minutes of the Prelude to the Afternoon of a Faun by Claude Debussy. It might not necessarily seem like it from our contemporary vantage point, but these pieces were really quite scandalous and upsetting for audiences when they first appeared. That only continued. Schoenberg, as I mentioned earlier, is really the figure that, in a sense, had the unfortunate historic luck, I suppose, to be the figure that took the next step. He himself had a very ambivalent attitude about this. He talks about the fact that he was fundamentally, in his deepest impulses, he was a conservative who was forced to become a radical, which is interesting, an interesting perspective, interesting reflection on his own creativity in the sense that he's somebody who's deeply indebted to the music of the past, deeply conscious of the weight of the past, of the value of his predecessors, but who also recognizes that music has come to such a point of development that it wasn't clear where to go next.

7.29 There was a vast new area of exploration waiting to be discovered. At first, he considered this new approach to be called pan-chromaticism, or pan-tonality. What he means by that is a music that is in not one key, but in many different keys, and the keys can change constantly. You can have very subtle but very fleeting references to a particular key center or a particular tonality, but it might only last for a few seconds, and then we're off to somewhere else. Again, with that, he starts to accept an increasing amount of dissonance as an integral component of his musical language. Perhaps one of the most emblematic pieces of this is the set of 3 piano pieces that he wrote, the Opus 11 piano pieces, which we'll hear an excerpt of.

7.30 But there's also a composition called the Second String Quartet that Schoenberg wrote, which dramatizes. It's a unique piece in the history of music in the sense that you can literally hear the fabric of tonality, of scales, of triadic harmony, falling apart in the piece. He's at a razor's edge point in his music, where it's becoming increasingly implausible to keep that going. The first two or three movements of the piece are still written in a basically, a very expanded and a very complex tonal style, but they're still basically based on triads and so on. In the last movement, the dam breaks. Suddenly, there's no more up, there's no more down. We're in a relativistic musical space where any tone can follow any other tone.

7.31 And it's music that's extraordinarily rich in imagination and new figures, new harmonies, new colors, much as the work of abstract painters, as I mentioned earlier, was in the domain of painting. But it resulted in an extremely conflictual relationship with the Viennese public of his time. So much so that composers like Schoenberg went so far as to create a private society for the performance of musical compositions. The Society for Private Musical Performances, which was basically a private club that was accessible only to a small number of people who paid a membership. No critics were

allowed. There was, in fact, a sign on the door at their performances: "Critics Must Turn Away."

7.32 The idea was that people who are interested, sincerely interested in art, who are not there to jeer, were not there to make fun of the pieces or cause a scandal or whatever it was, but just wanted to listen, hear what composers and artists were doing, could go and hear very high-quality performances that were meticulously rehearsed in optimal conditions. Applause was not allowed. You were not allowed to react after the pieces were performed. That's another interesting thing. No expression of approval, no expression of disapproval. You sit down, you listen to the piece. In the case of particularly difficult compositions, they would be played 2 or 3 times in the course of the concert, so the audiences could have a chance to hear them more than once, maybe hopefully understand them a little bit better.

7.33 And this was not an isolated occurrence. There were other private societies of this sort that start to emerge in Europe. And on the one hand, it allowed composers to have their pieces be performed in an atmosphere that was free of scandal, with high-quality performances. It could be guaranteed that there would be a fair showing of your piece. But as you can imagine, it also resulted in a restriction of the audience, a very considerable one. So let me play you the first of the three piano pieces, Opus 11, by Arnold Schoenberg.

7.34 [musical piece]

7.35 So that's the piano piece, Opus 11, no. 1, by Arnold Schoenberg. His attitude at this point is that it's a crisis in the sense that he feels compelled to explore this new musical world, but he doesn't know how to orient himself in it yet. He's saying that it was enormously difficult to write these pieces, but he had to do it. The thing is, when you're in this space, you have nothing except your feelings and sensibility and intuition to guide you. You no longer have reason, really. You no longer have the traditional tools of musical composition. Those things don't function as they used to. If you pull this piece apart, you do start to hear that there are, in fact, patterns in the music. There are themes that are varied, that are developed. There are certain melodic fragments that we hear multiple times the course of the piece, but it's less evident. It's harder to hear them, and you have to spend a bit more time with these pieces in order to appreciate that. That, of course, places greater demands on contemporary audiences.

7.36 Now, we've heard an example of Debussy's music. We've heard some Schoenberg. There's another figure that, in a sense, completely defines the music of the 20th century, along with Schoenberg, and that is Igor Stravinsky. Stravinsky, during his lifetime, was the Picasso of music, which is to say, a kind of emblematic figure who everybody had heard of. If you knew anything about the art world, you'd heard of Picasso, you'd heard of Stravinsky. He was a cultural lightning rod in a sense, but also a household name. And that just increased over the course of his life. When Disney used

an excerpt of the Rite of Spring in the movie Fantasia, he became vastly famous. Stravinsky's attitudes and his esthetic positioning became extraordinarily widespread.

- 7.37** What Stravinsky did was, in a sense, he pulled music away from Romanticism, from the late Romantic ethos that is still actually present in the Schoenberg piece. That piece, for all that the harmony is new and strange-sounding, is still very tied to a late Romantic, a mixture of Brahms and Wagner in its chromaticism and its continuous development of musical ideas and still got this very flowing, continuous, very expressive musical language. Stravinsky didn't want that. He went in a very, very different direction. There's a couple of things that happen in Stravinsky.
- 7.38** One is that he starts to reintegrate the concept of the popular and the vernacular into his music. There's no trace of that in Schoenberg. We're as far as we can possibly get from the popular in that. Schoenberg, along with Wagner and a few other composers of this era, Richard Strauss and others, are starting to position themselves as polar opposites to the popular. There's no trace of the vernacular, no trace of a common vulgar language. It's a music of transcendence, of sublimity, of constant pushing forward. Stravinsky, on the other hand, takes popular elements such as Russian traditional folk music and integrates it into his music. And not only that, he does so without a trace of sentimentality, without a trace of the hyper-expressive language of Schoenberg and his contemporaries.
- 7.39** The music of Stravinsky has an object-like quality to it. It's not necessarily attempting to move you emotionally. It has a kind of esthetic quality that is autonomous at this point. It's got a sharpness to it, an almost sarcastic dimension to it. It's not attempting to remind you of things that you found expressive in the past. It's a very, very different sort of attitude. There's a kind of dryness to it, kind of objectivity. This becomes quite characteristic of a lot of the music in the Weimar years, the interwar period.
- 7.40** So let's hear an example of a comparatively early composition by Igor Stravinsky. This piece is called Symphonies of Wind Instruments, symphonies in the plural. What is interesting about this piece is a couple of things. First of all, there's absolutely no notion of development anymore in his music. Remember we were talking about the principle of sonata form, where you take a bunch of different musical ideas that contrast with one another and you have them dialogue with each other. You have them interact, you combine them, you superimpose them, you put them through their paces, and maybe you end up with a synthesis at the end. Stravinsky rejects all of that. There's no synthesis. There's no development of musical ideas. He presents an idea, presents it for its autonomous qualities, whatever they are, and when he's done with the idea, he moves on to the next one. Then he makes a mosaic or a collage out of these different musical ideas.
- 7.41** The other thing is the characteristic sonority of this music. Stravinsky is an absolutely profoundly gifted orchestrator. He's able to come up with sounds and timbres and colors that we've never heard before. Entirely new conception of writing for traditional

instruments. Every piece by Stravinsky, it doesn't matter what style he's writing in—he lived a long time, died in 1971 —but every piece he wrote has this thumbprint where it could only have been written by him. This particular attitude to sound. It's got a slight stridency to it, slight edge to it, but it's very powerful. So here's the Symphony of Wind Instruments of Igor Stravinsky.

7.42 [musical piece]

7.43 OK, so that's a little overview of some of the major figures that defined the modernist period in musical history, with extremely contrasting attitudes, as you can see. The Stravinsky piece that we just heard is characterized by, well, a complete absence of synthesis of these different elements. It's predicated entirely on oppositions and contrast and sudden ruptures and breaks and this completely new approach to color.

7.44 So that's a little introduction to musical modernism. From there, it just gets stranger and stranger. But we'll talk about that a little bit tomorrow. First, I'd like to address any questions that you may have.

Lecture 8: Musical Evolution

At a glance — Musical evolution continued into the modern era.

- 8.1** We've covered a lot of ground already. As I mentioned at the outset, I've had to compress basically 1000 years of music history into a few hours, which, as you would expect, I've had to cover some things more than others. It hasn't been comprehensive. That's just the nature of the game. What's interesting now is we're getting into the post-war era. We're getting into the era that follows the initial impulse of modernism in music.
- 8.2** What's interesting there is, even though this is a comparatively short time span, we're talking about maybe 30 years, approximately, the sheer amount of things that changed during that era is so vast, it's so colossal, and the rate of change, the acceleration of change in music, is so dramatic, it's exponential, that in a sense, I could spend certainly 8 hours just talking about this very short little period within these 1000 years. We're going to talk a little bit about that today, about the period leading up to and immediately following the Second World War.
- 8.3** The first thing that I'll observe about that is that we talked about the advent of modernism, which has a number of core concepts. One of them is that things are accelerating. There's rapid technological progress. Society is changing extremely quickly. With that, as you would expect, art is also changing, and the attitudes of artists towards their audience are changing as well. There's a generalized sense within modernism that tomorrow will be better than today. That the world is improving in a certain sense. It's improving technologically. Living conditions for the population as a whole, generally, are improving. Industrialization is completely changing society. We have the possibility through technological mediation to produce things in art that would have been absolutely unthinkable before.
- 8.4** You have artists with absolutely insane levels of ambition. You have writers such as James Joyce, for example, writing a book like *Ulysses*, which is already extraordinarily challenging. It's a very lengthy book in which every single chapter is written according to a different concept of some kind. There's a chapter in *Ulysses* that's written in the style of a newspaper article. There's a chapter in *Ulysses* that's written in the style of a poem. There's chapters that replicate the experience of reading a textbook and so on. Every chapter has a different stylistic approach within this book. But then that wasn't enough. He went even farther in his next novel, which is called *Finnegan's Wake*, and that's a book that took him over 20 years to write. This is a book that is so extreme in its demand upon readers that the risk at that point becomes that the artwork becomes almost a hermetic object. It becomes almost an unreadable thing.
- 8.5** Now, for those who have, in fact, read *Finnegan's Wake*, the experience is very different. Of course, it's challenging, and it takes a very long time to read this book. But

the rewards are also immense. So that's kind of the gamble in a lot of modernism, is the idea that this is going to be very challenging. This is not like reading a Michael Crichton novel. This is going to be really difficult. You'll have to get a shelf of dictionaries. You're going to have to look things up. You're going to have to make the attempt to understand what's being related in this book. But if you have the commitment to doing that, if you're able to get through the book, there will be commensurate rewards. In other words, your worldview will be changed. You'll be a different person afterwards, and you'll have a newfound appreciation for just the sheer range of what it's possible to do in art.

- 8.6** This attitude is very, very widespread. It's not just in literature, not just in poetry. You see it in music. You see it also in architecture. You see it in painting, certainly the visual arts. There's this generalized idea that art is going to somehow go farther than it has gone in the past. Not necessarily be better, because although, as I've attempted to show in these lectures, art is always changing, it's by no means obvious that it's improving. Obviously, we can say in a measurable sense that technology is improving, and we can look at all the ways that that's happening. The artistic endeavor, however, first of all, I'm not sure that it ever really fundamentally changes in its basic ambitions and impulses.
- 8.7** Secondly, you can't say that the Baroque style is an improvement over the Renaissance style. You can't say that the music of the post-war is superior to the music of the pre-war era. It's not obvious in any sense that art ever improves. However, you would expect it to reflect, at least to some degree, the changing conditions that human beings are living in. That's just to give a bit of a background to this. But I also have to point out that, although this modernist attitude is certainly quite widespread and generalized amongst a lot of different artistic disciplines, it's by no means universal. Modernism can take many, many, many different forms. One component of that, in a sense, is that, to a degree, artists can now choose their precedents. They can now choose what direction they're going to go in stylistically. That wasn't so much the case before.
- 8.8** If you were an Italian composer living in 1720, first of all, your knowledge of previous music would have been very limited. You would not, in all likelihood, have had much understanding of the music of the 1200s, for example. And your stylistic options also would have been limited simply by the fact that your opportunities for having a piece performed and played and the audiences that you might have had would have been far more limited than is the case today. But after modernity, after the Second World War, that changes dramatically. Not only do composers have access to recordings of almost anything in scores and a vast body of scholarship, but in addition to that, there's a new and much more diverse audience that starts to emerge.
- 8.9** So then that makes it not so much an inevitable thing that you'll write in this or that style. It becomes more a personal matter, in a sense. The work becomes, to a greater degree, I would argue, the reflection of the artist's particular proclivities, background,

interests, and also the set of circumstances that they're born into. There are some artists in this era that are choosing a more conservative style. There are others that are more radical in their orientation, and then there are some that fall somewhere in between. I can cite a few examples of that.

- 8.10** The German composer Paul Hindemith, for example, was very concerned with the idea of what he called *gebrauchsmusik*, or music that was intended for a specific function. It's no longer this abstract reaching for the stars, this *ad astrum* concept that you get in a lot of modernism. Hindemith was, like Bach, a pragmatist, and he's thinking more in terms of things that people can actually play, where they're going to play it, who's going to be listening, can it be played by students, all this sort of question. Then there are other artists who are completely uninterested in those questions and are really just thinking in terms of, "How far can I go in a given direction? How far is it possible to push art? How far can I drive complexity? How far can I drive imagination? What happens if you give free rein to imagination? What ends up happening?"
- 8.11** There are lots of different attitudes that start to emerge in this era. But the background against which all of this is happening is one of absolute catastrophe, one of calamity, in fact, the two great wars. I think that it's difficult from our present vantage point to grasp exactly what it would have been like to have been born in 1910, 1920, and to have been young in the post-war era. What it would have been like to grow up in France in around 1945, 1946, or indeed, Germany, around the same time—to be a young artist and to be confronting the prospect of a totally ruined civilization, a civilization that's been brought to its knees, that's been nearly completely destroyed. This was the situation for a lot of young artists.
- 8.12** There were a couple of things that came out of that, obviously. One is there was a need to rebuild. The second was artists were confronted with the question of, "How do we respond? How do we respond to the greatest calamities that mankind has ever had to face? How do we respond to the fact that civilization itself nearly came to an end, that our cities have been leveled, destroyed?" There's virtually nothing left. What is the artistic response to such a thing? Tens of millions of people killed. Absolute barbarism.
- 8.13** The German philosopher, Theodore Adorno, thought that the response was a kind of a, well, basically, you had a moral obligation to accept what had happened, to commit to reflecting that in the work. There's a quote by his that's quite famous where he says, "To write poetry after Auschwitz is barbaric." And that sort of attitude was not uncommon amongst composers. There was this sense of, "Well, what do we do now?" It was a complicated problem. But what you see in Germany, in particular, in the years following the war, is an extended and vastly ambitious campaign to rebuild the country, and same in lots of other European nations as well. And along with that comes a period of rapid economic growth and rapid technological change as well. And while there's obviously this vast civilizational trauma that's occurred, there's also at the same time a

certain, what would you say, momentum that comes from this extremely rapid rebuilding process, this extremely rapid progress.

- 8.14** There's a kind of technological utopianism that emerges out of that. Young composers are starting to reflect on this. One of the ideas they have is that, in a sense, World War Two represents perhaps the end of something. Maybe we don't want to go there again. Maybe we don't want to produce things that remind us of that period and of the things leading up to it. Maybe there were certain attitudes that were expressed in the art of the pre-war era that we want to get far away from. The other reflection was, the Nazis suppressed any form of modernism during the war, and also indeed before it, systematically. So they held exhibitions of what they called Entartete kunst, or degenerate art, in which they would hold up examples of paintings and sculptures that were produced by living artists and exposed them to public mockery as examples of degenerate thinking.
- 8.15** So one of the attitudes that emerged out of this after the war was, first of all, let's get as far away from that as possible. Secondly, what if we predicated our artistic enterprise on the idea of moving forward? Sort of like what the Romantics did, but in a more accelerated manner, a more intense manner, a manner that is brought to a new height of intensity in response to the calamity that's just occurred.
- 8.16** What's interesting is a very new vision of art starts to emerge after this point. I don't think it would be an exaggeration to say that the fundamental line, you might say the unbroken continuity of the Austro-German musical enterprise that, to a large degree, dominated European musical thinking for at least 200, maybe 300 years — at that point, it comes to an end. After World War II, there's a profound shift. One of the dimensions of that profound shift is that the world becomes a lot more global. We start to see a lot more exchange between countries. We start to see the rise of the United Nations, the rise of the European Union, all of these international, supranational entities.
- 8.17** This is reflected in music as well. So that, whereas someone like Debussy openly mocked the Germans. He called them "les boches," which is a kind of not very nice French term for Germans. It's a kind of insulting term. He was very open about this, about, "Oh, I'm a French composer, and over there is those people." He was very insulting about them. Those sorts of attitudes become less defensible, certainly after World War II. There's a new spirit of international exchange and collaboration that emerges out of that. There's this whole generation of composers born between, let's say, 1916 and maybe 1930, approximately, just to give a general time frame, that's born into this drastically new social situation, this new need to respond to a completely new world order.
- 8.18** They come from all different parts of the world. They come from Japan, they come from New York City, they come from Los Angeles, they come from Berlin, Paris, Rome, Venice, Amsterdam, whatever. These composers are committed to rethinking music from the ground up. It's a kinda of a tabula rasa concept. The idea is they're starting to

find out about some of the music that was not permitted to be played during the war and before it. The music of Arnold Schoenberg, for example. Now, Schoenberg was Jewish, and he had to flee to the United States. In fact, much of his career was spent in Los Angeles, where he was a professor of composition. So this whole repertoire that nobody was able to hear. You couldn't obtain the scores, you couldn't hear performances. It was banned. After the war, gradually, these sorts of things start to filter in. Younger composers are finding out about the first wave of modernists and the work that they did leading up to World War II.

8.19 This, for them, is the basis for a new grammar. It's not the music of Richard Strauss. It's not the late Romantic music of Gustave Mahler or other composers of this era. It's something else. It's this willingness, and indeed this intense desire to think of a new grammar for music that starts to take hold. So this is the background—international collaboration, composers from all different parts of the world, starting to write in not exactly a common language, but to have a greater degree of common interests than was the case before. And increasing complexity of society. Also, we start to see more social fragmentation, and also the gradual commodification of artworks. That's another dimension.

8.20 We talked about the desire to get as far away from the ideologies of fascism and excessive state control and so on, get as far away from that as possible, return to the individual, to the sovereignty of the individual creative mind. But there's another thing that starts to happen, which is the emergence of the cultural industry. You see the beginnings of that in the Industrial Revolution. You see the beginnings of that in the creation of these new concert halls devoted to secular music in the classical era and beyond. But it rapidly and dramatically accelerates during this era. Music becomes a gigantic industry, and we see the rise of mass culture through the radio, through recordings of music, through discs, through LPs, through increasingly widespread and enormously popular methods of distribution of artworks.

8.21 As this mass commercialized culture becomes more and more prevalent, artists start to think, "Well, how do we fit into this exactly?" Because it's difficult. Their art is not really predicated on the same fundamental impulses as this new commercial art. There's a tendency, once again, for art to become increasingly esoteric. Again, this is not a fundamentally new attitude. We talked about this with respect to the Romantics. There's a measure of that. Except that in the post-war era, that divide between the commercial, between mass culture and the so-called fine arts, becomes greater and greater. I say so-called fine arts because there's a degree of exchange between those two things. There's no dividing line, and it becomes extremely ambiguous in many instances. I'm not even sure it's really useful necessarily to think of things in those terms. But nevertheless, there is a, on the one hand, a mass cultural industry that emerges in this era. And then there are artists that tend to be working in a more artisanal direction, let's say.

- 8.22** Once again, it's this Keatsian idea of negative capability, the idea that the artist not only can but should open up spaces for reflection, for contemplation, for ambiguity to be part of people's lives, and also critical engagement. You can't be just a passive listener anymore. You can't be a passive reader. You have to engage, you have to reflect, you have to think about what it is that you're seeing and hearing. At this point, the divide between light music and serious music becomes almost absolute, which I would argue is not a healthy thing. I think it's better when there's an exchange between those two things, when there's a degree of connectedness. We'll get to that a little bit later because, in the present day, I think that that is quite clearly beginning to happen. But in the immediate postwar era, it was a different matter.
- 8.23** Along with this, there's this concept of music as a form of mass entertainment. People are not learning to play instruments quite as much as they were in the past. It's no longer an absolute requirement of a sort of average educated person to play an instrument. Instead, you have this widespread mechanical reproduction of music and this vast industry that grows up around that. At the same time, what you have is this generation of composers who not only are growing up in the post-war world, but in addition to that, there starts to be a massive change in the way art is being funded. You might think, "Well, who cares about that? The art is the thing that matters."
- 8.24** But the thing is, in the history of music, it does, in fact, have consequences. Who is paying for it? It makes a difference. It makes a difference if the art is being paid for by the nobility, by the aristocracy, by private patrons, or by the state. What starts to happen in the post-war era is, increasingly, art is being subsidized by the state. So that has a number of consequences. What some composers were thinking at the time was, we want to be free from the pressures of the marketplace. We don't want to have to be beholden to the same pressures, the same requirements as Walt Disney when he puts out a \$20-million movie or whatever was at the time. We want to be able to pursue these adventures of the imagination, unfettered by those sorts of concerns, in the same sense as a composer like Haydn was when he was employed as an employee of the court during his day. That was the thinking.
- 8.25** So what you start to see are these state subsidies that are paying not only for festivals, but are indeed covering commission fees for composers and are providing increasingly this whole infrastructure for a very new type of musical situation. Remember in the past, there wasn't this concept of classical music. That was the creation of the 19th century, as we've seen. That really emerged out of the creation of a canon, a repertoire. Well, what starts to happen here is, you start to see just a complete division between, on the one hand, the world of classical music, the world of famous soloists and conductors and the Leonard Bernsteins and the Eugene Ormanys and the Stokowskis, these huge stars, the Isaac Sterns, all these famous performers and singers and conductors.

- 8.26** Then on the other hand, the world of what starts to be called Neue Musik in German, new music, or musique contemporaine, in France, or new music, as it's referred to in the UK and also in the United States. And it starts to be a matter of two separate worlds going their own ways. There's the classical establishment, which is big business and very, very successful, and becomes a huge thing, particularly in the 1950s and 1960s in the US. There's a huge industry around classical music, gigantic stars that can command enormous fees for their concerts. Then there's this world of Neue Musik, which is state-subsidized, has a smaller audience, and has extremely different aims. It's not a populist movement. It's attempting to preserve, to some degree, and extend the autonomy of composers and artists.
- 8.27** There's a number of names that emerge during this era that profoundly shaped the way composers, in general, were thinking about their work. Probably the first two that come to mind are the Frenchman Pierre Boulez, that's B-O-U-L-E-Z, and the German Karlheinz Stockhausen. Stockhausen is a name you may have heard. He's an interesting figure. Stockhausen was born in 1928, and he was a war widow. Sorry, he was a war orphan. It's quite a terrible story. Stockhausen's father was killed on the front, I believe, when he was 5 years old, and his mother was officially killed by the Nazis because she was mentally ill. So he's an orphan. And not only that, he has to endure horrific suffering during the Second World War. He's tasked, among other things, with collecting bodies from the streets of Cologne as a child. Imagine what experiences like that can do to somebody, when you're an orphan and you have to live through this.
- 8.28** He was profoundly shaped by that. But Stockhausen was also a genius. He was also a profoundly inventive, profoundly searching musical mind. What Stockhausen does is he vastly expands the range of what music can do. Starting in the early 1950s, he writes a string of pieces that are so extravagant, so extreme in their novelty, that you couldn't possibly remain indifferent to them if you were a musician. He writes pieces like Gruppen, or "groups" in English, for 3 orchestras. Imagine this, a composition for 3 orchestras playing simultaneously on 3 different stages, each of them playing in different tempi at the same time, with 3 different conductors, and basing all of this on entirely new conceptions of music.
- 8.29** This new conception had a name, and the name was serialism. Serialism became a very, very, very widespread concept in the music world in the post-war era. What it has to do with, fundamentally, is responding to the dissolution of tonality of triadic harmony that we talked about yesterday. Composers wanted some other means of imagining a new musical grammar. This ties in with the tabula rasa concept, the idea of "we're in a new era, we have to think of things differently."
- 8.30** So this new concept had to do with taking the different musical parameters, rhythm, pitch, anything that can be quantized and put into some kind of an order —rhythm, pitch, duration, intensity, whatever—and assigning these different parameters a particular order that would be determined in advance by the composer. In other words,

putting them in a series. So if you have 12 tones, 12 notes, 12 equals semitones, then instead of working with the scale, instead of working with triads, you put them in a new order, like inventing your own scale, and you use that as the basis for your composition. In other words, it's trying to find a new grammar and a new syntax for music.

- 8.31** That allowed some quite extraordinary discoveries to be made. Some of this music is, and is still today, extremely striking. It might sound unusual if you've never heard it before, but it can be very exciting. Stockhausen writes these pieces, these vastly ambitious pieces, but he also is very interested in these new technologies that are arising. In different parts of Europe, we start to see the emergence not only of radio orchestras in Germany, Italy, and later on in France, but also in the UK. What these radio orchestras are, are state-subsidized orchestras that exist in order to play music on the radio. Because of their generous subsidies, they can take enormous risks. They can commission hugely expensive pieces and give them adequate rehearsal time and present them to a broad public. This starts to become a very generalized model. In fact, it's still a very widespread model today, especially in Europe.
- 8.32** Now, Stockhausen, in addition to the opportunities that he had writing for these radio orchestras, rights for electronic media. So along with all of these other changes that are happening, you start to see these electronic music studios in Italy, in the Netherlands, in Germany, in France. Studios that are generally run by these different radio stations, subsidized radio stations, that offer musicians or anybody else who is interested in these technologies, the possibilities of creating electronic music. Music, in other words, that has nothing to do anymore with acoustic instruments, no longer has anything to do with conventional performance. It's all entirely created in a studio. You have oscillators, tone generators, filters, all sorts of different devices that allow composers to generate sounds and to record them, to edit them, and to stitch them together into new types of compositions.
- 8.33** Stockhausen is completely fascinated by these possibilities. He sees that for the first time, you can write pieces where you can write any frequency you want. You're no longer constrained to the limitations of what humans can physically play. That's not an issue anymore. You can have any frequency. You can have way beyond the limits of the keyboard range, and you can have any tiny little interval in between those tones, and you can play them at any intensity and for any amount of time. You can create any timbre also through additive synthesis or through subtractive synthesis, where you take a recording of some white noise and then you filter it down into its subcomponents and you pull some individual tones or tone bands out of that. Or you can use synthesizers to generate sine tones and square waves and all of these different new concepts.
- 8.34** He makes a piece out of this. He spends thousands of hours in a studio in Cologne, creating a piece called *Gesang der Jünglinge*, The Song of the Youth. I'm going to play you an excerpt from that. So *Gesang der Jünglinge* is a composition from 1955 to 1956 that was unlike anything anybody had ever heard at that point. This is an entirely new

conception of music, entirely produced in an electronic music studio, requiring untold numbers of hours to produce. Now, at this time, this is before Ableton, this before ProTools, this is before all these technologies. You had to physically record each individual sound on a tape. In some cases, these sounds would be microseconds long. Then if you wanted to create a complex sound, you'd have to record maybe 50 or 100 or 200 individual sounds and layer them on top of each other to create a composite sound.

8.35 The sheer amount of labor involved in doing this, all mechanically, using splicing techniques and precise coordination of different layers in a composition and so on, is difficult to conceive. But Stockhausen was one of the musicians that was willing. He had this sort of obsession with work. He was a bit of a workhorse, and he could do things like that. He could work for 18 hours a day. So that's what comes out of that, is *Gesang der Jünglinge*. Now, *Gesang der Jünglinge*, interestingly, is not a secular work. In fact, it's a religious piece. It talks about the song of the youth in the fiery furnace, among other things. So he hired a young Polish boy soprano to sing various texts and tones that he had composed. They recorded this boy's voice, and they threaded it through this electronic soundscape. So let's hear an excerpt of *Gesang der Jünglinge* by Karlheinz Stockhausen.

8.36 [musical piece]

8.37 OK, so this composition, *Gesang der Jünglinge*, a Song of the Youth, had an extremely profound impact on not just composers, but on the world of popular music as well. If you take the cover of Sgt. Pepper's Lonely Hearts Club Band and look at the people, there's W. C. Fields, there's Marilyn Monroe is on the cover of that album, and Karlheinz Stockhausen is also on the cover. It's a photograph of Stockhausen in black and white that was taken during the creation of this piece. The reason Stockhausen is on the cover of Sergeant Pepper's is that the Beatles were huge fans of his work. They were listening to this piece. And the sorts of extraordinary innovations that you start to see in pop music around this time, 1966, have, in many instances, their origins and some of the experimentation that was being done in these European studios at that time. The idea that you could create an entirely new type of music in the studio, that you no longer had to be constrained by the limitations of performance, had very far-reaching consequences.

8.38 You see this also very much in the early work of Pink Floyd, for example. If you listen to the first two or three Pink Floyd albums, there's a similar kind of extraordinary inventivity and a willingness to try new sounds and new recording techniques. The studio, as has often been observed, becomes a musical instrument in its own right. You see the Beatles coming up with extraordinary new textures in their music around this time, and not just them, but all kinds of different bands. So a lot of these technologies, which were in fact developed during the war for tactical reasons, become used by

composers as a method of helping them establish a new musical universe, and then they get picked up also in the popular world.

8.39 What I find so fascinating is there's a brief flash, there's a brief moment in popular culture, where the avant-garde and the popular meet in the middle. It didn't last for very long, but in the late 1960s, you start to see this. The third album of Pink Floyd—I think it's the third one—it's called *Ummagumma*. There's no other way to describe that album than as a product of the musical avant-garde. It's so strange, and it's so unusual against the backdrop of popular music that I can't classify it in any other way. If you listen to it and compare it to some of the things that were being done in these electronic music studios, it's actually not that different. The difference is the dissemination is on a vastly greater scale. Pink Floyd is reaching tens of millions of people.

8.40 It's the same with the Beatles. There's a point at which these two worlds intersect, the world of popular music and the world of savant music and the avant-garde. You see that, particularly on *The White Album* with the—how many of you have heard *Revolution 9* by the Beatles? It's this sound collage that they put on *The White Album*. It's about a 9-minute-long track. Well, that's pure Stockhausen. That's where that comes from. It starts to filter into the maelstrom of different esthetic ideas and attitudes and the spirit of innovation and exploration and discovery that is so characteristic of the 1960s. It's this utopian atmosphere around all of that. In many respects, that all comes out of this post-war era. Out of this horrible calamity, you see this rapid acceleration and this almost intoxicated obsession with novelty and pushing the boundaries farther and farther.

8.41 There's this attitude that, yes, this art is alien-sounding. You've probably never heard anything like it. But in the future, people are going to be sharper. They're going to have more time on their hands. They're going to be more cultivated. They're going to be more advanced, and they're going to be more equipped to deal with challenging art. I'm going to come back to Theodor Adorno again. I quoted him previously. He's a figure that encapsulates a lot of the attitudes of this time. One of his concepts that became quite widespread, not just as a concept, but also as an attitude, was the idea of the message in a bottle. That's what he sees artists as doing at this period. You create your message to some degree in isolation, and you throw it into the ocean because there's no audience for it yet. You don't know who's going to receive this message. You have no idea. But you throw it out there, and you hope that, at some point down the line, someone finds it and is able to decipher it and understand what it is you've said.

8.42 So all of this occurs along with the advent of these more specialized audiences. We talked about that a little bit yesterday, how there's no longer this 19th-century concept of "the audience," this generalized middle-class audience going to the opera and the symphony. That still exists, but you have this audience that's extremely interested in the new, extremely interested in artistic novelty in all of its forms. You have the

audience that wants nothing to do with that, that cannot stand this music and only wants to hear their favorite compositions by Haydn and Mozart and Schubert and so on. Then there's jazz, there's new music, there's old music, there's various types of popular music and different subgenres that emerge. Today, there's just such a cacophony of different styles that you cannot possibly keep up with them all. You can list them. They're endless, different subgenres. You can go on to Spotify and find countless playlists of all these different micro subgenres of music.

8.43 There's a few other figures that I just want to briefly touch upon before we get to the next lecture. There were a number of figures in this era that took on the status of giants, people that had an outsized reputation and impact. This is very characteristic of this era. You get these figures like Marcel Duchamp, for example, in the visual arts, who everybody knows who that is in the 50s and 60s. He's just a colossally influential figure. In music, it's people like Stockhausen and a few others that I'm going to mention. The thing is, in the post-war era, there just weren't that many people writing music compared to other periods. A lot of them were dead. And you don't have hundreds and hundreds and hundreds of major composers emerging out of that era. There was a demographic calamity. This is pre-Baby Boom, right? So you get these giants emerging out of that, these figures that get an enormous amount of attention.

8.44 Another one is Pierre Boulez. Boulez was born in 1925. He's a French composer, and he's more or less single-handedly responsible for the complete rebuilding of French musical life in the post-war era. He has an almost Wagnerian dimension to him, in the sense that he's not only a composer of great renown, but he was also a founder of institutions, and he had a public role in shaping French musical life after the war. He was also a theorist, an author, a very gifted writer, in fact, and also a recording artist. He had a brilliant career as a conductor that took him to the Cleveland Orchestra, the New York Philharmonic, the BBC Symphony Orchestra, and many others besides. He was an international star of the conducting world. He did all of these things simultaneously.

8.45 What Boulez did was he took the innovations of Schoenberg and his disciples, the Austrian composers, Anton von Webern and Alban Berg, among others—this concept of music that no longer has a key, music that is, what would you say, that has a radical new set of precepts, and this idea of serialism that I talked about earlier that influenced Stockhausen so much. And he pushes that quite far. And he creates this music that is new on virtually every level. In the same sense as Stockhausen did in his electronic compositions, Boulez is doing that in his instrumental pieces. He's drawing influences from all sorts of things. He's drawing influences from French symbolism, from the Surrealists, people like Henri Michaux, and certainly Rimbaud, and Mallarmé, and all of these wonderful symbolist poets of the late 19th century in France.

8.46 And he turns this into an art of explosive intensity, an art that is confrontational in its nature, that is radically new. And he makes a number of pieces that are so new, so

shocking in their ambition, that it forces people to pay attention to him. You can't be indifferent to Boulez. Young musicians of the generation that came after him were not indifferent. It's like you had to choose your camp in a certain sense. There was this idea of you either are interested in what this guy is doing, in which case you're going to be marked by that, or it's not possible for you. You can't imagine a conception of music that's like this.

8.47 Probably his most famous piece is called *Le Marteau Sans Maître*, which translates as the Hammer Without a Master, which is an interesting poetic image. It's this image of unbridled violence of hammer that nobody's controlling. These texts come from actually a very great French poet called René Char, C-H-A-R. It's a cantata featuring an alto singer, so a low female voice, and a small mixed chamber ensemble that includes some unusual instruments. The orchestra and the conventional chamber music groupings that are familiar from the classical and Romantic eras, composers aren't so interested in doing that anymore. They're looking for new sounds, new combinations of instruments.

8.48 Boulez, in *Le Marteau Sans Maître*, creates an ensemble that sounds really like nothing else. In fact, if anything, it sounds probably more like Balinese gamelan music than anything you might imagine from the repertory of chamber music. There's xylophones in that piece. There are alto flutes. There's a guitar. It's a very, very odd mixture of instruments, but it has a particular sound to it. So let's listen to the first movement of *Le Marteau Sans Maître* of Pierre Boulez.

8.49 [musical piece]

8.50 OK, so we talked about the high classical style previously, this apogee of order and symmetry and clarity and and so on. This is the high modernist style. The high modernist style comes out of the 1950s and a good part of the 1960s as well. As with anything, this extreme development in a particular direction doesn't last forever, as you can imagine. But nevertheless, the attitudes that emerged out of high modernism are still very much present with us today. Now, I've played you examples of two European composers, Stockhausen, the German, and Boulez, the Frenchman, and America was an extremely important center of new musical ideas. You could argue in many respects, that the musical 20th century belongs to the USA. Americans invented jazz, the blues, all of these fantastic new forms of musical expression.

8.51 There's also, along with the avant-garde generation born in the 1920s in Europe, there's also a generation of figures that emerges in the United States that has an equally profound impact, but in a very different direction. There's a number of figures, for example, the composer Carl Ruggles, who was a New Englander, and quite a remarkable musical figure who was also a painter and composed some extraordinary music in the 1920s. There's Charles Ives, who's one of the great figures of American music who writes these extraordinary symphonies in the 1910s. There's also Ruth Crawford Seeger, absolutely remarkable figure. She was, incidentally, the stepmother

of Pete Seeger. She comes from this musical family that's extremely interesting. Had all sorts of adventures as a modernist in the late 1920s and early 1930s. There's this whole generation of American ultra-moderns that emerges in the interwar period.

8.52 But after the Second World War, you start to see the emergence of figures like John Cage. John Cage is a figure who, in a similar sense to Stockhausen, becomes a massive media star, a massively controversial but also popular and very widespread figure in America, especially in the 60s and 70s and beyond. He lived until 1992, I believe. And Cage's ideas are diametrically opposed to those of the European avant-garde. Whereas Stockhausen and Boulez were obsessed with a new grammar and a new structure for music and coming up with these serial concepts and making these elaborate electronic compositions, Cage was more interested in the process of making something and in the sound of the piece itself. He was also very interested in chance as an integral component of the artistic process. In other words, you're not attempting to control everything and come up with a new method of construction. You're leaving those things up to pure chance. He would literally throw a dice in order to determine the parameters of a composition.

8.53 He comes up with other ideas as well. Cage, at the beginning of his career, was rather impoverished. He didn't really have any means at his disposal. He was living somewhere in New York City in a loft, and someone asked him to write some percussion music for a ballet. But the only thing that he had available was a piano. And so he thought, "Well, what can I do with a piano that would sound percussive?" So he comes up with this idea of what he calls the prepared piano. And the prepared piano consists of taking a keyboard and sticking various implements between the strings—bits of rubber, bits of foam, bits of nails, wood—in order to alter the attack and resonance and timbral characteristics of the piano and turn it into a kind of gamelan, a kind of orchestra of different sounds. And his idea was, "Well, you can do that with just a piano. You don't need an orchestra. It's very small and compact and it's cheap. You just need some nails and some bits of rubber, and there you go. You've got this completely new sound world available to you."

8.54 He writes a series of pieces called Sonatas and Interludes for Prepared Piano. These pieces have enduring popularity to this day. They're very striking. They have a very beautiful sound aspect. They're also more conspicuously rhythmic than some of the pieces we heard previously. In the electronic pieces of Stockhausen, there's certainly no sense of pulse. I mean, the sounds are very abstract in a sense. And in the Boulez, the rhythms are so complex, they're so complicated, that it's very difficult to find a pattern within them. So let's listen to the Sonata No. 5 from the Sonatas and Interludes for Prepared Piano by John Cage.

8.55 [musical piece]

8.56 OK, so hard to believe that's a piano, but, well, that's the 20th century for you. So that's just 3 examples out of this vast body of work that emerges in this era. I can't possibly

do it justice in one lecture because, well, it's too complex a period in a sense. But those are 3 instances that cover a very small bit of terrain.

8.57 History of Western Music

Lecture 9: Music Today

At a glance — Music today, the final lecture.

- 9.1 So now we come to the final lecture in this series. And first, I want to express my gratitude for the invitation to be a part of this academy, and certainly to everybody who's come out to be in the audience. I want to thank all of you for your engagement and your fascinating questions. I'm very excited to round up this series of lectures today. What we're going to talk about now, we've done a very broad chronological survey of development of music in the Western world. I suppose the question is, where are we today? What does music look like now? What are the conditions for people creating music and people listening to it?
- 9.2 My sense is that we are most definitely in an era of extremely rapid change. The first point that I think has to be made is the advent of the Internet has completely changed the game for everybody for, well, multiple reasons. The Internet has had so many far-reaching, far-reaching consequences for music that it's, in fact, hard to know where to begin. The first, though, is that finding a recording of almost anything is no longer an issue. It's not a barrier. Let me tell you an anecdote about this.
- 9.3 So when I was a student living in Canada in the late 1990s, if I wanted to hear a recording of *Le Marteau Sans Maitre*, or those John Cage pieces, or that Stockhausen electronic piece, that wasn't something that you could just go and find in a store very easily. You had to either go to a huge university library somewhere or a media tech that had tens of thousands of recordings available, and often they wouldn't even have what you were looking for, or you had to send away for the recording. You would actually have to go and order it. And imports in those days were really quite expensive. I remember actually, I wanted to study a particular Stockhausen piece when I was 18 and really interested in his music. I had to save up for two months to buy a copy of the score, and I had to order away from Germany for a recording of the piece. And it took weeks to come in, and it was this whole big deal.
- 9.4 And that was a common experience for composers who were trying to broaden their musical horizons at the time. It was difficult and expensive and inconvenient. All of that's just gone. That doesn't exist anymore. You can find anything. You can find a PDF of just about any piece on the Internet if you know where to look. You can find recordings of anything on Spotify, on YouTube. When I say anything, I mean, I have a friend, a poet, who says that "if it's not on the Internet, it doesn't exist," basically. It's a bit of an exaggeration, but the point is well taken. The repository of human knowledge and human cultural production that is available on the Internet is virtually comprehensive. It's astounding.
- 9.5 That barrier of access is just gone. The barrier for entry, in fact, in the cultural world, is lower than it's ever been in history. There's simply no reason, if you have the slightest

interest in it, not to engage with it. It's free. It's hugely abundant, and it's available to anyone on the planet at any time. So that's a profound revolution. In addition to the access of recordings and archive materials and so on, concerts themselves of classical music of all kinds, and indeed, new compositions, and all sorts of genres of music, are available all over the world in many, many different parts of the world, many different countries. There's all kinds of orchestras now, hundreds of them in the United States alone, and often playing at an extremely high standard, a generalized professional standard that is far higher than it would have been even 60 or 70 years ago.

- 9.6** These concerts are not expensive. Often, they're, in fact, free. So there's simply no barrier. When I was a student in Paris, we could go to the Opera, the Opera Bastille or the Opera Garnier in Paris for €5. We got student tickets to go to these grand operas. It was amazing. It was so cheap. So there's no barrier to access anymore in any realistic sense to just about anybody. And I think one of the consequences of this is we're limited only by our curiosity and our willingness to engage with music. We're going to talk about that a little bit. But I also want to talk about the fact that, along with this astonishing proliferation of music, of every conceivable description—popular, esoteric, avant-garde, savant, jazz, blues, opera, whatever, you name it, it doesn't matter, world music, Tibetan music, Korean music, Arabic music—it's all there. It's all available.
- 9.7** What does a composer do? That's the question because I had a talk about this a few weeks ago with a friend and colleague of mine, the British composer Julian Anderson. We were talking about the situation of music today. Julian is a great teacher. One of the things he observed was it's very difficult being a teacher of composition today because there's no longer any common language to teach. A student can be influenced by absolutely anything, from Indian ragas to hip hop to the music of Brahms. You don't know what they're going to turn up with when they come into a composition lesson. There's no longer any common language to teach. This musical culture no longer really has a center to it, in many respects.
- 9.8** The idea of Bach taking these Lutheran hymn tunes from 1524 and setting them 200 years later, and everyone still knows them—who can do that now? What would be the musical cultural equivalent of that? We can't imagine what music will be like in 10 generations, let alone what it will be like in 10 years. The rate of change has become so fast. This represents, in a certain sense, a great opportunity for composers. I think you could argue that the conditions for being an artist in the Western world have never been better, by far. Artists love to complain. They love to say, "Oh, it was better back then, or it was better in the 1950s, or the standards of this or that were higher, or audiences were more interested in classical music back then," etc, etc. Don't listen to them. None of it's true.
- 9.9** This is a golden age for being a musician. There's no question. The professional conditions, the range of opportunities, are just immeasurably greater than they have been in the past. I think what's different is that we talked a little bit about the fact that,

in the post-war era, there was an entire generation of young men that was slaughtered in the war. There just simply, demographically speaking, there weren't as many young composers around at that time. Well, that's changed a lot. One of the things that we've seen in the past few decades is there's an enormous increase in the number of people attending colleges and universities. And along with that, there's been an enormous increase in the number of people studying music composition at a high level. And these music schools are just filled to the brim with young people who have the ambition of becoming composers.

9.10 Some people see that as a consequence of a profound change in the way society is structured. Some people see it as a negative development. Some people will argue that there are no longer any gatekeepers. There's no longer anybody determining a cultural standard that we have to adhere to. In the age of YouTube, anyone can be a musician. Absolutely anyone can produce a piece of music on your own. All you need is an iPhone. You don't even need an iPhone, in fact, in many instances. You just need something that you can record something on, upload it to YouTube, and there you go. You can share it potentially with the entire world. There's no barrier to access anymore.

9.11 But along with that comes a new condition for artists, which is, well, there are no gatekeepers, so who's to say what we should be paying attention to? What should be filtered out? What are the things that are trivial? What are the things that are important? Well, that becomes a harder and harder determination to make, in many respects. In addition to that, with the rise of these new media, the traditional media have largely abandoned their role that they played in the past of providing musical criticism and providing a bird's-eye view of the cultural world for the average person. They don't fill that role anymore. You can open up major newspapers in the United States or go on to cable news, and you'll find extremely little about classical music, let alone new compositions. I mean, in most instances, it simply doesn't exist. There's a couple of outlets that still do exist, but they've abandoned that role for some reason.

9.12 But the new media is filling the void and, well, things like this, of course. You can go on to YouTube now, and there are channels that have hundreds of thousands, if not millions, of subscribers that provide all of this information. In a sense, it's perhaps more difficult to have, on a societal level, a coherent idea of what exactly is going on in the arts and trying to understand who the major figures might be and who you might want to pay attention to. It becomes a somewhat more interactive process, I would argue. You have to take the initiative of searching things out and finding out where to look. But it's all there. It's all there if you're curious enough and interested enough to look for it.

9.13 There's a number of things that are changing presently and that I think will continue to change. I mentioned the fact that there's been a dramatic increase in the number of young people studying composition in conservatories and in universities. We can talk about that for a minute. The European model is largely predicated upon conservatories

for teaching and studying music. The American model tends to be more focused on universities. Composers tend to study music in an academic setting in the USA, and they tend to study it in conservatories in Europe. Now, there's some degree of overlap between the two. In the UK, for example, they have both systems going on simultaneously. In France and in Germany, it's possible to get a master's or even a doctoral degree in composition, in addition to the option of studying at a conservatory. But they're two very different approaches.

9.14 The conservatory—what that really is, as an institution—is a place where you can learn hands-on everyday skills, practical skills, practical musical skills. It's oriented towards forming musicians who will become performers, composers who will write pieces that will be played in public. The university, as you would expect, has a different orientation. It's more focused on research and on musicology and writing papers about what you're doing. In many respects, that's kind of an odd fit for artists. It's an odd fit for composition, generally. You might wonder, how did we get there? How did it become a standard thing that a composer would try to get a doctoral degree in musical composition?

9.15 Well, what happened is, in the post-war era, for a while, composers were OK because after World War I and also after World War II, there was this whole generation of wealthy war widows in the United States and in other parts of the world who would take on artists like James Joyce and support them. And also many composers were supported this way. There were some outliers like Charles Ives, who I mentioned earlier. Charles Ives was one of the great innovators and one of the great imaginative musical minds in American musical history. But he was also independently wealthy. He had an enormously successful insurance business, and so he didn't have to worry about material concerns. He could write what he wanted, and it wasn't an issue.

9.16 But artists, nevertheless, they had to find some way of making a living, some way of supporting themselves as they produced this work. In some instances, it was private patronage. It came from these widows and other people who were wealthy patrons of the arts. But that didn't last forever. There were some composers in the United States in the 1950s, 1960s, such as the composer Milton Babbitt, for example, who's an emblematic figure of this era, who thought about things very differently. For Milton Babbitt, the university offered the possibility of maintaining his independence, having essentially a proxy for the support of the nobility that existed in the time of Haydn, that would allow him to be sheltered from the vagaries of the marketplace.

9.17 In other words, you could become a tenured professor. You'd be protected. You wouldn't have to worry. You wouldn't have to worry about earning your livelihood. You could write whatever you wanted, and the university ensembles or orchestras would play your pieces. That seemed to work well. It seemed to work well for him, but it had a severe limitation, which is that suddenly this music had no organic audience. It existed within universities. It became a kind of academic subject of its own with no real contact

with the air of everyday life. In a sense, we're still seeing the aftermath of that era, where there's a kind of a disembodied new music culture that exists within universities, exists on campuses, and composers with doctoral degrees, which I always think is a little bit strange. It's sort of like an elephant with a bicycle. An artist doesn't really need a doctoral degree. It can be useful for certain things, but it's not necessary to be a composer.

9.18 The other problem, though, is that there are so many people doing doctoral degrees in composition now. They're going to graduate into a void because there is no job market for composers with doctoral degrees. It doesn't exist. This is simply being perpetuated as a way of perhaps staving off the inevitable, which is the confrontation with reality. At a certain point, you need to have an audience. You need to be able to earn a living. You need to be able to put your work out there in the world. This is still very ingrained in the imaginations of composers, that this pathway towards professional success will consist something like that.

9.19 Maybe you go to a good university, maybe you get a degree, maybe you go to a conservatory, and then hopefully—this is the thinking—you will end up with a position somewhere. You'll have financial stability, maybe, and maybe you'll get a few performances and festivals here or there, or maybe the local ensemble in the institution you're teaching at will play your pieces. But I'd like to suggest that there's another way forward, which is that, with these new technologies, we have the possibility, potentially, theoretically, of reaching vastly more people than has ever been the case in the past. Now, that's a potential audience. It's not an actual audience, so you have to bear that in mind. The fact that billions of people watch YouTube doesn't mean that if you put a video up, billions of people will watch it. Obviously, that doesn't happen most of the time.

9.20 But what I've observed is that, if you're able to present your work in a way that is direct and interact with your audience and straightforward and compelling, then people will watch it and they'll engage with it. And you can have a relationship with your audience that is extraordinarily rewarding and compelling on both sides. This I see as fundamentally a net positive. I would like to encourage musical people, people who are involved in the arts. I'm not saying don't go to university, but I am saying that I think that the way forward is to figure out how you can engender a kind of sense of connection with an audience, a real organic audience, not the captive, pre-constituted audience of music festivals, avant-garde music festivals in Europe and other parts of the world. But rather, can you put together your own audience somehow? That might be the great challenge of this new technological age, so that you don't get lost in the cacophony of individual voices out there.

9.21 I think that one thing that's starting to change is you're seeing increasingly that composers have to be responsible for the entire chain of production, from the conception of the works to having them performed, recorded, broadcast, disseminated,

and possibly even explained. In some senses, I think that's a very positive thing, but it also represents a challenge. Not every composer is necessarily gifted or adept at marketing. Not every composer knows how to communicate. Not every composer has the wherewithal or the inclination to make YouTube videos. But that's, well, it's a changing reality, I suppose, to which artists will have to adapt.

9.22 There is still this infrastructure of ensembles, orchestras, and festivals, and institutions, conservatories, universities, and so on that still exists. There are still publishers, and there are still record labels and so on. But I think that these sorts of structures are not as robust as they were in the past. They're by no means a guarantee that you'll be able to get your work out there, that you'll be able to be heard. It's certainly not a guarantee to audiences that they'll be able to figure out what's interesting or what might they enjoy. I'd like to propose that, well, we're obviously in the early stages of this, but that this new media is probably going to profoundly change the way forward for musicians.

9.23 It's possible now for an artist to be truly independent in a way that would have been unthinkable even 20 years ago. It's by no means easy. It's never been easy, in a sense, to be an artist because the value of art is implicit, but it's not explicit. We don't know how much a piece of music is worth, especially if it's a piece of not explicitly commercial music. How do you put a value on that? How do you determine what a musician should be paid? How do you determine any of these things? It's very complicated. But that potential exists now. You don't even necessarily need an employer. You don't necessarily need to be a professor. You don't need to do any of those things. You can have a Patreon account, or you can find some way to monetize your social media. Not easy, again. Most channels do not generate any revenue whatsoever, but it's a possibility. It's possible to be an independent artist, a truly independent artist.

9.24 And maybe that's a beautiful thing. We'll have to see. Come with it the possibility, again, of those who are so inclined to make work that nobody else can meddle with. No middlemen, no intermediaries. You can produce the work exactly the way you want. Nobody's going to say, "Can you do that a little bit differently? Adjust that. That piece is too long. That piece is too short." You can do what you want, in a sense. That is also potentially a beautiful thing for artists. I see this as a time of extraordinary opportunity for people in creative fields, but you have to know how to harness these technologies and figure out how to use them. So today, art is really for everybody. It's there, it's accessible. There really is no barrier to entry whatsoever, either for audiences or for creators. And the price we have to pay for that, I suppose, is the fact that, well, we don't have this centralized cultural world anymore. I don't know if there's a dialing back from that. We'll have to see.

9.25 The other effect of this is that—we talked about how in the era of modernism, there were these giants, these these hugely impactful figures—Andy Warhol, Marcel Duchamp, John Cage—these figures who everybody knows their names, they're

household words. I look at the landscape, the cultural landscape today, I don't see that. It's not obvious to me, or to a lot of people, who are the great painters living today? Is there a Picasso today? Is there a Kandinsky? Is there an equivalent to these sorts of names? What about in the music world? Do we have a Stockhausens anymore? Do we have a Brahms anymore? Do we have these sorts of figures? A lot of people would say, "Not really."

- 9.26** We moved from a situation where you had these mountain peaks, you had these colossal figures with huge international reputations. Everybody knows who they are. Household words. Remember Jackson Pollock, the painter? How did Jackson Pollock become famous? There was an article in Life magazine that came out, showing his drip paintings and asking the question, "Is this the world's greatest contemporary painter?" And everybody saw that article. Everybody. Millions and millions of people. Jackson Pollock became a household name. We're not really seeing that today because, well, for the reasons I've laid out already, there's really not really a center to this culture. There's not a centralized media that is either interested in or that is disseminating ideas about art and culture in the same way as may have been the case in the 50s and 60s.
- 9.27** But there are also just way more artists active today than was the case then. How does anyone keep up with that? How do you determine what you should listen to? The cultural landscape today is not a landscape of massive mountain peaks. We don't have the Mont-Blanc and the Matterhorn and the Everest and these massive peaks. We have foothills stretching out as far as the eye can see. Figures who will have an impact, perhaps more locally, or maybe it'll be smaller, I don't know. But they don't seem to be reaching the same level of notoriety as may have been the case in the past. Now, that could be just a simple question of our present-day vantage point, that we don't have the perspicacity, we don't have the the ability to extract ourselves out of the present moment and see it from a distance and see what's actually going on and figure out who the major figures are and all of that.
- 9.28** To some degree, that's true. These things reveal themselves with time. But the way it looks right now is it's just not obvious. It's not obvious who the Picassos might be. Some would say that's a good thing. Well, maybe we don't want our culture to be dominated by 4 or 5 massive people like this, getting a disproportionate amount of attention, or perhaps it is proportionate to their genius. I don't know. Others would say that this is unfortunate because, well, we don't know where to look. There's no mountain peaks. How do you orient your vision? But I don't have an answer to this. So we'll see.
- 9.29** So I mentioned that the conditions for artists in many respects have never been better than they are today. There are obviously challenges of a sort that are entirely new. In order to get your work out there and succeed in this environment, you have to have a range of skills that is very, very broad. You have to obviously be a master of your craft, but you also have to know how to communicate. You have to know about new

technologies and stay on top of those. You might need to assemble a team around you. I mean, there's all sorts of new things that composers have to do that they didn't have to do in the past. If you were a successful composer of Italian opera in the 19th century, you would have had people taking care of these things for you. You would have had agents and publishers. But that's not necessarily the case anymore. The artist is responsible for all of this now.

9.30 But our conditions are just so much better. I mentioned the Austrian composer, Anton von Webern, earlier. Webern was a student of Arnold Schoenberg and born in Vienna in 1883. And an absolutely remarkable figure. He's one of my favorite composers. But the life story of Webern is unspeakably tragic, and not just him, but so many composers in the first part of the 20th century. Edgar Varèse is another one. He's a composer who had an absolutely profound impact on 20th-century music. He had a completely new conception of sound, a completely new conception of what an orchestra could be. He was writing for electronic instruments, new inventions like the theremin, and all of these incredible new things. But Varèse also had a lousy life. So did Béla Bartók, the Hungarian composer. Died in absolute poverty and obscurity in New York City.

9.31 Webern was born into the Austro-Hungarian Empire in 1883, and by the time he died in 1945, Austria was a tiny little ruined state, just a sliver of what it had once been. And Webern had the misfortune of being a modern-oriented composer throughout both world wars, and he lost everything. His music was banned by the Nazis. He progressively lost any ability to make a living. His son was killed in World War II, terrible thing to live through. He himself fell very ill with dysentery shortly after that. And then the culmination of this tragic life is that he was accidentally shot to death by an American soldier in 1945, who mistakenly took him, so the story goes, as a belligerent. This was after the end of the war, but there were still a presence of American soldiers in Austria at that time, and he just happened to be in the wrong place at the wrong time and was shot to death. 1945. It's a horrible, tragic story.

9.32 The number of performances that Webern had of his music during his lifetime is tiny. A lot of the pieces that he wrote, he never even got to hear. It was suppressed. It's not just Webern. There are all kinds of composers in the Communist countries, in Hungary, for example, who were absolutely sealed off from developments in the West. They weren't even allowed to know about it. You couldn't bring the scores in. You couldn't bring recordings in. You couldn't hear things on the radio. It was all blocked out. There's one composer, another Hungarian, György Ligeti, who managed to escape Communist Hungary and escape into the West and talks about what that was like and how he could very easily have been killed. And "I had to take that risk" because there was no way for him to be a musician in the environment that he was in.

9.33 So there's lots of stories like this in the first part of the 20th century. Just terrible, terrible stories about artists basically having no audience, no ability to get their work performed, or very little, and living in terrible circumstances. We could also talk about

Dmitri Shostakovich and his experiences contending with the Russian authorities during the course of his career. There are many, many instances of this. Today, if you're a composer living in the Western world, it's extremely unlikely you're going to be dealing with anything like this. It's extremely unlikely. At a baseline level, you're going to be able to survive. You're going to be able to feed yourself. You're not going to be systematically shut out of the profession. You have means to get your work out onto the Internet. All of this exists.

- 9.34** In that sense, if I compare the opportunities that I've had with some of the things that my musical heroes had to endure in the 20th century, it's hard not to feel ridiculously fortunate. It's hard not to feel grateful for the opportunities that we have now. This is why I spend a lot of time reading biographies of musicians. I have to be reminded of these things, of what it was like being a composer in 1936 in Germany or Austria or France or all kinds of other parts of the world. So there's a lot of problems that we just don't have anymore. We don't have to worry about. I suppose you could argue that there are other things that are emerging, but at least on that level.
- 9.35** So this is my vision of where we're at right now. It's a strange time, I suppose, in many respects, to be an artist. We have opportunities ahead of us that are, and would have been, unimaginable to the artists of even two generations ago. I think we're going to see the rise of the independent artist, and I hope that that will continue. I hope artists have the courage to do that and also to follow their artistic visions with absolute fidelity and see where they lead and connect with people, connect with audiences, and see what emerges out of that.
- 9.36** So against this backdrop, the last thing I would say about that is we've also observed the expansion of the Western conception of music to basically the entire world, which is not to say that Western music has dominated over other musical cultures or that they don't exist anymore, but simply that —well, I can give you an anecdote about this. When I was a student in Paris, the composition class that I was a part of was totally international. We had students from Korea, from Japan, from the Czech Republic. I was Canadian. Obviously, there were French people in the class, and it was completely international. There were Australians, people from all over the world. If you go into any composition program, pretty much anywhere in the Western world now, you'll see something similar.
- 9.37** It's possible to be born pretty much anywhere and take an interest in this approach and want to study composition. What I'm seeing also, I work with a lot of different students in my private practice as a teacher, and I'm seeing students from all over the world who are interested in creating a synthesis of their own musical traditions and cultures and then aspects of the Western tradition, which has been extremely interesting to see. I suppose I have this vantage point because I work with so many students, of just seeing this incredible level of creativity just coming from everywhere. It can really just come

from anywhere now. And that really fills me with gratitude and excitement and optimism for the future.

- 9.38** I'm going to round off this lecture with a few words about the creative process, about how a composer actually goes about their work. I think that for a lot of people, this is a very abstract idea. We don't really understand what goes on when a composer is inventing a piece of music. What does that actually look like? What's that process like? I get asked this all the time. I remember I used to get questions like, "What sort of music do you write? How does that work? And do you work at a piano? Do you write with a pen or a pencil?" And all this. So the creative process is obviously a very individual thing.
- 9.39** But this is something that I teach quite a lot when I'm working with students, because one of the things I've noticed is, in order to be an artist, you have to develop a way of fitting the work into your life. You have to develop a routine, in other words, that is sustainable and that you can iterate across time. Think of it as a practice, the same way that a lawyer has a practice or a dentist has a practice. You show up and you go to work. And artists need that, too. We need that regularity.
- 9.40** So first thing that I would say about the creative process is a lot of it has to do with figuring out how you can have a sustainable artistic practice and endeavor that you can have as part of your life on an ongoing basis, and not just from time to time, because if you do it on the occasional weekend, it's going to be very hard to make any kind of progress. In my case, what has emerged is that I do quite a lot of different things. I'm involved in teaching, I'm involved in YouTube, I have a podcast, I write a lot of compositions as well. I have premieres in different parts of the world, and I write orchestral pieces and solo pieces, chamber pieces, and it's a very busy life.
- 9.41** But what I've observed, and other composers have observed this as well, incidentally, is that the creative process is cognitively demanding. It's hard. It's hard work. I believe that the optimal way to approach it is to figure out, first of all, what is the amount of time that you can spend writing something, where you are absolutely focused, you have a high quality of attention that you can bring to bear upon the creative process? You're not distracted. You're not checking your phone every 5 minutes. You're not getting bored. What's the amount of time you can realistically sustain where you're absolutely there in the flow of generating ideas and working on your compositions? Figure out what that span is.
- 9.42** It is different for different people. There are composers who claim to be able to do that for 6 or 8 hours a day. There are people who can sustain it for an hour or two. That's an individual thing. I think on average, based upon my experiences and the range of students that I've worked with, most people can manage about 3 hours of sustained focus on some kind of a creative endeavor. And 3 hours, by the way, is really the upper limit. And beyond that point, it starts to be a question of diminishing returns because it's very, very difficult to sustain.

- 9.43 What I tend to advocate for creative people or anybody who's interested in creating music is to figure out what that span is. Can you do it for 15 minutes? Can you do it for 3 hours? Can you do it for 8 hours? What is it? And what can you realistically fit into your life? And then try to make that a habit. Try to make it a regular thing. Make sure that you keep to that. A lot of creativity is actually a lot less mysterious when you start looking at it from that perspective, that there has to be a dimension that's sustainable. If it's too irregular, it's too fitful, if you're able to do 10 minutes one day and 3 hours the next day, and then maybe you take a break for 2 weeks, work on it for 10 hours in a huge panicked rush—
- 9.44 I know there are people that work that way, and there are some people that claim that that's working for them. If that's great, if that's true, then great. But in most instances, most people I've worked with, it's better to have a routine of some kind. That's the first thing, is you have to carve out that space to do your work and make sure that nobody's interrupting you and make sure that you're not being distracted all the time.
- 9.45 Then the second thing is the process of generating ideas, and I can only talk about my own approach, what that looks like. In the beginning, I don't know what I'm going to do, and I really enjoy that. That, to me, is a precious part of the process because it's like you have this potential to create a world ahead of you. You don't know what that world is going to be populated with. The metaphor that I often use is, it's sort of like going deep-sea diving, in a sense. You're in this environment where you've got your intuitions and you can sense there's something there under the water. There are forms that you can sense. Maybe there's a texture, and you can gradually figure out what the contours are of this thing that you're trying to create, you're trying to bring into the world. That's often how it appears to me at the beginning, is there's a desire to create, and there's a creative drive, a creative volition, and there's the intuition that there's something that needs to be said, but you haven't quite formulated it yet.
- 9.46 And so the beginning stages are slow. The very early stages of a piece, I tend not to do those in 3 or 4 hours. Tend to sit and reflect on what I'd like to do for actually a relatively short amount of time. Then as I get into the flow of the piece, then the working days get longer and longer. And then by the end, it can be easily 10 hours or more at a stretch. It gets very intense in the last few weeks, in particular. But the beginning stage is so lovely because you have the possibility of pulling something out of the ether, as it were, something that's never been said before, something that's never been expressed, something that nobody's ever given a form to. Isn't that just the greatest thing in the world? It's so exciting. That's my greatest joy in art, is to surprise myself and come up with things that I wouldn't have imagined.
- 9.47 I had a conversation with a Korean composer, Unsuk Chin, a couple years ago. She said something that I thought was really fascinating. She said that after she's finished writing a piece, she no longer has any memory of having written it. She says, "It's like I was in a trance. The piece just emerges, and then at the end, it's like, 'Oh, there it is.'"

And I know exactly what she's talking about. There's a level upon which, not to get too mystical about it, but there's a dimension of creativity that is involuntary, that you don't know what you're going to be given to say or what is going to emerge out of this process. You can really surprise yourself. I just love that. I love that aspect of it.

- 9.48** The Romantics had this idea, and you see this in Wagner and Mahler and Schoenberg and a few other composers of this era, late 19th century, early 20th century. They had this idea that the composer, in a sense, is like a radio capturing transmissions from somewhere else and merely relaying them. The composer is merely taking dictation. Mahler talked about that with the composition of his Eighth Symphony. We heard a couple of excerpts of that earlier, and this colossally huge piece that he wrote in this kind of exalted state in about 2 weeks. He did this draft of this massive symphony. Two weeks, it's mind-boggling. He talks about the notes just flying out of his pen, just this absolutely spontaneous eruption of creativity.
- 9.49** I wish that happened to me all the time. That would be nice. Most of the time it's not like that, I can tell you. But that's a kind of wonderful ideal. It implies that there's a dimension to the artistic process that is involuntary and compulsive. I suppose on a level, maybe that makes me a Romantic, but I adhere to that idea. Once again, you say the things as an artist that you can't prevent yourself from saying. If it's voluntary, then I'm not so interested. It has to be something you're compelled to say.
- 9.50** So there we are. Music will always be with us. Music is an indelible part of human civilization. As I said at the outset of this series of lectures, it's always been with us. It's present in virtually every musical culture around the world. I don't believe it's going away anytime soon. What I want to impress upon you is that there is a vast ocean of music out there that you can dive into, and it's inexhaustible, and it's there for everybody. It's beyond possession. You just have to pay attention.
- 9.51** So I want to encourage anyone listening to adopt this new relationship with art and engage with it. See what its limits are. See what your limits are also. What are the limits of your taste? What are the limits of your curiosity? Explore a little bit more. Don't be limited by the things that you know you enjoy already. See what else there is. I think there are great rewards for doing that. That's it. Thank you for listening and invite your questions.